

ELITE IGCSE ACADEMY

Edexcel IGCSE Mathematics A - Unit 2 (4WM2)

# Higher Tier

## Elite IGCSE Classified Unit 2 Expertise Answers (Q20+)

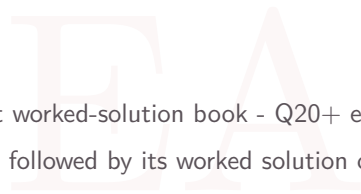
133 Questions | Question page then worked-solution page

PREPARED & CLASSIFIED BY

**Dr. Eslam Ahmed**

*Assistant Lecturer, Cairo University Faculty of Engineering*

Modular palette | Unit 2 / 4WM2



Student worked-solution book - Q20+ expertise.  
Each question is followed by its worked solution on the next page.

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**25 Vectors****265**

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# Direct & Inverse Proportion

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**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jan 2022

Q#: 20

Marks: 4

TOPIC: Direct &amp; Inverse Proportion

Source: Jan 2022 P2HR

- 20**  $y$  is inversely proportional to  $\sqrt{x}$   
 $x$  is directly proportional to  $T^3$

Given that  $y = 8$  when  $T = 25$

find the exact value of  $T$  when  $y = 27$

$T = \dots\dots\dots$

(Total for Question 20 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2022 P2HR | Question 20

Direct &amp; Inverse Proportion

**Topic check.** Direct & Inverse Proportion is correct.**Method**

$$y \propto \frac{1}{\sqrt{x}}$$

and

$$x \propto T^3$$

So

$$\sqrt{x} \propto T^{3/2}$$

Therefore

$$y \propto \frac{1}{T^{3/2}}$$

So  $yT^{3/2}$  is constant.When  $y = 8$  and  $T = 25$ ,

$$8(25)^{3/2} = 8(125) = 1000$$

When  $y = 27$ ,

$$27T^{3/2} = 1000$$

$$T^{3/2} = \frac{1000}{27} = \left(\frac{10}{3}\right)^3$$

$$T = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$

**FINAL ANSWER**

$$\frac{100}{9}$$

# Algebraic Fractions

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**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: Specimen

Q#: 21

Marks: 3

TOPIC: **Algebraic Fractions**

Source: Specimen 4WM2H

21  $a = \frac{14}{3x-7} \quad x = \frac{7}{4y-3}$

Express  $a$  in the form  $\frac{py+q}{ry+s}$  where  $p, q, r$  and  $s$  are integers.

Give your answer in its simplest form.

$$a = \dots\dots\dots$$

(Total for Question 21 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Specimen 4WM2H | Question 21

Algebraic Fractions

**Topic check.** Algebraic Fractions. The tag is correct because the task is substitution and simplification of algebraic fractions.

## Solution

$$a = \frac{14}{3x - 7}, \quad x = \frac{7}{4y - 3}$$

Substitute for  $x$ :

$$a = \frac{14}{3\left(\frac{7}{4y-3}\right) - 7}$$

Simplify the denominator:

$$\begin{aligned} 3\left(\frac{7}{4y-3}\right) - 7 &= \frac{21}{4y-3} - 7 \\ &= \frac{21 - 7(4y-3)}{4y-3} = \frac{42 - 28y}{4y-3} \end{aligned}$$

So

$$\begin{aligned} a &= \frac{14}{\frac{42-28y}{4y-3}} = \frac{14(4y-3)}{42-28y} \\ a &= \frac{4y-3}{3-2y} \end{aligned}$$

## FINAL ANSWER

$$a = \frac{4y-3}{3-2y}.$$

# Algebraic Proof

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**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: May2025

Q#: 20

Marks: 3

TOPIC: **Algebraic Proof**

Source: May 2025 4WM2H

**20** |  $a, b$  and  $c$  are three consecutive even numbers where  $a < b < c$

Prove algebraically that  $b^2 = ac + 4$

**(Total for Question 20 is 3 marks)**



## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM2H | Question 20

Algebraic Proof

**Solution**

Let

$$a = 2n, \quad b = 2n + 2, \quad c = 2n + 4$$

Then

$$ac + 4 = (2n)(2n + 4) + 4$$

$$= 4n^2 + 8n + 4$$

$$= (2n + 2)^2 = b^2$$

So  $b^2 = ac + 4$ .

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jan 2020

Q#: 25

Marks: 3

TOPIC: **Algebraic Proof**

Source: Jan 2020 P2HR

**25**  $N$  is a multiple of 5

$$A = N + 1$$

$$B = N - 1$$

Prove, using algebra, that  $A^2 - B^2$  is always a multiple of 20

---

**(Total for Question 25 is 3 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 25

Algebraic Proof

**Topic check.** Corrected from Right-Angled Triangles to Algebraic Proof.**Method**Since  $N$  is a multiple of 5, let

$$N = 5k$$

where  $k$  is an integer.

Given

$$A = N + 1, \quad B = N - 1$$

So

$$A^2 - B^2 = (N + 1)^2 - (N - 1)^2$$

Expand:

$$(N + 1)^2 = N^2 + 2N + 1$$

$$(N - 1)^2 = N^2 - 2N + 1$$

Therefore

$$A^2 - B^2 = (N^2 + 2N + 1) - (N^2 - 2N + 1)$$

$$= 4N$$

Substitute  $N = 5k$ :

$$4N = 4(5k) = 20k$$

Since  $k$  is an integer,  $20k$  is a multiple of 20.**FINAL ANSWER** $A^2 - B^2$  is always a multiple of 20.

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2024

Q#: 23

Marks: 4

TOPIC: **Algebraic Proof**

Source: Nov 2024 P2H

23 Show that  $\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x) = n$

where  $n$  is an integer to be found.  
Show clear algebraic working.

(Total for Question 23 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2024 P2H | Question 23

Algebraic Proof

Topic check. Correct.

## Method

$$\frac{16x^2 - 36}{x - 7} \div \frac{2x^2 + 7x + 6}{x^2 - 5x - 14} - (7 + 8x)$$

Factorise:

$$16x^2 - 36 = 4(2x - 3)(2x + 3)$$

$$2x^2 + 7x + 6 = (2x + 3)(x + 2)$$

$$x^2 - 5x - 14 = (x - 7)(x + 2)$$

So

$$\begin{aligned} & \frac{4(2x - 3)(2x + 3)}{x - 7} \div \frac{(2x + 3)(x + 2)}{(x - 7)(x + 2)} - (7 + 8x) \\ &= \frac{4(2x - 3)(2x + 3)}{x - 7} \times \frac{(x - 7)(x + 2)}{(2x + 3)(x + 2)} - (7 + 8x) \\ &= 4(2x - 3) - (7 + 8x) \\ &= 8x - 12 - 7 - 8x \\ &= -19 \end{aligned}$$

So  $n = -19$ , which is an integer.

## FINAL ANSWER

$$n = -19$$

# Solving Inequalities

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**CLASSIFIED PROBLEM**

Paper: 4WM2HR

Session: May2025

Q#: 20

Marks: 3

TOPIC: **Solving Inequalities**

Source: May 2025 4WM2HR

**20** Solve the inequality  $2x^2 - 7x - 15 > 0$

Show clear algebraic working.

**(Total for Question 20 is 3 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM2HR | Question 20

Solving Inequalities

**Required:**Solve  $2x^2 - 7x - 15 > 0$ . Show clear algebraic working.**Step:**

Factorise the quadratic:

$$2x^2 - 7x - 15 = (2x+3)(x-5)$$

$$\text{Check: } (2x+3)(x-5) = 2x^2 - 10x + 3x - 15 = 2x^2 - 7x - 15$$

**Step:**

Roots: set each factor to zero:

$$2x + 3 = 0 \implies x = -\frac{3}{2}; \quad x - 5 = 0 \implies x = 5$$

**Step:**

Since the coefficient of  $x^2$  is  $+2 > 0$ , the parabola opens **upward**. The expression is  $> 0$  *outside* the roots:

**FINAL ANSWER**

$$x < -\frac{3}{2} \text{ or } x > 5$$



# Simultaneous Equations

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**CLASSIFIED PROBLEM**

Paper: 4WM2HR

Session: May2025

Q#: 22

Marks: 5

TOPIC: Simultaneous Equations

Source: May 2025 4WM2HR

**22** Solve the simultaneous equations

$$y^2 + 5y + x^2 = 12$$

$$x = y + 5$$

Show clear algebraic working.

---

(Total for Question 22 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**May 2025 4WM2HR** | **Question 22**  
Simultaneous Equations

**Required:**

$$\text{Solve } y^2 + 5y + x^2 = 12 \text{ and } x = y + 5.$$

**Step:**

*Substitute  $x = y + 5$  into the first equation :*

$$y^2 + 5y + (y + 5)^2 = 12$$

**Step:**

$$\text{Expand } (y + 5)^2 = y^2 + 10y + 25 :$$

$$y^2 + 5y + y^2 + 10y + 25 = 12 \implies 2y^2 + 15y + 13 = 0$$

**Step:**

Factorise:

$$(2y + 13)(y + 1) = 0 \implies y = -\frac{13}{2} \text{ or } y = -1$$

**Step:**

*Find corresponding  $x$  values using  $x = y + 5$  :*

**FINAL ANSWER**

$$(x, y) = (4, -1) \text{ and } \left(-\frac{3}{2}, -\frac{13}{2}\right)$$

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: Nov2025

Q#: 20

Marks: 5

TOPIC: Simultaneous Equations

Source: Nov 2025 4WM2H

**20** Solve the simultaneous equations

$$\begin{aligned}x^2 + y^2 &= 3xy - 11 \\ y &= x + 2\end{aligned}$$

Show clear algebraic working.

---

**(Total for Question 20 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 20

Simultaneous Equations

**1** Substitute  $y = (x+2)$  into the quadratic equation.

$$x^2 + (x+2)^2 = 3x(x+2) - 11$$

**2** Expand all brackets.

$$x^2 + (x^2 + 4x + 4) = 3x^2 + 6x - 11$$

$$2x^2 + 4x + 4 = 3x^2 + 6x - 11$$

**3** Move all terms to one side to set the equation to zero.

$$0 = 3x^2 - 2x^2 + 6x - 4x - 11 - 4$$

$$x^2 + 2x - 15 = 0$$

**4** Factorise the quadratic equation.

$$(x+5)(x-3) = 0$$

So,  $x = -5$  or  $x = 3$ .

**5** Find corresponding  $y$  values using  $y = x + 2$ .

$$\text{If } x = -5 \implies y = -5 + 2 = -3$$

$$\text{If } x = 3 \implies y = 3 + 2 = 5$$

**FINAL ANSWER**

(-5,-3) and (3,5)

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: Specimen

Q#: 20

Marks: 5

TOPIC: Simultaneous Equations

Source: Specimen 4WM2H

**20** Solve the simultaneous equations

$$x - 2y = 3$$

$$x^2 - y^2 + 2x = 10$$

Show clear algebraic working.

---

(Total for Question 20 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Specimen 4WM2H | Question 20

Simultaneous Equations

**Topic check.** Simultaneous Equations. The tag is correct; this is a linear and quadratic simultaneous equation.

**Solution**

From

$$x - 2y = 3$$

$$x = 2y + 3$$

Substitute into  $x^2 - y^2 + 2x = 10$ :

$$(2y + 3)^2 - y^2 + 2(2y + 3) = 10$$

$$4y^2 + 12y + 9 - y^2 + 4y + 6 = 10$$

$$3y^2 + 16y + 5 = 0$$

$$(3y + 1)(y + 5) = 0$$

So

$$y = -\frac{1}{3} \quad \text{or} \quad y = -5$$

If  $y = -\frac{1}{3}$ ,

$$x = 2\left(-\frac{1}{3}\right) + 3 = \frac{7}{3}$$

If  $y = -5$ ,

$$x = 2(-5) + 3 = -7$$

**FINAL ANSWER**

$\left(\frac{7}{3}, -\frac{1}{3}\right)$  and  $(-7, -5)$ .

# Solving Inequalities

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**CLASSIFIED PROBLEM**

Paper: P2H

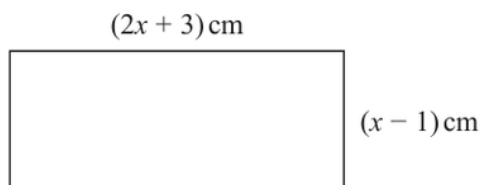
Session: Jan 2022

Q#: 22

Marks: 5

TOPIC: **Solving Inequalities**

Source: Jan 2022 P2H

**22** Here is a rectangle.Diagram **NOT**  
accurately drawnGiven that the area of the rectangle is less than  $75\text{ cm}^2$ find the range of possible values of  $x$ 

---

**(Total for Question 22 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 22  
Solving Inequalities

**Topic check.** Corrected from Functions to Solving Inequalities.

**Method**

The rectangle has side lengths

$$2x + 3$$

and

$$x - 1$$

For the side lengths to be possible,

$$x > 1$$

The area is less than 75:

$$(2x + 3)(x - 1) < 75$$

$$2x^2 + x - 3 < 75$$

$$2x^2 + x - 78 < 0$$

Factorise:

$$2x^2 + x - 78 = (2x + 13)(x - 6)$$

So

$$-\frac{13}{2} < x < 6$$

Combining with  $x > 1$ :

$$1 < x < 6$$

**FINAL ANSWER**

$$1 < x < 6$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2024

Q#: 20

Marks: 4

TOPIC: **Solving Inequalities**

Source: Jun 2024 P2HR

- 20** Solve  $6x^2 - 7x - 20 > 0$   
Show clear algebraic working.

---

(Total for Question 20 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2024 P2HR** | **Question 20**  
Solving Inequalities

**Topic check.** Corrected from Solving Quadratic Equations to Solving Inequalities.

**Method**

$$6x^2 - 7x - 20 > 0$$

Factorise:

$$6x^2 - 7x - 20 = (3x + 4)(2x - 5)$$

The critical values are

$$x = -\frac{4}{3} \quad \text{and} \quad x = \frac{5}{2}$$

Since the coefficient of  $x^2$  is positive, the quadratic is greater than 0 outside the roots.

**FINAL ANSWER**

$$x < -\frac{4}{3} \quad \text{or} \quad x > \frac{5}{2}$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2025

Q#: 21

Marks: 3

TOPIC: **Solving Inequalities**

Source: Jun 2025 P2HR

**21** Solve the inequality  $2x^2 - 7x - 15 > 0$ 

Show clear algebraic working.

---

(Total for Question 21 is 3 marks)

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Jun 2025 P2HR** | **Question 21**  
Solving Inequalities

**Topic check.** Correct.

**Method**

$$2x^2 - 7x - 15 > 0$$

Factorise:

$$2x^2 - 7x - 15 = (2x + 3)(x - 5)$$

The critical values are

$$x = -\frac{3}{2} \quad \text{and} \quad x = 5$$

The quadratic is greater than 0 outside the roots.

**FINAL ANSWER**

$$x < -\frac{3}{2} \quad \text{or} \quad x > 5$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2024

Q#: 20

Marks: 3

TOPIC: **Solving Inequalities**

Source: Nov 2024 P2H

- 20** Solve the inequality  $10x^2 + 11x - 21 < 0$   
Show clear algebraic working.

---

(Total for Question 20 is 3 marks)

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Nov 2024 P2H | Question 20**  
Solving Inequalities

**Topic check.** Correct.

**Method**

$$10x^2 + 11x - 21 < 0$$

Factorise:

$$10x^2 + 11x - 21 = (10x + 21)(x - 1)$$

The critical values are

$$x = -\frac{21}{10} \quad \text{and} \quad x = 1$$

The quadratic is less than 0 between the roots.

**FINAL ANSWER**

$$-\frac{21}{10} < x < 1$$



**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2025

Q#: 25

Marks: 3

TOPIC: **Solving Inequalities**

Source: Nov 2025 P1H

- 25** Solve the inequality  $2x^2 + x - 28 > 0$   
Show clear algebraic working.

---

(Total for Question 25 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Nov 2025 P1H** | **Question 25**  
Solving Inequalities

**Topic check.** Correct.

**Method**

$$2x^2 + x - 28 > 0$$

Factorise:

$$2x^2 + x - 28 = (2x - 7)(x + 4)$$

The critical values are

$$x = \frac{7}{2} \quad \text{and} \quad x = -4$$

The quadratic is greater than 0 outside the roots.

**FINAL ANSWER**

$$x < -4 \text{ or } x > \frac{7}{2}$$

# Sequences

---

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: May2025

Q#: 24

Marks: 4

TOPIC: Sequences

Source: May 2025 4WM2H

- 24** The 21st term of an arithmetic series is 109  
The sum of the first 52 terms of the series is 4381  
Work out the 5th term of the series.  
Show clear algebraic working.

.....  
(Total for Question 24 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 24  
Sequences

**Solution**

For an arithmetic series,

$$a + 20d = 109$$

and

$$\frac{52}{2}(2a + 51d) = 4381$$

$$26(2a + 51d) = 4381$$

Solving gives

$$d = -4.5, \quad a = 199$$

The fifth term is

$$a + 4d = 199 + 4(-4.5) = 181$$

**FINAL ANSWER**

181.

**CLASSIFIED PROBLEM**

Paper: 4WM2HR

Session: May2025

Q#: 24

Marks: 6

TOPIC: Sequences

Source: May 2025 4WM2HR

**24** An arithmetic series  $P$  has first term  $a$  and common difference  $d$

The sum of the first 40 terms of  $P$  is 2070

An arithmetic series  $Q$  has first term  $2a$  and common difference  $3d$

The 25th term of  $Q$  is 186

Work out the sum of the first 40 terms of  $Q$   
Show clear algebraic working.

---

(Total for Question 24 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2HR | Question 24  
Sequences

**Required:**

Series  $P$  : first term  $a$ , c.d.  $d$ ;  $S_{40}(P) = 2070$ . Series  $Q$  : first term  $2a$ , c.d.  $3d$ ; 25th term of  $Q = 186$ . Find  $S_{40}(Q)$ .

**Step:**

Form equation from  $S_{40}(P) = 2070$  :

$$S_{40} = \frac{40}{2}[2a + 39d] = 20(2a + 39d) = 2070 \implies 2a + 39d = 103.5 \quad (1)$$

**Step:**

Form equation from 25th term of  $Q = 186$  :

$$T_{25}(Q) = 2a + 24(3d) = 2a + 72d = 186$$

**Step:**

Solve simultaneously – subtract (1) from (2):

$$33d = 82.5 \implies d = 2.5$$

**Step:**

Substituted  $d = 2.5$  into (1) :

$$2a + 39(2.5) = 103.5 \implies 2a = 6 \implies a = 3$$

**Step:**

Compute  $S_{40}(Q)$ , with first term  $= 2a = 6$  and c.d.  $= 3d = 7.5$  :

$$S_{40}(Q) = \frac{40}{2}[2(6) + 39(7.5)] = 20[12 + 292.5] = 20 \times 304.5 = 6090$$

**FINAL ANSWER**

$$S_{40}(Q) = 6090$$

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: Nov2025

Q#: 26

Marks: 5

TOPIC: Sequences

Source: Nov 2025 4WM2H

- 26** The first term of an arithmetic series is 10  
The 20th term of the series is 86
- The sum of the first  $N$  terms of the series is 5194

Work out the value of  $N$   
Show your working clearly.

 $N = \dots\dots\dots$ 

---

(Total for Question 26 is 5 marks)

---



## WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 26  
Sequences

**1 Find the common difference  $d$ .**

The  $n^{\text{th}}$  term is given by  $u_n = a + (n-1)d$ .

$$u_{20} = 10 + 19d = 86$$

$$19d = 76$$

$$d = \frac{76}{19} = 4$$

**2 Use the sum formula to form an equation in terms of  $N$ .**

$$S_N = \frac{N}{2} [2a + (N-1)d]$$

Substitute the known values  $S_N = 5194$ ,  $a = 10$ , and  $d = 4$ :

$$5194 = \frac{N}{2} [2(10) + (N-1)(4)]$$

**3 Simplify and solve the quadratic equation.**

Multiply by 2 to clear the fraction:

$$10388 = N[20 + 4N - 4]$$

$$10388 = N(4N + 16)$$

$$4N^2 + 16N - 10388 = 0$$

Divide the entire equation by 4 to make it simpler:

$$N^2 + 4N - 2597 = 0$$

**4 Use the quadratic formula to find  $N$ .**

$$N = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$N = \frac{-4 \pm \sqrt{16 - 4(1)(-2597)}}{2}$$

$$N = \frac{-4 \pm \sqrt{16 + 10388}}{2}$$

$$N = \frac{-4 \pm \sqrt{10404}}{2}$$

$$N = \frac{-4 \pm 102}{2}$$

Since  $N$  must be a positive integer, we discard the negative root:

$$N = \frac{-4 + 102}{2} = \frac{98}{2} = 49$$

**FINAL ANSWER**

$$N = 49$$

Always double-check that your  $N$  (number of terms) evaluates to a positive whole integer. If it ends up as a decimal or fraction, recheck your arithmetic progression formulas.

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: Specimen

Q#: 24

Marks: 5

TOPIC: Sequences

Source: Specimen 4WM2H

- 24** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.

Show clear algebraic working.

.....

---

(Total for Question 24 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Specimen 4WM2H | Question 24  
Sequences**Topic check.** Sequences. The tag is correct.**Solution**Let the first term be  $a$  and the common difference be  $d$ .

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

So

$$S_{10} = 5(2a + 9d)$$

and

$$S_5 = \frac{5}{2}(2a + 4d)$$

Given  $S_{10} = 4S_5$ :

$$5(2a + 9d) = 4\left(\frac{5}{2}(2a + 4d)\right)$$

$$10a + 45d = 20a + 40d$$

$$5d = 10a$$

$$d = 2a$$

The 8th term is 45:

$$a + 7d = 45$$

Substitute  $d = 2a$ :

$$a + 14a = 45$$

$$15a = 45$$

$$a = 3$$

**FINAL ANSWER**

the first term is 3.

# Simultaneous Equations

---

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2022

Q#: 21

Marks: 5

TOPIC: Simultaneous Equations

Source: Jun 2022 P2HR

**21** Solve the simultaneous equations

$$\begin{aligned}x - 2y &= 3 \\ x^2 - y^2 + 2x &= 10\end{aligned}$$

Show clear algebraic working.

---

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2022 P2HR | Question 21**  
Simultaneous Equations

**Topic check.** Correct.

**Method**

$$x - 2y = 3$$

So

$$x = 2y + 3$$

Substitute into

$$x^2 - y^2 + 2x = 10$$

$$(2y + 3)^2 - y^2 + 2(2y + 3) = 10$$

$$4y^2 + 12y + 9 - y^2 + 4y + 6 = 10$$

$$3y^2 + 16y + 5 = 0$$

$$(3y + 1)(y + 5) = 0$$

So

$$y = -\frac{1}{3} \quad \text{or} \quad y = -5$$

Find  $x$ :

$$y = -\frac{1}{3} \Rightarrow x = \frac{7}{3}$$

$$y = -5 \Rightarrow x = -7$$

**FINAL ANSWER**

$$(x, y) = \left(\frac{7}{3}, -\frac{1}{3}\right) \text{ or } (-7, -5)$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2023

Q#: 21

Marks: 5

TOPIC: Simultaneous Equations

Source: Jun 2023 P2H

**21** Solve the simultaneous equations

$$\begin{aligned}2x^2 + 3y^2 &= 11 \\ x &= 3y - 1\end{aligned}$$

Show clear algebraic working.

---

**(Total for Question 21 is 5 marks)**



## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2023 P2H | Question 21

Simultaneous Equations

Topic check. Correct.

## Method

$$2x^2 + 3y^2 = 11$$

$$x = 3y - 1$$

Substitute:

$$2(3y - 1)^2 + 3y^2 = 11$$

$$18y^2 - 12y + 2 + 3y^2 = 11$$

$$21y^2 - 12y - 9 = 0$$

$$7y^2 - 4y - 3 = 0$$

$$(7y + 3)(y - 1) = 0$$

So

$$y = -\frac{3}{7} \quad \text{or} \quad y = 1$$

Find  $x$ :

$$y = -\frac{3}{7} \Rightarrow x = -\frac{16}{7}$$

$$y = 1 \Rightarrow x = 2$$

## FINAL ANSWER

$$(x, y) = \left(-\frac{16}{7}, -\frac{3}{7}\right) \text{ or } (2, 1)$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2024

Q#: 22

Marks: 5

TOPIC: Simultaneous Equations

Source: Jun 2024 P2HR

**22** Solve the simultaneous equations

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

Show clear algebraic working.

---

**(Total for Question 22 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2024 P2HR | Question 22**  
Simultaneous Equations

**Topic check.** Correct.

**Method**

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

Substitute:

$$x^2 + (3x - 1)^2 = 3x - 1 + 11$$

$$x^2 + 9x^2 - 6x + 1 = 3x + 10$$

$$10x^2 - 9x - 9 = 0$$

$$(5x + 3)(2x - 3) = 0$$

So

$$x = -\frac{3}{5} \quad \text{or} \quad x = \frac{3}{2}$$

Find  $y$ :

$$x = -\frac{3}{5} \Rightarrow y = -\frac{14}{5}$$

$$x = \frac{3}{2} \Rightarrow y = \frac{7}{2}$$

**FINAL ANSWER**

$$(x, y) = \left(-\frac{3}{5}, -\frac{14}{5}\right) \text{ or } \left(\frac{3}{2}, \frac{7}{2}\right)$$

**CLASSIFIED PROBLEM****Paper: P1H****Session: Jun 2025****Q#: 22****Marks: 5**TOPIC: **Simultaneous Equations**

Source: Jun 2025 P1H

**22** Solve the simultaneous equations

$$\begin{aligned}x^2 + 3y + y^2 &= 7 \\ y &= x + 2\end{aligned}$$

Show clear algebraic working.

---

**(Total for Question 22 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2025 P1H | Question 22

Simultaneous Equations

Topic check. Correct.

## Method

$$x^2 + y^2 + 3y = 7$$

$$y = x + 2$$

Substitute:

$$x^2 + (x + 2)^2 + 3(x + 2) = 7$$

$$x^2 + x^2 + 4x + 4 + 3x + 6 = 7$$

$$2x^2 + 7x + 3 = 0$$

$$(2x + 1)(x + 3) = 0$$

So

$$x = -\frac{1}{2} \quad \text{or} \quad x = -3$$

Find  $y$ :

$$x = -\frac{1}{2} \Rightarrow y = \frac{3}{2}$$

$$x = -3 \Rightarrow y = -1$$

## FINAL ANSWER

$$(x, y) = \left(-\frac{1}{2}, \frac{3}{2}\right) \text{ or } (-3, -1)$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jun 2025

Q#: 22

Marks: 5

TOPIC: Simultaneous Equations

Source: Jun 2025 P1HR

**22** Solve the simultaneous equations

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

Show clear algebraic working.

---

**(Total for Question 22 is 5 marks)**

---

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2025 P1HR | Question 22**  
Simultaneous Equations

**Topic check.** Correct.

**Method**

$$x^2 + y^2 = 41$$

$$2x + y = 3$$

So

$$y = 3 - 2x$$

Substitute:

$$x^2 + (3 - 2x)^2 = 41$$

$$x^2 + 9 - 12x + 4x^2 = 41$$

$$5x^2 - 12x - 32 = 0$$

$$(5x + 8)(x - 4) = 0$$

So

$$x = -\frac{8}{5} \quad \text{or} \quad x = 4$$

Find  $y$ :

$$x = -\frac{8}{5} \Rightarrow y = \frac{31}{5}$$

$$x = 4 \Rightarrow y = -5$$

**FINAL ANSWER**

$$(x, y) = \left(-\frac{8}{5}, \frac{31}{5}\right) \text{ or } (4, -5)$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2024

Q#: 22

Marks: 5

TOPIC: Simultaneous Equations

Source: Nov 2024 P2H

**22** Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

---

**(Total for Question 22 is 5 marks)**



## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2024 P2H | Question 22

Simultaneous Equations

Topic check. Correct.

## Method

$$x^2 + y^2 + y = 3$$

$$y = x + 2$$

Substitute:

$$x^2 + (x + 2)^2 + (x + 2) = 3$$

$$x^2 + x^2 + 4x + 4 + x + 2 = 3$$

$$2x^2 + 5x + 3 = 0$$

$$(2x + 3)(x + 1) = 0$$

So

$$x = -\frac{3}{2} \quad \text{or} \quad x = -1$$

Find  $y$ :

$$x = -\frac{3}{2} \Rightarrow y = \frac{1}{2}$$

$$x = -1 \Rightarrow y = 1$$

## FINAL ANSWER

$$(x, y) = \left(-\frac{3}{2}, \frac{1}{2}\right) \text{ or } (-1, 1)$$

# Forming & Solving Equations

---

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2025

Q#: 20

Marks: 3

TOPIC: **Forming & Solving Equations**

Source: Nov 2025 P2H

**20** Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by  $x = \frac{-4 \pm 2\sqrt{39}}{10}$

Find the value of  $a$  and the value of  $c$

Show your working clearly.

 $a = \dots\dots\dots$  $c = \dots\dots\dots$ 

---

(Total for Question 20 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 P2H | Question 20

Forming &amp; Solving Equations

Topic check. Correct.

## Method

For

$$ax^2 + 4x + c = 0$$

the quadratic formula gives

$$x = \frac{-4 \pm \sqrt{4^2 - 4ac}}{2a}$$

The given solutions are

$$x = \frac{-4 \pm 2\sqrt{39}}{10}$$

Compare denominators:

$$2a = 10$$

$$a = 5$$

Compare the square root parts:

$$\sqrt{16 - 4ac} = 2\sqrt{39}$$

Square both sides:

$$16 - 4ac = 156$$

Use  $a = 5$ :

$$16 - 20c = 156$$

$$-20c = 140$$

$$c = -7$$

## FINAL ANSWER

$$a = 5, c = -7$$

# Functions

---

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: May2025

Q#: 21

Marks: 7

TOPIC: **Functions**

Source: May 2025 4WM2H

**21** The function  $g$  is defined as

$$g(x) = \frac{7x+20}{2x} \quad x \neq 0$$

- (a) Solve  $g(x) = 6$   
Show clear algebraic working.

$$x = \dots\dots\dots (3)$$

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The function  $h$  is defined as

$$h(x) = 5x^2 + 30x - 7 \quad x \geq -3$$

- (b) Express the inverse function  $h^{-1}$  in the form  $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots (4)$$

**(Total for Question 21 is 7 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM2H | Question 21

Functions

**Solution**(a) If  $g(g(x)) = 6$ , then

$$\frac{7g(x) + 20}{2g(x)} = 6$$

$$7g(x) + 20 = 12g(x)$$

$$g(x) = 4$$

So

$$\frac{7x + 20}{2x} = 4$$

$$7x + 20 = 8x$$

$$x = 20$$

(b)

$$h(x) = 5x^2 + 30x - 7 = 5(x + 3)^2 - 52$$

Let  $y = 5(x + 3)^2 - 52$ . Since  $x \geq -3$ , use the positive square root:

$$x + 3 = \sqrt{\frac{y + 52}{5}}$$

$$h^{-1}(x) = \sqrt{\frac{x + 52}{5}} - 3$$

**FINAL ANSWER**

$$x = 20; h^{-1}(x) = \sqrt{\frac{x + 52}{5}} - 3.$$

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: Nov2025

Q#: 21

Marks: 7

TOPIC: Functions

Source: Nov 2025 4WM2H

**21** The functions  $f$  and  $g$  are defined as

$$f(x) = 5x - 3$$

$$g(x) = 4x^2 + 16x - 9 \quad \text{where } x \geq -2$$

(a) Find  $g(-1)$ .....  
(1)(b) Find  $fg(x)$ 

Give your answer in its simplest form.

$$fg(x) = \dots\dots\dots$$

(2)

$$g(x) = 4x^2 + 16x - 9 \quad \text{where } x \geq -2$$

(c) Express the inverse function  $g^{-1}$  in the form  $g^{-1}(x) = \dots$ 

$$g^{-1}(x) = \dots\dots\dots$$

(4)

**(Total for Question 21 is 7 marks)**



## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 4WM2H | Question 21

Functions

**(a) Find  $g(-1)$ :**

$$g(-1) = 4(-1)^2 + 16(-1) - 9 = 4 - 16 - 9 = -21$$

**FINAL ANSWER**

$$a - 21$$

**(b) Find  $fg(x)$ :**

$$\begin{aligned} fg(x) &= f(g(x)) = 5(4x^2 + 16x - 9) - 3 \\ &= 20x^2 + 80x - 45 - 3 = 20x^2 + 80x - 48 \end{aligned}$$

**FINAL ANSWER**

$$b \ 20x^2 + 80x - 48$$

Let  $y = 4x^2 + 16x - 9$ . We must rearrange to make  $x$  the subject.

**1 Complete the square for the  $x$  terms.**

$$\begin{aligned} y &= 4(x^2 + 4x) - 9 \\ y &= 4((x+2)^2 - 4) - 9 \\ y &= 4(x+2)^2 - 16 - 9 \\ y &= 4(x+2)^2 - 25 \end{aligned}$$

**2 Rearrange for  $x$ .**

$$\begin{aligned} y + 25 &= 4(x + 2)^2 \\ \frac{y + 25}{4} &= (x + 2)^2 \\ x + 2 &= \pm \sqrt{\frac{y + 25}{4}} \end{aligned}$$

Since  $x \geq -2$ ,  $x + 2$  is positive, so we take the positive root.

$$\begin{aligned} x + 2 &= \frac{\sqrt{y + 25}}{2} \\ x &= \frac{\sqrt{y + 25}}{2} - 2 \end{aligned}$$

**3 Swap  $y$  for  $x$  to write the inverse function.****FINAL ANSWER**

$$g^{-1}(x) = \frac{\sqrt{x+25}}{2} - 2$$

When finding the inverse of a quadratic, always complete the square. The restricted domain ( $x \geq -2$ ) tells you whether to take the positive or negative square root!

# Sequences

---

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2020

Q#: 25

Marks: 3

TOPIC: Sequences

Source: Jan 2020 P2H

**25** Mario is going to save \$50 in the year 2021

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$ $k$

Mario will save a total of \$33 125 from 2021 to 2070

Work out the value of  $k$ .

$k =$  .....

(Total for Question 25 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 25  
Sequences

**Topic check.** Corrected from Right-Angled Triangles to Sequences.

**Method**

Mario saves from 2021 up to and including 2070.

Number of years:

$$2070 - 2021 + 1 = 50$$

This is an arithmetic sequence.

First term:

$$50$$

Common difference:

$$k$$

Last term:

$$50 + 49k$$

Use the arithmetic series formula:

$$S_n = \frac{n}{2}(a + l)$$

$$33125 = \frac{50}{2} (50 + (50 + 49k))$$

$$33125 = 25(100 + 49k)$$

$$1325 = 100 + 49k$$

$$49k = 1225$$

$$k = 25$$

**FINAL ANSWER**

$$k = 25$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jan 2020

Q#: 22

Marks: 4

TOPIC: Sequences

Source: Jan 2020 P2HR

- 22** The first term of an arithmetic series  $S$  is  $-6$   
The sum of the first 30 terms of  $S$  is 2865

Find the 9th term of  $S$ .

---

(Total for Question 22 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 22  
Sequences

**Topic check.** Correct.

**Method**

First term:

$$a = -6$$

$$S_{30} = 2865$$

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$2865 = \frac{30}{2}(2(-6) + 29d)$$

$$2865 = 15(-12 + 29d)$$

$$191 = -12 + 29d$$

$$29d = 203$$

$$d = 7$$

The 9th term is

$$u_9 = a + 8d$$

$$u_9 = -6 + 8(7) = 50$$

**FINAL ANSWER**

50

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jan 2021

Q#: 24

Marks: 4

TOPIC: Sequences

Source: Jan 2021 P1H

**24** An arithmetic series has first term  $a$  and common difference  $d$ .

The sum of the first  $2n$  terms of the series is four times the sum of the first  $n$  terms of the series.

Find an expression for  $a$  in terms of  $d$ .  
Show your working clearly.

 $a = \dots\dots\dots$ 

(Total for Question 24 is 4 marks)

---

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 24  
Sequences

**Topic check.** Correct.

**Method**

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{2n} = n(2a + (2n-1)d)$$

Given

$$S_{2n} = 4S_n$$

$$n(2a + (2n-1)d) = 4 \cdot \frac{n}{2}(2a + (n-1)d)$$

Divide by  $n$ :

$$2a + (2n-1)d = 2(2a + (n-1)d)$$

$$2a + 2nd - d = 4a + 2nd - 2d$$

$$d = 2a$$

$$a = \frac{d}{2}$$

**FINAL ANSWER**

$$a = \frac{d}{2}$$



**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jan 2021

Q#: 21

Marks: 6

TOPIC: Sequences

Source: Jan 2021 P1HR

- 21** The  $n$ th term of an arithmetic series is  $u_n$  where  $u_n > 0$  for all  $n$   
The sum to  $n$  terms of the series is  $S_n$

Given that  $u_4 = 6$  and that  $S_{11} = (u_6)^2 + 18$

find the value of  $u_{20}$

(Total for Question 21 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1HR | Question 21  
Sequences

**Topic check.** Correct.

**Method**

Let  $u_6 = v$ .

For an arithmetic series,

$$S_{11} = \frac{11}{2}(u_1 + u_{11})$$

The average of  $u_1$  and  $u_{11}$  is  $u_6$ , so

$$S_{11} = 11u_6 = 11v$$

Given

$$S_{11} = (u_6)^2 + 18$$

$$11v = v^2 + 18$$

$$v^2 - 11v + 18 = 0$$

$$(v - 2)(v - 9) = 0$$

So

$$u_6 = 2 \quad \text{or} \quad u_6 = 9$$

But  $u_4 = 6$ . If  $u_6 = 2$ , then the common difference is negative and later terms would not all be positive. Therefore,

$$u_6 = 9$$

$$d = \frac{u_6 - u_4}{2} = \frac{9 - 6}{2} = \frac{3}{2}$$

$$u_{20} = u_6 + 14d$$

$$u_{20} = 9 + 14\left(\frac{3}{2}\right) = 30$$

**FINAL ANSWER**

$$u_{20} = 30$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2022

Q#: 26

Marks: 5

TOPIC: Sequences

Source: Jan 2022 P2H

- 26** An arithmetic series has first term  $a$  and common difference  $d$ , where  $d$  is a prime number.

The sum of the first  $n$  terms of the series is  $S_n$  and

$$S_m = 39$$

$$S_{2m} = 320$$

Find the value of  $d$  and the value of  $m$   
Show clear algebraic working.

$$d = \dots\dots\dots$$

$$m = \dots\dots\dots$$

---

(Total for Question 26 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 26  
Sequences

**Topic check.** Correct.

**Method**

$$S_m = 39$$

$$S_{2m} = 320$$

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

Then

$$2S_m = m(2a + (m-1)d) = 78$$

and

$$S_{2m} = m(2a + (2m-1)d) = 320$$

Subtract:

$$S_{2m} - 2S_m = m[(2m-1)d - (m-1)d]$$

$$320 - 78 = m(md)$$

$$242 = m^2 d$$

$$242 = 2 \times 11^2$$

Since  $d$  is prime and  $m^2$  is a square factor,

$$d = 2, \quad m^2 = 121$$

$$m = 11$$

**FINAL ANSWER**

$$d = 2, \quad m = 11$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jun 2022

Q#: 25

Marks: 5

TOPIC: Sequences

Source: Jun 2022 P1HR

- 25** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.  
Show clear algebraic working.

---

(Total for Question 25 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1HR | Question 25  
Sequences

**Topic check.** Correct.

**Method**

$$S_{10} = 4S_5$$

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{10} = 5(2a + 9d)$$

$$S_5 = \frac{5}{2}(2a + 4d) = 5(a + 2d)$$

So

$$5(2a + 9d) = 4 \cdot 5(a + 2d)$$

$$2a + 9d = 4a + 8d$$

$$d = 2a$$

The 8th term is 45:

$$u_8 = a + 7d = 45$$

$$a + 7(2a) = 45$$

$$15a = 45$$

$$a = 3$$

**FINAL ANSWER**

The first term is 3

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2023

Q#: 23

Marks: 5

TOPIC: Sequences

Source: Jun 2023 P2HR

**23** Here are the first three terms of an arithmetic sequence.

$$8p \qquad 7p - 3 \qquad 4p + 2$$

The sum of the first  $n$  terms of the sequence is  $-1914$

Work out the value of  $n$   
Show your working clearly.

$$n = \dots\dots\dots$$

(Total for Question 23 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 23  
Sequences

**Topic check.** Correct.

**Method**

The first three terms are

$$8p, \quad 7p - 3, \quad 4p + 2$$

For an arithmetic sequence, consecutive differences are equal:

$$(7p - 3) - 8p = (4p + 2) - (7p - 3)$$

$$-p - 3 = -3p + 5$$

$$2p = 8$$

$$p = 4$$

So the first term is

$$8p = 32$$

and the common difference is

$$(7p - 3) - 8p = 25 - 32 = -7$$

The sum of the first  $n$  terms is

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

$$-1914 = \frac{n}{2}(64 - 7(n - 1))$$

$$-1914 = \frac{n}{2}(71 - 7n)$$

$$-3828 = n(71 - 7n)$$

$$7n^2 - 71n - 3828 = 0$$

$$n = \frac{71 \pm 335}{14}$$

Only the positive value is possible:

$$n = 29$$

**FINAL ANSWER**

$$n = 29$$



**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jun 2025

Q#: 24

Marks: 5

TOPIC: Sequences

Source: Jun 2025 P1H

**24** An arithmetic series has 30 terms.

The first term is  $a$

The common difference is  $d$

The 20th term is 123

The sum of the 30 terms is 2880

Work out the value of  $a$  and the value of  $d$

Show clear algebraic working.

$a =$  .....

$d =$  .....

**(Total for Question 24 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1H | Question 24  
Sequences

Topic check. Correct.

**Method**

$$u_{20} = a + 19d = 123$$

$$S_{30} = \frac{30}{2}(2a + 29d) = 2880$$

$$2a + 29d = 192$$

Double the first equation:

$$2a + 38d = 246$$

Subtract:

$$9d = 54$$

$$d = 6$$

$$a + 19(6) = 123$$

$$a = 9$$

**FINAL ANSWER**

$$a = 9, d = 6$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jun 2025

Q#: 24

Marks: 6

TOPIC: Sequences

Source: Jun 2025 P1HR

**24** The first 3 terms of an arithmetic series are

$$(2x + 5) \quad (3y - 4) \quad (4x - 2)$$

where  $x$  and  $y$  are constants.

The sum of the first 9 terms of the series is 216

Find the value of  $x$  and the value of  $y$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

**(Total for Question 24 is 6 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 24  
Sequences

**Topic check.** Correct.

**Method**

The first 3 terms are

$$2x + 5, \quad 3y - 4, \quad 4x - 2$$

Equal differences give

$$(3y - 4) - (2x + 5) = (4x - 2) - (3y - 4)$$

$$6y - 6x = 11$$

For an arithmetic series,

$$S_9 = 9(a + 4d)$$

Here

$$a = 2x + 5$$

$$d = (3y - 4) - (2x + 5) = 3y - 2x - 9$$

Given  $S_9 = 216$ :

$$9[(2x + 5) + 4(3y - 2x - 9)] = 216$$

$$12y - 6x = 55$$

Now solve

$$6y - 6x = 11$$

$$12y - 6x = 55$$

Subtracting gives

$$6y = 44$$

$$y = \frac{22}{3}$$

Then

$$6\left(\frac{22}{3}\right) - 6x = 11$$

$$x = \frac{11}{2}$$

**FINAL ANSWER**

$$x = \frac{11}{2}, y = \frac{22}{3}$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2021

Q#: 23

Marks: 5

TOPIC: Sequences

Source: May 2021 P1H

- 23** The sum of the first  $N$  terms of an arithmetic series,  $S$ , is 292  
The 2nd term of  $S$  is 8.5  
The 5th term of  $S$  is 13

Find the value of  $N$ .  
Show clear algebraic working.

 $N = \dots\dots\dots$ 

---

(Total for Question 23 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 23  
Sequences

**Topic check.** Correct.

**Method**

The 2nd term is

$$a + d = 8.5$$

The 5th term is

$$a + 4d = 13$$

Subtract:

$$3d = 4.5$$

$$d = 1.5$$

$$a = 8.5 - 1.5 = 7$$

Use

$$S_N = \frac{N}{2}(2a + (N - 1)d)$$

$$292 = \frac{N}{2}(14 + 1.5(N - 1))$$

$$292 = \frac{N}{2}(1.5N + 12.5)$$

$$3N^2 + 25N - 1168 = 0$$

$$N = \frac{-25 + 121}{6} = 16$$

**FINAL ANSWER**

$$N = 16$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2023

Q#: 20

Marks: 6

TOPIC: Sequences

Source: May 2023 P1H

**20** The sum of the first 80 terms of an arithmetic series,  $S$ , is 470

The 75th term of  $S$  is 14.5

The sum of the first  $X$  terms of  $S$  is 171

Work out the value of  $X$

Show your working clearly.

---

---

 $X = \dots\dots\dots$ 

---

(Total for Question 20 is 6 marks)



## WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1H | Question 20  
Sequences

**Topic check.** Correct.

**Method**

$$S_{80} = 470$$

$$40(2a + 79d) = 470$$

$$2a + 79d = \frac{47}{4}$$

The 75th term is

$$a + 74d = 14.5 = \frac{29}{2}$$

Double it:

$$2a + 148d = 29$$

Subtract:

$$69d = 29 - \frac{47}{4} = \frac{69}{4}$$

$$d = \frac{1}{4}$$

$$a + 74\left(\frac{1}{4}\right) = \frac{29}{2}$$

$$a = -4$$

Now

$$S_X = 171$$

$$171 = \frac{X}{2} \left( 2(-4) + (X-1)\frac{1}{4} \right)$$

$$171 = \frac{X(X-33)}{8}$$

$$X^2 - 33X - 1368 = 0$$

$$X = \frac{33 + 81}{2} = 57$$

**FINAL ANSWER**

$$X = 57$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2024

Q#: 24

Marks: 6

TOPIC: Sequences

Source: May 2024 P1H

**24** A polygon has  $n$  sides, where  $n > 5$ 

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is  $84^\circ$ The common difference of the sequence is  $4^\circ$ 

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

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.....  
 o

---

**(Total for Question 24 is 6 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1H | Question 24  
Sequences

**Topic check.** Correct.

**Method**

The interior angles form an arithmetic sequence with first term 84 and common difference 4.  
So the sum of the angles is

$$\begin{aligned} & \frac{n}{2}(2(84) + (n-1)4) \\ &= \frac{n}{2}(168 + 4n - 4) \\ &= 2n^2 + 82n \end{aligned}$$

The sum of the interior angles of an  $n$ -sided polygon is

$$180(n-2)$$

So

$$2n^2 + 82n = 180(n-2)$$

$$2n^2 - 98n + 360 = 0$$

$$n^2 - 49n + 180 = 0$$

$$(n-45)(n-4) = 0$$

Since  $n > 5$ ,

$$n = 45$$

Therefore the sum of the interior angles is

$$180(45-2) = 7740$$

**FINAL ANSWER**

7740 degrees

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: May 2024

Q#: 23

Marks: 4

TOPIC: Sequences

Source: May 2024 P1HR

**23** Here are the first three terms of an arithmetic sequence.

$$(4x-14) \quad , \quad (x+2) \quad , \quad (7x-9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.  
Show clear algebraic working.

---

(Total for Question 23 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2024 P1HR | Question 23

Sequences

**Topic check.** Correct.**Method**

The first three terms are

$$4x - 14, \quad x + 2, \quad 7x - 9$$

Equal differences give

$$(x + 2) - (4x - 14) = (7x - 9) - (x + 2)$$

$$-3x + 16 = 6x - 11$$

$$9x = 27$$

$$x = 3$$

So the sequence is

$$-2, 5, 12, \dots$$

with common difference 7.

$$S_{40} = \frac{40}{2}(2(-2) + 39(7))$$

$$S_{40} = 20(-4 + 273) = 5380$$

**FINAL ANSWER**

5380

**CLASSIFIED PROBLEM**

Paper: P1H

Session: MayNov 2020

Q#: 24

Marks: 4

TOPIC: Sequences

Source: May-Nov 2020 P1H

**24** Here are the first five terms of an arithmetic sequence.

8      15      22      29      36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

---

(Total for Question 24 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1H | Question 24  
Sequences

**Topic check.** Correct.

**Method**

The arithmetic sequence is

$$8, 15, 22, 29, 36, \dots$$

So

$$a = 8, \quad d = 7$$

The  $n$ th term is

$$u_n = 8 + 7(n - 1) = 7n + 1$$

$$u_{50} = 7(50) + 1 = 351$$

$$u_{100} = 7(100) + 1 = 701$$

There are

$$100 - 50 + 1 = 51$$

terms from the 50th to the 100th inclusive.

$$\text{sum} = \frac{51}{2}(351 + 701)$$

$$= 26826$$

**FINAL ANSWER**

26826

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2021

Q#: 20

Marks: 5

TOPIC: Sequences

Source: Nov 2021 P2H

**20** Here are the first four terms of an arithmetic series.

$$k \quad \frac{3k}{4} \quad \frac{k}{2} \quad \frac{k}{4}$$

Given that the 15th term of the series is  $(90 + 2k)$ ,

calculate the sum of the first 30 terms of the series.

---

(Total for Question 20 is 5 marks)



## WORKED SOLUTION

Dr. Eslam Ahmed

**Nov 2021 P2H** | **Question 20**  
Sequences

**Topic check.** Correct.

**Method**

The first four terms are

$$k, \quad \frac{3k}{4}, \quad \frac{k}{2}, \quad \frac{k}{4}$$

The common difference is

$$d = -\frac{k}{4}$$

The 15th term is

$$u_{15} = k + 14\left(-\frac{k}{4}\right)$$

$$u_{15} = k - \frac{7k}{2} = -\frac{5k}{2}$$

Given

$$-\frac{5k}{2} = 90 + 2k$$

$$-5k = 180 + 4k$$

$$k = -20$$

So

$$a = -20, \quad d = 5$$

$$S_{30} = \frac{30}{2}(2(-20) + 29(5))$$

$$S_{30} = 15(105) = 1575$$

**FINAL ANSWER**

1575

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2025

Q#: 25

Marks: 5

TOPIC: Sequences

Source: Nov 2025 P2H

- 25** The first term of an arithmetic series is 10  
The 20th term of the series is 86
- The sum of the first  $N$  terms of the series is 5194
- Work out the value of  $N$   
Show your working clearly.

 $N = \dots\dots\dots$ 

---

(Total for Question 25 is 5 marks)

---

## WORKED SOLUTION

Dr. Eslam Ahmed

**Nov 2025 P2H** | **Question 25**  
Sequences

**Topic check.** Correct.

**Method**

$$a = 10$$

$$u_{20} = 86$$

$$10 + 19d = 86$$

$$d = 4$$

Use

$$S_N = \frac{N}{2}(2a + (N - 1)d)$$

$$5194 = \frac{N}{2}(20 + 4(N - 1))$$

$$5194 = \frac{N}{2}(4N + 16)$$

$$5194 = 2N(N + 4)$$

$$N^2 + 4N - 2597 = 0$$

$$N = \frac{-4 + 102}{2} = 49$$

**FINAL ANSWER**

$$N = 49$$

# Functions

---

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jan 2020

Q#: 21

Marks: 5

TOPIC: **Functions**

Source: Jan 2020 P1H

**21** The functions  $f$  and  $g$  are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function  $h$  is such that  $h(x) = fg(x)$  for  $x \geq -2$ Express the inverse function  $h^{-1}(x)$  in the form  $h^{-1}(x) = \dots$ 

$$h^{-1}(x) = \dots\dots\dots$$

---

**(Total for Question 21 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1H | Question 21  
Functions

**Topic check.** Correct.

**Method**

$$f(x) = x^2 - 2x$$

$$g(x) = x + 3$$

$$h(x) = fg(x) = f(g(x))$$

$$h(x) = (x + 3)^2 - 2(x + 3)$$

$$h(x) = x^2 + 4x + 3$$

$$h(x) = (x + 2)^2 - 1$$

Let

$$y = (x + 2)^2 - 1$$

Since the domain is  $x \geq -2$ , take the positive square root:

$$x + 2 = \sqrt{y + 1}$$

$$x = \sqrt{y + 1} - 2$$

**FINAL ANSWER**

$$h^{-1}(x) = \sqrt{x + 1} - 2$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jan 2021

Q#: 22

Marks: 3

TOPIC: **Functions**

Source: Jan 2021 P1H

**22** The function  $f$  is such that  $f(x) = x^2 - 8x + 5$  where  $x \leq 4$

Express the inverse function  $f^{-1}$  in the form  $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots\dots\dots$

**(Total for Question 22 is 3 marks)**

---

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 22  
Functions

**Topic check.** Correct.

**Method**

$$f(x) = x^2 - 8x + 5$$

$$f(x) = (x - 4)^2 - 11$$

Let

$$y = (x - 4)^2 - 11$$

$$y + 11 = (x - 4)^2$$

Since the domain is  $x < 4$ , take the negative square root:

$$x - 4 = -\sqrt{y + 11}$$

$$x = 4 - \sqrt{y + 11}$$

**FINAL ANSWER**

$$f^{-1}(x) = 4 - \sqrt{x + 11}$$



**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2022

Q#: 25

Marks: 5

TOPIC: Functions

Source: Jan 2022 P2H

**25** The function  $g$  is defined as

$$g : x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x : x \geq 3\}$$

(a) Express the inverse function  $g^{-1}$  in the form  $g^{-1} : x \mapsto \dots$ 

$$g^{-1} : x \mapsto \dots$$

(4)

(b) State the domain of  $g^{-1}$ 

$$\dots$$

(1)

**(Total for Question 25 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 25  
Functions

**Topic check.** Correct.

**Method**

$$g(x) = 5 + 6x - x^2$$

Complete the square:

$$g(x) = 14 - (x - 3)^2$$

Let

$$y = 14 - (x - 3)^2$$

$$(x - 3)^2 = 14 - y$$

The domain of  $g$  is  $x \geq 3$ , so take the positive square root:

$$x - 3 = \sqrt{14 - y}$$

$$x = 3 + \sqrt{14 - y}$$

Therefore

$$g^{-1} : x \mapsto 3 + \sqrt{14 - x}$$

The range of  $g$  is  $x \leq 14$ , so this is the domain of  $g^{-1}$ .

**FINAL ANSWER**

(a)  $g^{-1} : x \mapsto 3 + \sqrt{14 - x}$ , (b) domain  $x \leq 14$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jan 2022

Q#: 23

Marks: 4

TOPIC: Functions

Source: Jan 2022 P2HR

**23** The functions  $f$  and  $g$  are such that

$$f(x) = x + 25 \qquad g(x) = x^2 - 12x$$

The function  $h$  is such that  $h(x) = fg(x)$

The domain of  $h$  is  $\{x : x \leq 6\}$

Express the inverse function  $h^{-1}$  in the form  $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

---

(Total for Question 23 is 4 marks)

---

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2HR | Question 23  
Functions

**Topic check.** Correct.

**Method**

$$f(x) = x + 25$$

$$g(x) = x^2 - 12x$$

$$h(x) = fg(x) = f(g(x))$$

$$h(x) = x^2 - 12x + 25$$

$$h(x) = (x - 6)^2 - 11$$

Let

$$y = (x - 6)^2 - 11$$

$$y + 11 = (x - 6)^2$$

Since the domain of  $h$  is  $x < 6$ , take the negative square root:

$$x - 6 = -\sqrt{y + 11}$$

$$x = 6 - \sqrt{y + 11}$$

**FINAL ANSWER**

$$h^{-1}(x) = 6 - \sqrt{x + 11}$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2021

Q#: 24

Marks: 6

TOPIC: **Functions**

Source: May 2021 P1H

**24** The functions  $f$  and  $g$  are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find  $fg(2)$ .....  
(2)(b) Express the inverse function  $f^{-1}$  in the form  $f^{-1}(x) = \dots$ 

$$f^{-1}(x) = \dots\dots\dots$$

(4)

**(Total for Question 24 is 6 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 24  
Functions

**Topic check.** Correct.

**Method**

(a)

$$g(2) = 7(2) - 6 = 8$$

$$fg(2) = f(8)$$

$$f(8) = 5(8)^2 - 10(8) + 7$$

$$f(8) = 247$$

(b)

$$f(x) = 5x^2 - 10x + 7$$

$$f(x) = 5(x - 1)^2 + 2$$

Let

$$y = 5(x - 1)^2 + 2$$

$$(x - 1)^2 = \frac{y - 2}{5}$$

Since the domain is  $x \geq 1$ , take the positive square root:

$$x = 1 + \sqrt{\frac{y - 2}{5}}$$

**FINAL ANSWER**

$$(a) 247, (b) f^{-1}(x) = 1 + \sqrt{\frac{x - 2}{5}}$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2024

Q#: 25

Marks: 4

TOPIC: **Functions**

Source: Nov 2024 P1H

**25** The function  $f$  is such that  $f(x) = 2x^2 - 24x + 7$  where  $x \geq 6$

Find the inverse function  $f^{-1}(x)$

$f^{-1}(x) = \dots\dots\dots$

(Total for Question 25 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P1H | Question 25  
Functions

**Topic check.** Correct.

**Method**

$$f(x) = 2x^2 - 24x + 7$$

Complete the square:

$$f(x) = 2(x^2 - 12x) + 7$$

$$f(x) = 2[(x - 6)^2 - 36] + 7$$

$$f(x) = 2(x - 6)^2 - 65$$

Let  $y = f(x)$ .

$$y = 2(x - 6)^2 - 65$$

$$y + 65 = 2(x - 6)^2$$

$$\frac{y + 65}{2} = (x - 6)^2$$

Since  $x \geq 6$ , take the positive square root:

$$x - 6 = \sqrt{\frac{y + 65}{2}}$$

$$x = 6 + \sqrt{\frac{y + 65}{2}}$$

Replace  $y$  by  $x$ .

**FINAL ANSWER**

$$f^{-1}(x) = 6 + \sqrt{\frac{x + 65}{2}}$$



# Graphs of Functions

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**CLASSIFIED PROBLEM**

Paper: 4WM2HR

Session: May2025

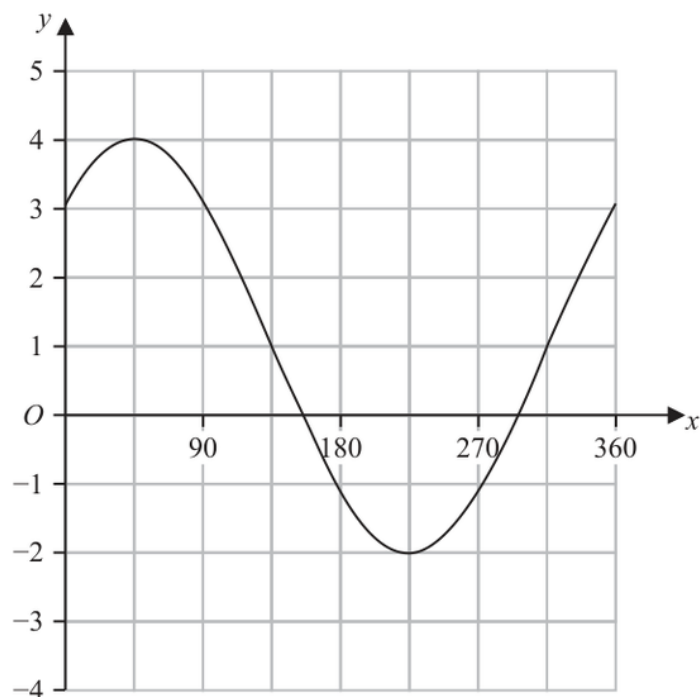
Q#: 23

Marks: 3

TOPIC: Graphs of Functions

Source: May 2025 4WM2HR

23 The graph of  $y = a \sin(x + b)^\circ + c$  for  $0 \leq x \leq 360$  is drawn on the grid below.



Find a suitable value for  $a$ , for  $b$  and for  $c$

$a =$  .....

$b =$  .....

$c =$  .....

(Total for Question 23 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM2HR | Question 23

Graphs of Functions

**Required:**

The graph of  $y = a \sin (x+b)^{\circ} + c$  for  $0 \leq x \leq 360$  is shown. Find suitable values for  $a$ ,  $b$ , and  $c$ .

**Step:**

Read the maximum and minimum values from the graph to find the **amplitude**  $a$  and **vertical shift**  $c$ :

$$a = \frac{\max - \min}{2} = \frac{4 - (-2)}{2} = 3 \quad c = \frac{\max + \min}{2} = \frac{4 + (-2)}{2} = 1$$

**Step:**

Find the **phase shift**  $b$ . The standard sin has its peak at  $x = 90^{\circ}$ . From the graph, the maximum occurs at  $x \approx 45^{\circ}$ :

$$x + b = 90^{\circ} \implies 45^{\circ} + b = 90^{\circ} \implies b = 45^{\circ}$$

**Step:**

Verify at  $x = 0$ :  $y = 4 \sin(0 + 45)^{\circ} = 4 \times \frac{\sqrt{2}}{2} \approx 2.83$  – consistent with graph

**FINAL ANSWER**

$$a = 3, b = 45, c = 1$$

**Note:**

The period of the graph is still  $360^{\circ}$  (unchanged), so no horizontal stretch is applied.

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: Nov2025

Q#: 25

Marks: 3

TOPIC: Graphs of Functions

Source: Nov 2025 4WM2H

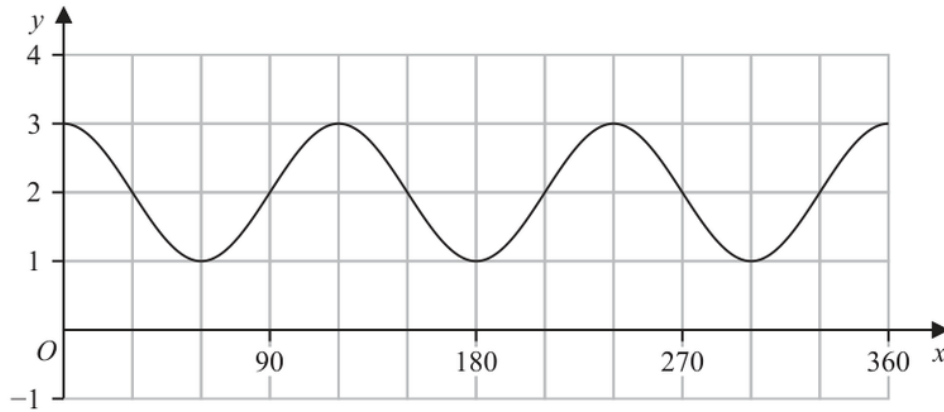
**25** A curve has equation  $y = f(x)$ 

There is one minimum point on this curve.

The coordinates of this minimum point are  $(8, 1)$ 

- (a) Write down the coordinates of the minimum point on the curve with equation  $y = f(4x)$

(..... , .....)  
(1)

The diagram shows the graph of  $y = \cos(ax) + b$  for  $0 \leq x \leq 360$ 

- (b) (i) Find the value of  $a$

$a = \dots\dots\dots$   
(1)

- (ii) Find the value of  $b$

$b = \dots\dots\dots$   
(1)

(Total for Question 25 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 4WM2H | Question 25

Graphs of Functions

The transformation  $y = f(4x)$  represents a horizontal stretch by a scale factor of  $\frac{1}{4}$ . The x-coordinate is multiplied by  $\frac{1}{4}$  while the y-coordinate stays the same.

$$\text{New minimum} = \left(8 \times \frac{1}{4}, 1\right) = (2, 1)$$

## FINAL ANSWER

(2,1)

**1 Find the vertical shift  $b$ .**

The standard  $y = \cos(x)$  graph fluctuates between -1 and 1, centered at  $y = 0$ . The given graph fluctuates between a minimum of  $y = 1$  and a maximum of  $y = 3$ . The center line is  $\frac{3+1}{2} = 2$ . Therefore,  $b = 2$ .

**2 Find the frequency multiplier  $a$ .**

The period of the given graph completes one full cycle between  $x = 0$  and  $x = 90^\circ$  (it starts at 3, dips to 1, and returns to 3 at  $90^\circ$ ). The standard cosine period is  $360^\circ$ .

$$\begin{aligned}\text{Period} &= \frac{360^\circ}{a} \\ 90^\circ &= \frac{360^\circ}{a} \implies a = 4\end{aligned}$$

## FINAL ANSWER

 $a = 4, b = 2$

# Transformations of Graphs

---

**CLASSIFIED PROBLEM****Paper: 4WM2H****Session: May2025****Q#: 27****Marks: 4****TOPIC: Transformations of Graphs**

Source: May 2025 4WM2H

- 27 A curve has equation
- $y = f(x)$

There is one maximum point on this curve.

The coordinates of this maximum point are  $(6, -5)$ 

- (a) Write down the coordinates of the maximum point on the curve with equation

(i)  $y = f(x - 4)$

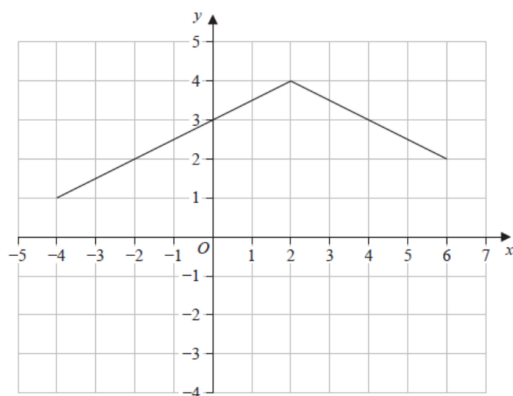
(....., .....)  
(1)

(ii)  $y = f(3x)$

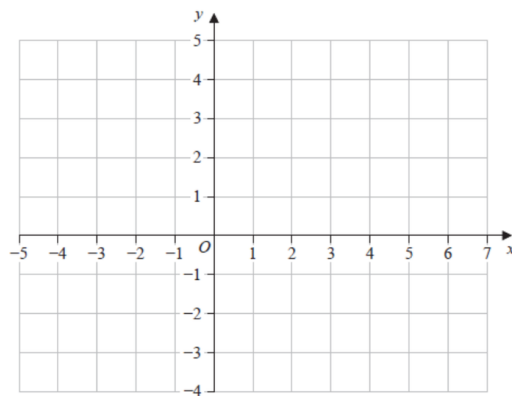
(....., .....)  
(1)

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Here is the graph of  $y = g(x)$ 

- (b) On the grid below, draw the graph of
- $y = g(2x) - 3$



(2)

(Total for Question 27 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM2H | Question 27

Transformations of Graphs

**Solution**

(a)

For  $y = f(x - 4)$ , move the graph 4 units right:

$$(6, -5) \rightarrow (10, -5)$$

For  $y = f(3x)$ , divide the  $x$ -coordinate by 3:

$$(6, -5) \rightarrow (2, -5)$$

(b) For  $y = g(2x) - 3$ , halve each  $x$ -coordinate and subtract 3 from each  $y$ -coordinate.

Useful transformed points are

$$(-4, 1) \rightarrow (-2, -2), \quad (2, 4) \rightarrow (1, 1), \quad (6, 2) \rightarrow (3, -1)$$

Join these points with straight line segments.

**FINAL ANSWER**

(10, -5), (2, -5), graph through (-2, -2), (1, 1), (3, -1).



# Differentiation

---

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: May2025

Q#: 22

Marks: 4

TOPIC: **Differentiation**

Source: May 2025 4WM2H

- 22 Work out the range of values of  $x$  for which the graph of  $y = x^3 + \frac{1}{2}x^2 - 10x + 5$  has a positive gradient.  
Show clear algebraic working.

.....  
(Total for Question 22 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM2H | Question 22

Differentiation

## Solution

$$\frac{dy}{dx} = 3x^2 + 2x - 10$$

For a positive gradient,

$$3x^2 + 2x - 10 > 0$$

The roots are

$$x = \frac{-1 \pm \sqrt{31}}{3}$$

Since the quadratic opens upwards,

$$x < \frac{-1 - \sqrt{31}}{3} \quad \text{or} \quad x > \frac{-1 + \sqrt{31}}{3}$$

**FINAL ANSWER**

$$x < \frac{-1 - \sqrt{31}}{3} \quad \text{or} \quad x > \frac{-1 + \sqrt{31}}{3}.$$

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: Nov2025

Q#: 22

Marks: 6

TOPIC: **Differentiation**

Source: Nov 2025 4WM2H

**22** Curve **C** has equation  $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form  $y = ax + b$

Work out the two possible values of  $b$

Show clear algebraic working.

---

(Total for Question 22 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 22

Differentiation

**1** Differentiate the curve to find the gradient expression.

$$\frac{dy}{dx} = 3x^2 - 16$$

**2** Set the gradient to 11 to find the x-coordinates.

$$3x^2 - 16 = 11$$

$$3x^2 = 27$$

$$x^2 = 9 \implies x = 3 \text{ or } x = -3$$

**3** Find the corresponding y-coordinates on the curve.

$$\text{For } x = 3: y = (3)^3 - 16(3) + 7 = 27 - 48 + 7 = -14$$

**Point 1: (3,-14)**

$$\text{For } x = -3: y = (-3)^3 - 16(-3) + 7 = -27 + 48 + 7 = 28$$

**Point 2: (-3,28)****4** Find the y-intercept ( $b$ ) for each tangent ( $y = 11x + b$ ).

Using (3,-14):

$$-14 = 11(3) + b \implies -14 = 33 + b \implies b = -47$$

Using (-3,28):

$$28 = 11(-3) + b \implies 28 = -33 + b \implies b = 61$$

**FINAL ANSWER**

$$b = -47 \text{ and } 61$$

# Transformations of Graphs

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**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jan 2020

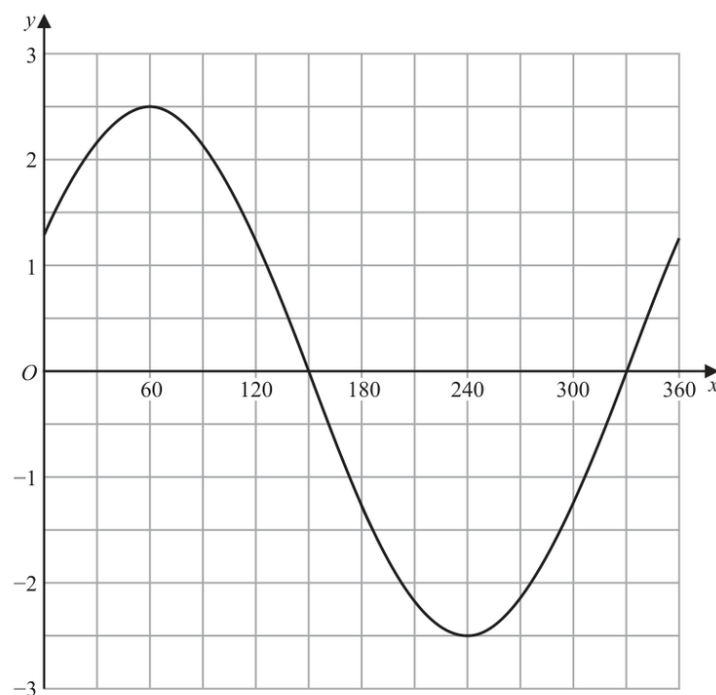
Q#: 22

Marks: 4

TOPIC: Transformations of Graphs

Source: Jan 2020 P1HR

22 The graph of  $y = a \cos(x + b)^\circ$  for  $0 \leq x \leq 360$  is drawn on the grid.



(a) Find the value of  $a$  and the value of  $b$ .

$a = \dots\dots\dots$

$b = \dots\dots\dots$   
(2)

Another curve  $C$  has equation  $y = f(x)$

The coordinates of the minimum point of  $C$  are  $(4, 5)$

(b) Write down the coordinates of the minimum point of the curve with equation

(i)  $y = f(2x)$

(....., .....)

(ii)  $y = f(x) - 7$

(....., .....)  
(2)

(Total for Question 22 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2020 P1HR | Question 22

Transformations of Graphs

**Topic check.** Correct.**Method**

For part (a), the graph has maximum 2.5 and minimum  $-2.5$ , so

$$a = 2.5$$

The maximum occurs at  $x = 60^\circ$ . For  $y = a \cos(x + b)^\circ$ , a maximum occurs when

$$x + b = 360$$

So

$$60 + b = 360$$

$$b = 300$$

For part (b), the minimum point of  $y = f(x)$  is  $(4, 5)$ .

For

$$y = f(2x)$$

the  $x$ -coordinate is divided by 2:

$$(4, 5) \rightarrow (2, 5)$$

For

$$y = f(x) - 7$$

the  $y$ -coordinate decreases by 7:

$$(4, 5) \rightarrow (4, -2)$$

**FINAL ANSWER**

$a = 2.5$ ,  $b = 300$ ;  $(2, 5)$  and  $(4, -2)$



**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2020

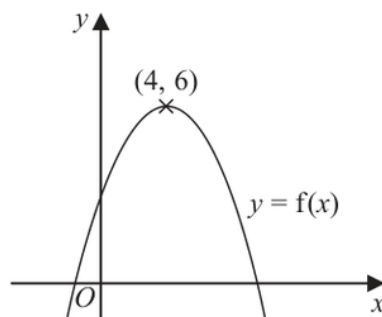
Q#: 21

Marks: 4

TOPIC: Transformations of Graphs

Source: Jan 2020 P2H

21 The diagram shows a sketch of part of the curve with equation  $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are (4, 6)

(a) Write down the coordinates of the maximum point on the curve with equation

(i)  $y = f(x + 4)$

(....., .....)

(ii)  $y = f(2x)$

(....., .....)

(2)

The equation of a curve **C** is  $y = x^2 + 3x + 4$

The curve **C** is transformed to curve **S** under the translation  $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

(b) Find an equation of curve **S**.

*You do not need to simplify the equation.*

(2)

(Total for Question 21 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2020 P2H | Question 21

Transformations of Graphs

**Topic check.** Correct.**Method**The maximum point of  $y = f(x)$  is  $(4, 6)$ .

For

$$y = f(x + 4)$$

the graph moves 4 units left:

$$(4, 6) \rightarrow (0, 6)$$

For

$$y = f(2x)$$

the  $x$ -coordinate is divided by 2:

$$(4, 6) \rightarrow (2, 6)$$

For part (b),

$$y = x^2 + 3x + 4$$

The translation vector is

$$\begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

So the graph moves 4 units right and 6 units up.

Replace  $x$  by  $x - 4$ , then add 6:

$$y = (x - 4)^2 + 3(x - 4) + 4 + 6$$

$$y = (x - 4)^2 + 3(x - 4) + 10$$

**FINAL ANSWER** $(0, 6)$ ,  $(2, 6)$ , and  $y = (x - 4)^2 + 3(x - 4) + 10$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jan 2021

Q#: 21

Marks: 4

TOPIC: Transformations of Graphs

Source: Jan 2021 P1H

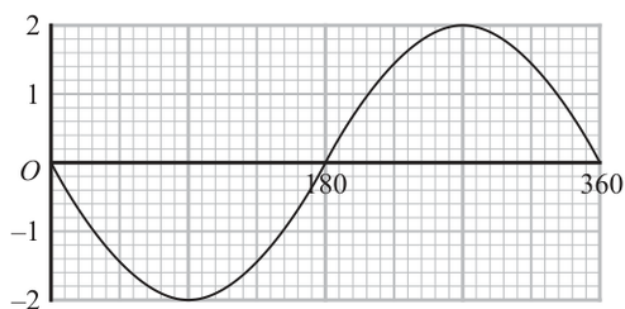
**21** A curve has equation  $y = f(x)$ The coordinates of the minimum point on this curve are  $(-9, 15)$ 

(a) Write down the coordinates of the minimum point on the curve with equation

(i)  $y = f(x + 3)$

(....., .....)

(ii)  $y = \frac{1}{3}f(x)$

(....., .....)  
(2)The graph of  $y = a \cos(x + b)^\circ$  for  $0 \leq x \leq 360$  is drawn on the grid below.Given that  $a > 0$  and that  $0 < b < 360$ (b) find the value of  $a$  and the value of  $b$ . $a = \dots\dots\dots$  $b = \dots\dots\dots$ 

(2)

(Total for Question 21 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2021 P1H | Question 21

Transformations of Graphs

**Topic check.** Correct.**Method**The minimum point of  $y = f(x)$  is  $(-9, 15)$ .

For

$$y = f(x + 3)$$

the graph moves 3 units left:

$$(-9, 15) \rightarrow (-12, 15)$$

For

$$y = \frac{1}{3}f(x)$$

the  $y$ -coordinate is divided by 3:

$$(-9, 15) \rightarrow (-9, 5)$$

For part (b), the cosine graph has maximum 2 and minimum  $-2$ , so

$$a = 2$$

The graph has a minimum at  $x = 90^\circ$ . For  $a \cos(x + b)^\circ$ , a minimum occurs when

$$x + b = 180$$

So

$$90 + b = 180$$

$$b = 90$$

**FINAL ANSWER** $(-12, 15), (-9, 5), a = 2, b = 90$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jan 2022

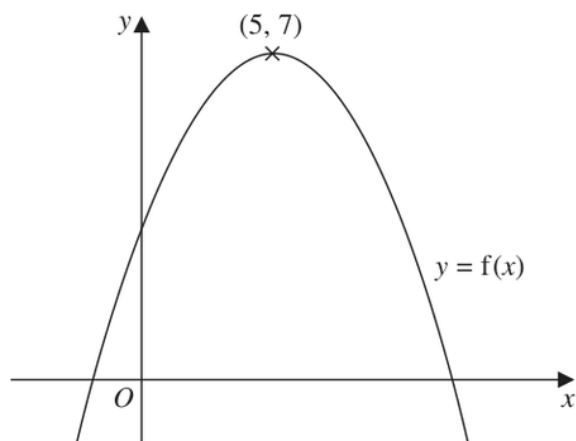
Q#: 23

Marks: 2

TOPIC: Transformations of Graphs

Source: Jan 2022 P1H

23 The diagram shows a sketch of the curve with equation  $y = f(x)$



There is only one maximum point on the curve.  
The coordinates of this maximum point are (5, 7)

Write down the coordinates of the maximum point on the curve with equation

(i)  $y = f(x + 9)$

(....., .....)

(ii)  $y = f(x) + 3$

(....., .....)

(Total for Question 23 is 2 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2022 P1H | Question 23

Transformations of Graphs

**Topic check.** Correct.**Method**The maximum point of  $y = f(x)$  is  $(5, 7)$ .

For

$$y = f(x + 9)$$

the graph moves 9 units left:

$$(5, 7) \rightarrow (-4, 7)$$

For

$$y = f(x) + 3$$

the graph moves 3 units up:

$$(5, 7) \rightarrow (5, 10)$$

**FINAL ANSWER** $(-4, 7)$  and  $(5, 10)$

**CLASSIFIED PROBLEM****Paper: P1HR****Session: Jan 2022****Q#: 20****Marks: 6****TOPIC: Transformations of Graphs**

Source: Jan 2022 P1HR

20 The curve with equation  $y = f(x)$  has one turning point.

The coordinates of this turning point are  $(-6, -4)$

(a) Write down the coordinates of the turning point on the curve with equation

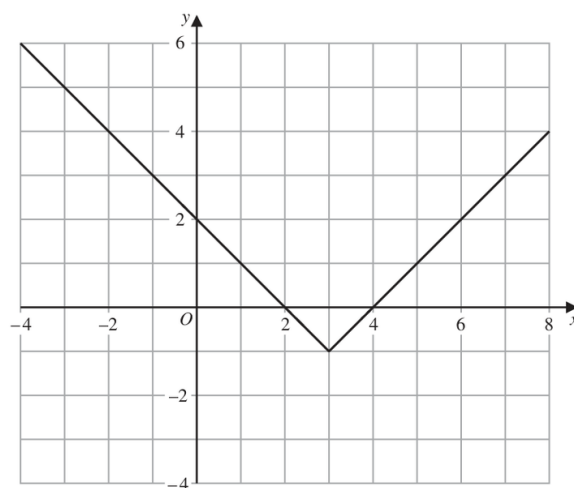
(i)  $y = f(x) + 5$

(....., .....)

(ii)  $y = f(3x)$

(....., .....)  
(2)

The graph of  $y = g(x)$  is shown on the grid below.



(b) On the grid, sketch the graph of  $y = 2g(x)$  for  $-1 \leq x \leq 7$

(2)

The graph of  $y = h(x)$  intersects the  $x$ -axis at two points.

The coordinates of the two points are  $(-1, 0)$  and  $(6, 0)$

The graph of  $y = h(x + a)$  passes through the point with coordinates  $(2, 0)$ , where  $a$  is a constant.

(c) Find the two possible values of  $a$

(2)

(Total for Question 20 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2022 P1HR | Question 20

Transformations of Graphs

**Topic check.** Correct.**Method**The turning point of  $y = f(x)$  is  $(-6, -4)$ .

For

$$y = f(x) + 5$$

the graph moves 5 units up:

$$(-6, -4) \rightarrow (-6, 1)$$

For

$$y = f(3x)$$

the  $x$ -coordinate is divided by 3:

$$(-6, -4) \rightarrow (-2, -4)$$

For the sketch of  $y = 2g(x)$ , keep every  $x$ -coordinate the same and double every  $y$ -coordinate.  
Useful points are

$$(-1, 3) \rightarrow (-1, 6)$$

$$(2, 0) \rightarrow (2, 0)$$

$$(3, -1) \rightarrow (3, -2)$$

$$(4, 0) \rightarrow (4, 0)$$

$$(7, 3) \rightarrow (7, 6)$$

Draw straight line segments through these points for  $-1 \leq x \leq 7$ .For part (c),  $h(x)$  has roots at  $x = -1$  and  $x = 6$ .For  $h(x + a)$  to pass through  $(2, 0)$ ,

$$2 + a = -1 \quad \text{or} \quad 2 + a = 6$$

$$a = -3 \quad \text{or} \quad a = 4$$

**FINAL ANSWER** $(-6, 1)$ ,  $(-2, -4)$ , sketch through the doubled points, and  $a = -3$  or  $a = 4$



**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jun 2022

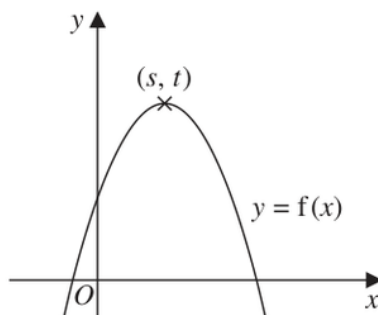
Q#: 20

Marks: 2

TOPIC: Transformations of Graphs

Source: Jun 2022 P1H

20 The diagram shows a sketch of part of the curve with equation  $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are  $(s, t)$

Find, in terms of  $s$  and  $t$ , the coordinates of the maximum point on the curve with equation

(i)  $y = f(x - 2)$

(....., .....)  
(1)

(ii)  $y = 3f(x)$

(....., .....)  
(1)

(Total for Question 20 is 2 marks)

**WORKED SOLUTION***Dr. Eslam Ahmed***Jun 2022 P1H | Question 20**

Transformations of Graphs

**Topic check.** Correct.**Method**The maximum point of  $y = f(x)$  is  $(s, t)$ .

For

$$y = f(x - 2)$$

the graph moves 2 units right:

$$(s, t) \rightarrow (s + 2, t)$$

For

$$y = 3f(x)$$

the  $y$ -coordinate is multiplied by 3:

$$(s, t) \rightarrow (s, 3t)$$

**FINAL ANSWER** $(s + 2, t)$  and  $(s, 3t)$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2022

Q#: 22

Marks: 4

TOPIC: Transformations of Graphs

Source: Jun 2022 P2HR

22 The point  $A$  with coordinates  $(-3, 2)$  lies on the straight line with equation  $y = f(x)$

(a) Find the coordinates of the image of the point  $A$  on the straight line with equation

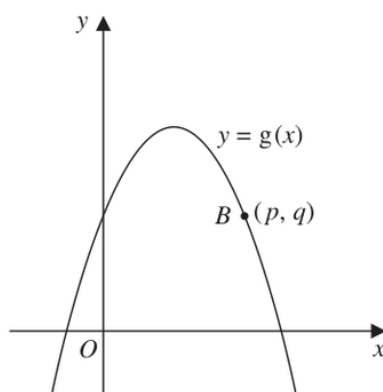
(i)  $y = f(x) - 3$

(....., .....)  
(1)

(ii)  $y = f\left(\frac{x}{2}\right)$

(....., .....)  
(1)

Here is a sketch of part of the curve with equation  $y = g(x)$



The point  $B$  with coordinates  $(p, q)$  lies on the curve.

(b) Find the coordinates of the image of the point  $B$  on the curve with equation

$$y = -g(x - c)$$

where  $c$  is a constant.

(....., .....)  
(2)

(Total for Question 22 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2022 P2HR | Question 22

Transformations of Graphs

**Topic check.** Correct.**Method**Point  $A = (-3, 2)$  lies on  $y = f(x)$ .

For

$$y = f(x) - 3$$

the graph moves 3 units down:

$$(-3, 2) \rightarrow (-3, -1)$$

For

$$y = f\left(\frac{x}{2}\right)$$

the  $x$ -coordinate is multiplied by 2:

$$(-3, 2) \rightarrow (-6, 2)$$

For part (b), point  $B = (p, q)$  lies on  $y = g(x)$ .

For

$$y = -g(x - c)$$

 $x - c$  moves the graph  $c$  units right, and the minus sign reflects the graph in the  $x$ -axis.

So

$$(p, q) \rightarrow (p + c, -q)$$

**FINAL ANSWER** $(-3, -1)$ ,  $(-6, 2)$ , and  $(p + c, -q)$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2024

Q#: 23

Marks: 2

TOPIC: Transformations of Graphs

Source: Jun 2024 P2HR

**23** A curve has equation  $y = f(x)$ The coordinates of the minimum point on this curve are  $(6, -3)$ 

Write down the coordinates of the minimum point on the curve with equation

(i)  $y = f(x) + 10$ (....., .....)  
(1)(ii)  $y = f(3x)$ (....., .....)  
(1)

(Total for Question 23 is 2 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2024 P2HR | Question 23

Transformations of Graphs

**Topic check.** Correct.**Method**The minimum point of  $y = f(x)$  is  $(6, -3)$ .

For

$$y = f(x) + 10$$

the graph moves 10 units up:

$$(6, -3) \rightarrow (6, 7)$$

For

$$y = f(3x)$$

the  $x$ -coordinate is divided by 3:

$$(6, -3) \rightarrow (2, -3)$$

**FINAL ANSWER** $(6, 7)$  and  $(2, -3)$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jun 2025

Q#: 21

Marks: 3

TOPIC: Transformations of Graphs

Source: Jun 2025 P1HR

**21** The curve with equation  $y = f(x)$  has one turning point.

The coordinates of this turning point are  $(-6, 9)$

- (a) Write down the coordinates of the turning point on the curve with equation  $y = f(3x)$

(....., .....)  
(1)

The curve **C** with equation  $y = g(x)$  is transformed to give the curve **S** with equation  $y = g(x + a) + b$

The point  $(4, -5)$  on **C** is mapped to the point  $(1, -16)$  on **S**

- (b) Write down the value of  $a$  and the value of  $b$

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(2)

(Total for Question 21 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2025 P1HR | Question 21

Transformations of Graphs

**Topic check.** Correct.**Method**The turning point of  $y = f(x)$  is  $(-6, 9)$ .

For

$$y = f(3x)$$

the  $x$ -coordinate is divided by 3:

$$(-6, 9) \rightarrow (-2, 9)$$

For part (b), a point  $(4, -5)$  on  $y = g(x)$  maps to  $(1, -16)$  on

$$y = g(x + a) + b$$

For the  $x$ -coordinate:

$$1 + a = 4$$

$$a = 3$$

For the  $y$ -coordinate:

$$-5 + b = -16$$

$$b = -11$$

**FINAL ANSWER**

$$(-2, 9), a = 3, b = -11$$



**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2023

Q#: 21

Marks: 2

TOPIC: Transformations of Graphs

Source: May 2023 P1H

**21** A curve has equation  $y = f(x)$ 

There is only one turning point on the curve.

The coordinates of this turning point are (6, 5)

Write down the coordinates of the turning point on the curve with equation

(a)  $y = f(x - 4)$

(....., .....)  
(1)

(b)  $y = f(3x)$

(....., .....)  
(1)**(Total for Question 21 is 2 marks)**

**WORKED SOLUTION***Dr. Eslam Ahmed***May 2023 P1H | Question 21**

Transformations of Graphs

**Topic check.** Correct.**Method**The turning point of  $y = f(x)$  is  $(6, 5)$ .

For

$$y = f(x - 4)$$

the graph moves 4 units right:

$$(6, 5) \rightarrow (10, 5)$$

For

$$y = f(3x)$$

the  $x$ -coordinate is divided by 3:

$$(6, 5) \rightarrow (2, 5)$$

**FINAL ANSWER** $(10, 5)$  and  $(2, 5)$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: May 2023

Q#: 20

Marks: 7

TOPIC: Transformations of Graphs

Source: May 2023 P1HR

**20** Curve **C** has equation  $y = f(x)$ The graph of curve **C** has one maximum point.

The coordinates of this maximum point are (3, 5)

(a) Write down the coordinates of the maximum point on the curve with equation

(i)  $y = 2f(x)$

(....., .....)  
(1)

(ii)  $y = f(x) - 7$

(....., .....)  
(1)

(iii)  $y = f(-x)$

(....., .....)  
(1)Curve **L** has equation  $y = x^2 + 7x + 20$ Curve **L** is transformed to curve **S** under the translation  $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ (b) Find an equation for **S**Give your answer in the form  $y = ax^2 + bx + c$  $y = \dots\dots\dots$   
(4)

(Total for Question 20 is 7 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2023 P1HR | Question 20

Transformations of Graphs

**Topic check.** Correct.**Method**The maximum point of  $y = f(x)$  is  $(3, 5)$ .

For

$$y = 2f(x)$$

the  $y$ -coordinate is multiplied by 2:

$$(3, 5) \rightarrow (3, 10)$$

For

$$y = f(x) - 7$$

the graph moves 7 units down:

$$(3, 5) \rightarrow (3, -2)$$

For

$$y = f(-x)$$

the graph reflects in the  $y$ -axis:

$$(3, 5) \rightarrow (-3, 5)$$

For part (b),

$$y = x^2 + 7x + 20$$

The translation vector is

$$\begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

So the graph moves 2 units right.

Replace  $x$  by  $x - 2$ :

$$y = (x - 2)^2 + 7(x - 2) + 20$$

$$y = x^2 + 3x + 10$$

**FINAL ANSWER** $(3, 10)$ ,  $(3, -2)$ ,  $(-3, 5)$ , and  $y = x^2 + 3x + 10$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2024

Q#: 21

Marks: 2

TOPIC: Transformations of Graphs

Source: May 2024 P1H

**21** A curve has equation  $y = f(x)$ 

There is one minimum point on this curve.

The coordinates of this minimum point are  $(5, -4)$ 

Write down the coordinates of the minimum point on the curve with equation

(i)  $y = f(x + 7)$

( ..... , ..... )  
(1)

(ii)  $y = f(x) - 6$

( ..... , ..... )  
(1)

---

**(Total for Question 21 is 2 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2024 P1H | Question 21

Transformations of Graphs

**Topic check.** Correct.**Method**The minimum point of  $y = f(x)$  is  $(5, -4)$ .

For

$$y = f(x + 7)$$

the graph moves 7 units left:

$$(5, -4) \rightarrow (-2, -4)$$

For

$$y = f(x) - 6$$

the graph moves 6 units down:

$$(5, -4) \rightarrow (5, -10)$$

**FINAL ANSWER** $(-2, -4)$  and  $(5, -10)$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: MayNov 2020

Q#: 25

Marks: 3

TOPIC: Transformations of Graphs

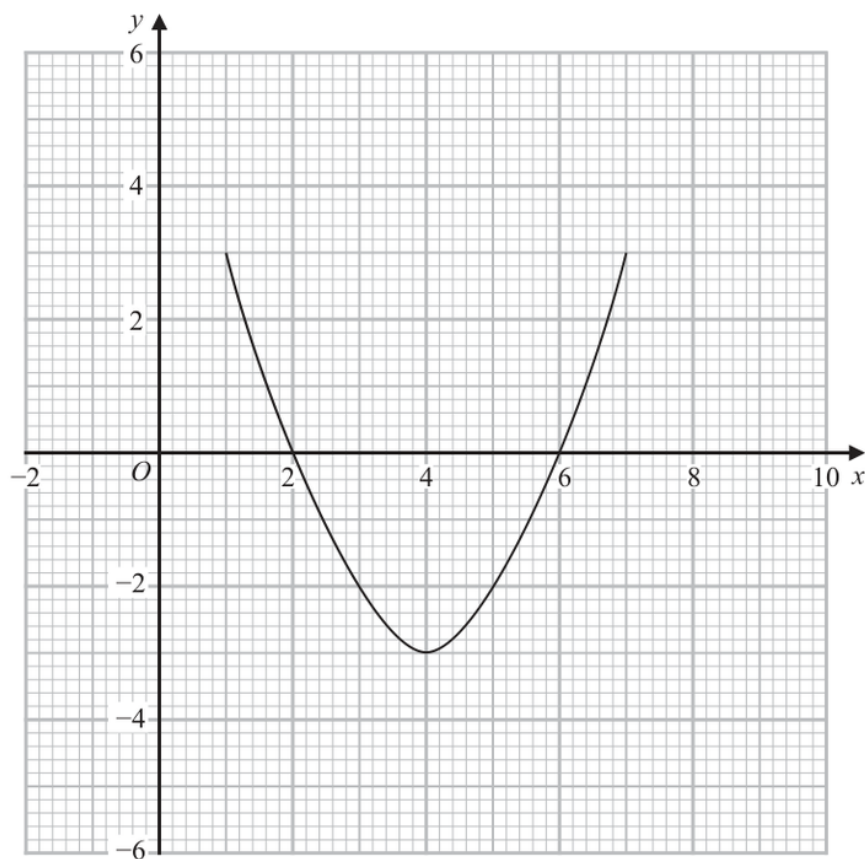
Source: May-Nov 2020 P1H

- 25 The curve with equation  $y = g(x)$  is transformed to the curve with equation  $y = -g(x)$  by the single transformation **T**.

(a) Describe fully the transformation **T**.

(1)

The diagram shows the graph of  $y = f(x)$



(b) On the grid, draw the graph of  $y = 2f(x - 1)$

(2)

(Total for Question 25 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May-Nov 2020 P1H | Question 25

Transformations of Graphs

**Topic check.** Correct.**Method**

For part (a),

$$y = g(x) \rightarrow y = -g(x)$$

The  $x$ -coordinates stay the same and the  $y$ -coordinates change sign.So the transformation is a reflection in the  $x$ -axis.

For part (b),

$$y = 2f(x - 1)$$

First,  $x - 1$  moves the graph 1 unit right. Then the factor 2 stretches the graph vertically by scale factor 2.

Use points from the original graph:

$$(1, 3) \rightarrow (2, 6)$$

$$(2, 0) \rightarrow (3, 0)$$

$$(4, -3) \rightarrow (5, -6)$$

$$(6, 0) \rightarrow (7, 0)$$

$$(7, 3) \rightarrow (8, 6)$$

Draw the transformed parabola through these points.

**FINAL ANSWER**reflection in the  $x$ -axis; transformed graph through  $(2, 6)$ ,  $(3, 0)$ ,  $(5, -6)$ ,  $(7, 0)$ ,  $(8, 6)$



**CLASSIFIED PROBLEM**

Paper: P1HR

Session: MayNov 2020

Q#: 20

Marks: 2

TOPIC: Transformations of Graphs

Source: May-Nov 2020 P1HR

**20** A curve has equation  $y = f(x)$ 

There is only one maximum point on the curve.

The coordinates of this maximum point are  $(-3, 4)$ 

Write down the coordinates of the maximum point on the curve with equation

(i)  $y = f(x) - 6$

(....., .....)

(ii)  $y = f(2x)$

(....., .....)

---

**(Total for Question 20 is 2 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## May-Nov 2020 P1HR | Question 20

Transformations of Graphs

**Topic check.** Correct.**Method**The maximum point of  $y = f(x)$  is  $(-3, 4)$ .

For

$$y = f(x) - 6$$

the graph moves 6 units down:

$$(-3, 4) \rightarrow (-3, -2)$$

For

$$y = f(2x)$$

the  $x$ -coordinate is divided by 2:

$$(-3, 4) \rightarrow \left(-\frac{3}{2}, 4\right)$$

**FINAL ANSWER** $(-3, -2)$  and  $\left(-\frac{3}{2}, 4\right)$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2021

Q#: 21

Marks: 6

TOPIC: Transformations of Graphs

Source: Nov 2021 P2H

- 21 The curve **C** has equation  $y = f(x)$  where  $f(x) = 9 - 3(x + 2)^2$   
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(....., .....)  
(1)

The curve **C** is transformed to the curve **S** by a translation of  $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

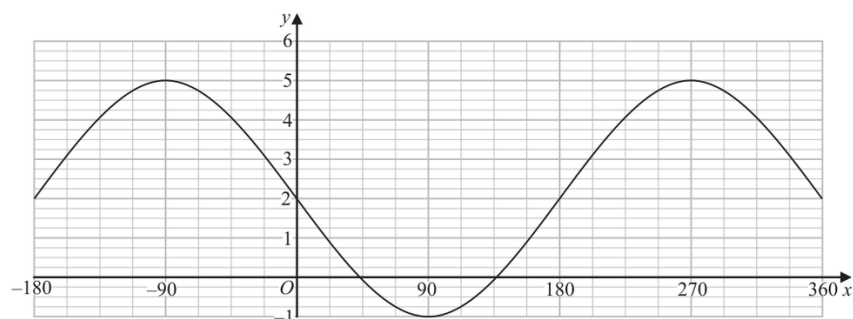
.....  
(1)

The curve **C** is transformed to the curve **T**.  
The curve **T** has equation  $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....  
(1)

The graph of  $y = a \cos(x - b)^\circ + c$  for  $-180 \leq x \leq 360$  is drawn on the grid below.



(d) Find the value of  $a$ , the value of  $b$  and the value of  $c$ .

$a =$  .....

$b =$  .....

$c =$  .....

(3)

(Total for Question 21 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2021 P2H | Question 21

Transformations of Graphs

**Topic check.** Correct.**Method**

$$f(x) = 9 - 3(x + 2)^2$$

This is in completed square form, so the maximum point is

$$(-2, 9)$$

For the translation

$$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

the graph moves 4 units right. Replace  $x$  by  $x - 4$ :

$$y = 9 - 3((x - 4) + 2)^2$$

$$y = 9 - 3(x - 2)^2$$

The curve  $T$  has equation

$$y = 3(x + 2)^2 - 9$$

This is

$$y = -\left(9 - 3(x + 2)^2\right)$$

So  $C$  is reflected in the  $x$ -axis.

For the cosine graph, the maximum is 5 and the minimum is  $-1$ .

$$a = \frac{5 - (-1)}{2} = 3$$

$$c = \frac{5 + (-1)}{2} = 2$$

For

$$y = a \cos(x - b)^\circ + c$$

a maximum occurs when  $x - b = 0$ . The graph has a maximum at  $x = 270^\circ$ , so

$$b = 270$$

**FINAL ANSWER**

$A = (-2, 9)$ ,  $y = 9 - 3(x - 2)^2$ , reflection in the  $x$ -axis,  $a = 3$ ,  $b = 270$ ,  $c = 2$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2023

Q#: 24

Marks: 4

TOPIC: Transformations of Graphs

Source: Nov 2023 P1H

24 The curve with equation  $f(x) = 5x^2 + 9x + 2$  is transformed to the curve with equation

$$g(x) = 5(x+4)^2 + 9(x+4) + 8 \text{ by the translation } \begin{pmatrix} a \\ b \end{pmatrix}$$

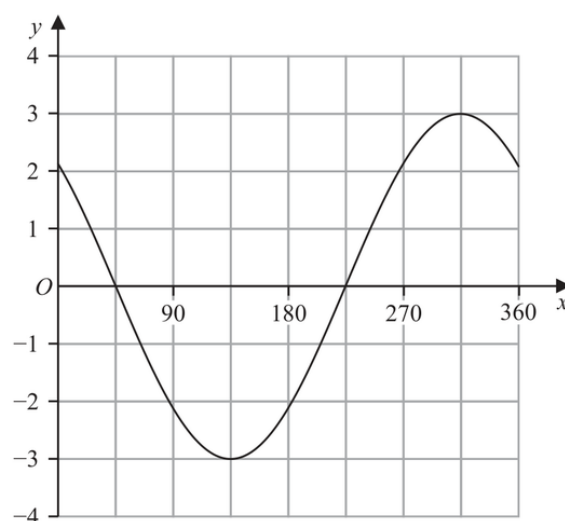
(a) Write down the value of  $a$  and the value of  $b$

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

(2)

The graph of  $y = p \cos(x+q)^\circ$  for  $0 \leq x \leq 360$  is drawn on the grid below.



Given that  $p > 0$  and  $0 < q < 360$

(b) find the value of  $p$  and the value of  $q$

$$p = \dots\dots\dots$$

$$q = \dots\dots\dots$$

(2)

(Total for Question 24 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2023 P1H | Question 24

Transformations of Graphs

**Topic check.** Moved from Functions to Transformations of Graphs.**Method**

For part (a),

$$f(x) = 5x^2 + 9x + 2$$

$$g(x) = 5(x + 4)^2 + 9(x + 4) + 8$$

Since

$$f(x + 4) = 5(x + 4)^2 + 9(x + 4) + 2$$

$$g(x) = f(x + 4) + 6$$

So the graph has moved 4 units left and 6 units up.

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}$$

For part (b), the graph has maximum value 3 and minimum value  $-3$ , so

$$p = 3$$

The minimum occurs when

$$x + q = 180$$

From the graph, the minimum is at  $x = 135$ , so

$$135 + q = 180$$

$$q = 45$$

**FINAL ANSWER**

$$a = -4, b = 6, p = 3, q = 45$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2025

Q#: 24

Marks: 3

TOPIC: Transformations of Graphs

Source: Nov 2025 P1H

**24** A curve has equation  $y = f(x)$ 

There is only one turning point on the curve.

The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i)  $y = f(x + 5)$

( ..... , ..... )  
(1)

(ii)  $y = f(x) + 7$

( ..... , ..... )  
(1)

(iii)  $y = f(2x)$

( ..... , ..... )  
(1)**(Total for Question 24 is 3 marks)**

**WORKED SOLUTION***Dr. Eslam Ahmed***Nov 2025 P1H | Question 24**

Transformations of Graphs

**Topic check.** Correct.**Method**The turning point of  $y = f(x)$  is  $(4, 3)$ .

For

$$y = f(x + 5)$$

the graph moves 5 units left:

$$(4, 3) \rightarrow (-1, 3)$$

For

$$y = f(x) + 7$$

the graph moves 7 units up:

$$(4, 3) \rightarrow (4, 10)$$

For

$$y = f(2x)$$

the  $x$ -coordinate is divided by 2:

$$(4, 3) \rightarrow (2, 3)$$

**FINAL ANSWER** $(-1, 3)$ ,  $(4, 10)$ , and  $(2, 3)$



# Differentiation

---

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jan 2020

Q#: 23

Marks: 6

TOPIC: Differentiation

Source: Jan 2020 P1HR

**23** A particle moves along a straight line.The fixed point  $O$  lies on this line.The displacement of the particle from  $O$  at time  $t$  seconds,  $t \geq 0$ , is  $s$  metres where

$$s = t^3 + 4t^2 - 5t + 7$$

At time  $T$  seconds the velocity of  $P$  is  $V$  m/s where  $V \geq -5$ Find an expression for  $T$  in terms of  $V$ .Give your expression in the form  $\frac{-4 + \sqrt{k + mV}}{3}$  where  $k$  and  $m$  are integers to be found. $T = \dots\dots\dots$ **(Total for Question 23 is 6 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2020 P1HR | Question 23

Differentiation

**Topic check.** Correct.**Method**

$$s = t^3 + 4t^2 - 5t + 7$$

Velocity is the derivative of displacement:

$$v = \frac{ds}{dt} = 3t^2 + 8t - 5$$

At time  $T$ , the velocity is  $V$ , so

$$V = 3T^2 + 8T - 5$$

$$3T^2 + 8T - (V + 5) = 0$$

Use the quadratic formula:

$$T = \frac{-8 \pm \sqrt{8^2 - 4(3)(-(V + 5))}}{2(3)}$$

$$T = \frac{-8 \pm \sqrt{64 + 12V + 60}}{6}$$

$$T = \frac{-8 \pm \sqrt{124 + 12V}}{6}$$

$$T = \frac{-4 \pm \sqrt{31 + 3V}}{3}$$

Since  $T \geq 0$ , take the positive root:

$$T = \frac{-4 + \sqrt{31 + 3V}}{3}$$

**FINAL ANSWER**

$$T = \frac{-4 + \sqrt{31 + 3V}}{3}$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jan 2021

Q#: 20

Marks: 5

TOPIC: **Differentiation**

Source: Jan 2021 P1HR

- 20** A particle  $P$  is moving along a straight line.  
The fixed point  $O$  lies on the line.

At time  $t$  seconds ( $t \geq 0$ ), the displacement of  $P$  from  $O$  is  $s$  metres where

$$s = t^3 - 9t^2 + 33t - 6$$

Find the minimum speed of  $P$ .

..... m/s

**(Total for Question 20 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1HR | Question 20  
Differentiation

**Topic check.** Moved from Linear Graphs to Differentiation.

**Method**

$$s = t^3 - 9t^2 + 33t - 6$$

Velocity is the rate of change of displacement:

$$v = \frac{ds}{dt}$$

$$v = 3t^2 - 18t + 33$$

Complete the square:

$$v = 3(t^2 - 6t + 11)$$

$$v = 3[(t - 3)^2 + 2]$$

$$v = 3(t - 3)^2 + 6$$

Since  $(t - 3)^2 \geq 0$ , the minimum value of  $v$  is

$$6$$

The velocity is always positive, so the minimum speed is also 6.

**FINAL ANSWER**

6 m/s

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jan 2022

Q#: 24

Marks: 6

TOPIC: Differentiation

Source: Jan 2022 P1H

**24** The curve **C** has equation  $y = ax^3 + bx^2 - 12x + 6$  where  $a$  and  $b$  are constants.

The point  $A$  with coordinates  $(2, -6)$  lies on **C**

The gradient of the curve at  $A$  is 16

Find the  $y$  coordinate of the point on the curve whose  $x$  coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

(Total for Question 24 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1H | Question 24  
Differentiation

**Topic check.** Correct.

**Method**

$$y = ax^3 + bx^2 - 12x + 6$$

Point  $A = (2, -6)$  lies on the curve:

$$-6 = 8a + 4b - 24 + 6$$

$$8a + 4b = 12$$

$$2a + b = 3$$

Differentiate:

$$\frac{dy}{dx} = 3ax^2 + 2bx - 12$$

The gradient at  $x = 2$  is 16:

$$16 = 12a + 4b - 12$$

$$12a + 4b = 28$$

$$3a + b = 7$$

Now solve

$$2a + b = 3$$

$$3a + b = 7$$

Subtracting gives

$$a = 4$$

Then

$$2(4) + b = 3$$

$$b = -5$$

For  $x = 3$ ,

$$y = 4(3)^3 - 5(3)^2 - 12(3) + 6$$

$$y = 108 - 45 - 36 + 6$$

$$y = 33$$

**FINAL ANSWER**

$$y = 33$$



**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jan 2022

Q#: 23

Marks: 6

TOPIC: Differentiation

Source: Jan 2022 P1HR

- 23** Two particles,  $P$  and  $Q$ , move along a straight line.  
The fixed point  $O$  lies on this line.

The displacement of  $P$  from  $O$  at time  $t$  seconds is  $s$  metres, where

$$s = t^3 - 4t^2 + 5t \quad \text{for } t > 1$$

The displacement of  $Q$  from  $O$  at time  $t$  seconds is  $x$  metres, where

$$x = t^2 - 4t + 4 \quad \text{for } t > 1$$

Find the range of values of  $t$  where  $t > 1$  for which both particles are moving in the same direction along the straight line.

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(Total for Question 23 is 6 marks)

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## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2022 P1HR | Question 23

Differentiation

**Topic check.** Corrected from Functions to Differentiation.**Method**Particle  $P$ :

$$s = t^3 - 4t^2 + 5t$$

$$\frac{ds}{dt} = 3t^2 - 8t + 5$$

$$\frac{ds}{dt} = (3t - 5)(t - 1)$$

Particle  $Q$ :

$$x = t^2 - 4t + 4$$

$$\frac{dx}{dt} = 2t - 4 = 2(t - 2)$$

For  $t > 1$ :

- $P$  moves in the negative direction when  $1 < t < \frac{5}{3}$ .
- $P$  moves in the positive direction when  $t > \frac{5}{3}$ .
- $Q$  moves in the negative direction when  $1 < t < 2$ .
- $Q$  moves in the positive direction when  $t > 2$ .

So both particles move in the same direction when both velocities have the same sign.

**FINAL ANSWER**

$$1 < t < \frac{5}{3} \text{ or } t > 2$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2022

Q#: 24

Marks: 5

TOPIC: Differentiation

Source: Jan 2022 P2H

**24** A particle  $P$  moves along a straight line that passes through the fixed point  $O$

The displacement,  $x$  metres, of  $P$  from  $O$  at time  $t$  seconds, where  $t \geq 0$ , is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of  $P$  reverses when  $P$  is at the point  $A$  on the line.

The acceleration of  $P$  at the instant when  $P$  is at  $A$  is  $a \text{ m/s}^2$

Find the value of  $a$

$a = \dots\dots\dots$

(Total for Question 24 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 24

Differentiation

**Topic check.** Corrected from Ratio Toolkit to Differentiation.**Method**

$$x = 4t^3 - 27t + 8$$

Velocity is

$$\frac{dx}{dt} = 12t^2 - 27$$

The direction of motion reverses when velocity is zero:

$$12t^2 - 27 = 0$$

$$12t^2 = 27$$

$$t^2 = \frac{9}{4}$$

Since  $t \geq 0$ ,  $t = \frac{3}{2}$ .

Acceleration is

$$\frac{d^2x}{dt^2} = 24t$$

At  $t = \frac{3}{2}$ ,

$$a = 24 \left( \frac{3}{2} \right) = 36$$

**FINAL ANSWER**

$$a = 36$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jun 2022

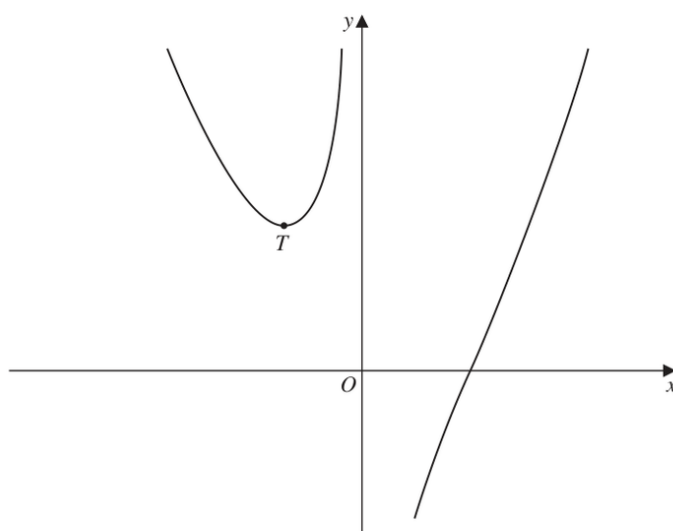
Q#: 22

Marks: 7

TOPIC: Differentiation

Source: Jun 2022 P1HR

- 22 The diagram shows a sketch of part of the curve with equation  $y = x^2 - \frac{p}{x}$  where  $p$  is a positive constant.

Diagram **NOT**  
accurately drawn

For all values of  $p$ , the curve has exactly one turning point and this turning point is a minimum shown as the point  $T$  in the sketch.

For the curve where the  $x$  coordinate of  $T$  is  $-3$

- (a) find the value of  $p$

$$p = \dots\dots\dots (4)$$

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The line with equation  $y = k$  is a tangent to the curve with equation  $y = x^2 - \frac{16}{x}$

- (b) Find the value of  $k$

$$k = \dots\dots\dots (3)$$

(Total for Question 22 is 7 marks)

---

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2022 P1HR | Question 22

Differentiation

**Topic check.** Moved from Estimating Gradients to Differentiation.**Method**

For part (a),

$$y = x^2 - \frac{p}{x}$$

$$\frac{dy}{dx} = 2x + \frac{p}{x^2}$$

At the minimum point,  $x = -3$  and the gradient is 0.

$$2(-3) + \frac{p}{(-3)^2} = 0$$

$$-6 + \frac{p}{9} = 0$$

$$p = 54$$

For part (b),

$$y = x^2 - \frac{16}{x}$$

$$\frac{dy}{dx} = 2x + \frac{16}{x^2}$$

At a horizontal tangent,

$$2x + \frac{16}{x^2} = 0$$

$$2x^3 + 16 = 0$$

$$x^3 = -8$$

$$x = -2$$

Now substitute  $x = -2$  into the curve:

$$k = (-2)^2 - \frac{16}{-2}$$

$$k = 4 + 8 = 12$$

**FINAL ANSWER**

$$p = 54, k = 12$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2025

Q#: 22

Marks: 5

TOPIC: **Differentiation**

Source: Jun 2025 P2HR

22 The diagram shows a solid cuboid.

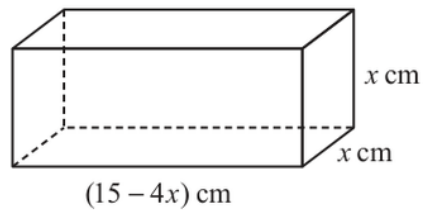


Diagram **NOT**  
accurately drawn

The volume of the cuboid is  $V \text{ cm}^3$

Find the maximum value of  $V$

---

(Total for Question 22 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | Question 22

Differentiation

**Topic check.** Correct.**Method**

The cuboid has dimensions

$$x, \quad x, \quad 15 - 4x$$

So

$$V = x^2(15 - 4x)$$

$$V = 15x^2 - 4x^3$$

Differentiate:

$$\frac{dV}{dx} = 30x - 12x^2$$

At a maximum,

$$30x - 12x^2 = 0$$

$$6x(5 - 2x) = 0$$

The useful value is

$$x = 2.5$$

Now substitute into the volume:

$$V = (2.5)^2(15 - 4(2.5))$$

$$V = 6.25 \times 5$$

$$V = 31.25$$

**FINAL ANSWER**

$$31.25 \text{ cm}^3$$



**CLASSIFIED PROBLEM**

Paper: P2H

Session: May 2021

Q#: 21

Marks: 5

TOPIC: **Differentiation**

Source: May 2021 P2H

- 21** The point  $A$  is the only stationary point on the curve with equation  $y = kx^2 + \frac{16}{x}$  where  $k$  is a constant.

Given that the coordinates of  $A$  are  $\left(\frac{2}{3}, a\right)$

find the value of  $a$ .

Show your working clearly.

$a = \dots\dots\dots$

(Total for Question 21 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2021 P2H | Question 21

Differentiation

**Topic check.** Correct.**Method**

$$y = kx^2 + \frac{16}{x}$$

Differentiate:

$$\frac{dy}{dx} = 2kx - \frac{16}{x^2}$$

At the stationary point,  $x = \frac{2}{3}$ , so the gradient is 0:

$$2k \left( \frac{2}{3} \right) - \frac{16}{\left( \frac{2}{3} \right)^2} = 0$$

$$\frac{4k}{3} - \frac{16}{\frac{4}{9}} = 0$$

$$\frac{4k}{3} - 36 = 0$$

$$\frac{4k}{3} = 36$$

$$k = 27$$

Now find  $a$ , the  $y$ -coordinate when  $x = \frac{2}{3}$ :

$$a = 27 \left( \frac{2}{3} \right)^2 + \frac{16}{\frac{2}{3}}$$

$$a = 27 \cdot \frac{4}{9} + 24$$

$$a = 12 + 24 = 36$$

**FINAL ANSWER**

$$a = 36$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: May 2023

Q#: 21

Marks: 6

TOPIC: Differentiation

Source: May 2023 P1HR

**21** The curve **T** has equation  $y = x^3 - 2x^2 - 9x + 15$

(a) Find  $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

(2)

(b) Find the range of values of  $x$  for which **T** has a positive gradient.  
Give your values correct to 3 significant figures.  
Show your working clearly.

(4)

(Total for Question 21 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2023 P1HR | Question 21

Differentiation

**Topic check.** Corrected from Functions to Differentiation.**Method**

(a)

$$y = x^3 - 2x^2 - 9x + 15$$

$$\frac{dy}{dx} = 3x^2 - 4x - 9$$

(b) The curve has a positive gradient when

$$3x^2 - 4x - 9 > 0$$

Solve

$$3x^2 - 4x - 9 = 0$$

$$x = \frac{4 \pm \sqrt{16 + 108}}{6}$$

$$x = \frac{2 \pm \sqrt{31}}{3}$$

To 3 significant figures, the roots are

$$x = -1.19, \quad x = 2.52$$

Since the quadratic opens upwards, it is positive outside the roots.

**FINAL ANSWER**(a)  $\frac{dy}{dx} = 3x^2 - 4x - 9$ , (b)  $x < -1.19$  or  $x > 2.52$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: MayNov 2020

Q#: 23

Marks: 4

TOPIC: **Differentiation**

Source: May-Nov 2020 P1H

**23** Curve **C** has equation  $y = px^3 - mx$  where  $p$  and  $m$  are positive integers.

Find the range of values of  $x$ , in terms of  $p$  and  $m$ , for which the gradient of **C** is negative.

---

(Total for Question 23 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May-Nov 2020 P1H | Question 23

Differentiation

**Topic check.** Corrected from Functions to Differentiation.**Method**

$$y = px^3 - mx$$

Differentiate:

$$\frac{dy}{dx} = 3px^2 - m$$

The gradient is negative when

$$3px^2 - m < 0$$

$$3px^2 < m$$

$$x^2 < \frac{m}{3p}$$

So

$$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

**FINAL ANSWER**

$$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2024

Q#: 20

Marks: 4

TOPIC: **Differentiation**

Source: Nov 2024 P1H

**20** The curve with equation  $y = 2x^4 - 64x$  has a minimum point.

Find an equation of the tangent to the curve at the minimum point.

Show clear algebraic working.

---

(Total for Question 20 is 4 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Nov 2024 P1H** | **Question 20**  
Differentiation

**Topic check.** Moved from Estimating Gradients to Differentiation.

**Method**

$$y = 2x^4 - 64x$$

Differentiate:

$$\frac{dy}{dx} = 8x^3 - 64$$

At the minimum point, the gradient is 0.

$$8x^3 - 64 = 0$$

$$8x^3 = 64$$

$$x^3 = 8$$

$$x = 2$$

Find the  $y$ -coordinate:

$$y = 2(2)^4 - 64(2)$$

$$y = 32 - 128 = -96$$

The tangent at a minimum point is horizontal, so its equation is

$$y = -96$$

**FINAL ANSWER**

$$y = -96$$



**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2025

Q#: 22

Marks: 6

TOPIC: **Differentiation**

Source: Nov 2025 P2H

**22** Curve **C** has equation  $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form  $y = ax + b$

Work out the two possible values of  $b$

Show clear algebraic working.

(Total for Question 22 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 P2H | Question 22

Differentiation

**Topic check.** Moved to Differentiation.**Method**

$$y = x^3 - 16x + 7$$

Differentiate:

$$\frac{dy}{dx} = 3x^2 - 16$$

The gradient of the tangent is 11, so

$$3x^2 - 16 = 11$$

$$3x^2 = 27$$

$$x^2 = 9$$

$$x = 3 \quad \text{or} \quad x = -3$$

When  $x = 3$ :

$$y = 3^3 - 16(3) + 7 = 27 - 48 + 7 = -14$$

The tangent has equation  $y = 11x + b$ , so

$$-14 = 11(3) + b$$

$$b = -47$$

When  $x = -3$ :

$$y = (-3)^3 - 16(-3) + 7 = -27 + 48 + 7 = 28$$

$$28 = 11(-3) + b$$

$$b = 61$$

**FINAL ANSWER**

$$b = -47 \text{ or } b = 61$$

# Angles in Polygons & Parallel Lines

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**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jun 2022

Q#: 23

Marks: 6

TOPIC: Angles in Polygons &amp; Parallel Lines

Source: Jun 2022 P1H

**23** A polygon has  $n$  sides, where  $n > 5$ 

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177      175      173      171      169

Find the value of  $n$

Show clear algebraic working.

**Question 23 continued**

$n = \dots\dots\dots$

**(Total for Question 23 is 6 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2022 P1H | Question 23

Angles in Polygons &amp; Parallel Lines

**Topic check.** Correct.**Method**

The interior angles form an arithmetic sequence:

$$177, 175, 173, 171, 169, \dots$$

So

$$a = 177, \quad d = -2$$

For an  $n$ -sided polygon, the sum of the interior angles is

$$180(n - 2)$$

The sum of the arithmetic sequence is

$$\begin{aligned} & \frac{n}{2} (2(177) + (n - 1)(-2)) \\ &= \frac{n}{2} (354 - 2n + 2) \\ &= \frac{n}{2} (356 - 2n) \\ &= n(178 - n) \end{aligned}$$

Set this equal to the polygon angle sum:

$$n(178 - n) = 180(n - 2)$$

$$178n - n^2 = 180n - 360$$

$$n^2 + 2n - 360 = 0$$

Factorise:

$$(n + 20)(n - 18) = 0$$

Since  $n > 5$ ,

$$n = 18$$

**FINAL ANSWER**

$$n = 18$$

# Bearings, Scale Drawing & Constructions

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**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2024

Q#: 24

Marks: 6

TOPIC: Bearings, Scale Drawing &amp; Constructions

Source: Nov 2024 P2H

24  $A$ ,  $B$  and  $C$  are three points on horizontal ground.

$AB = 8.4$  metres  $BC = 9.2$  metres

$B$  is on a bearing of  $067^\circ$  from  $A$

$C$  is on a bearing of  $129^\circ$  from  $B$

Calculate the bearing of  $A$  from  $C$

Give your answer correct to the nearest degree.

---



---

o

(Total for Question 24 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2024 P2H | Question 24

Bearings, Scale Drawing &amp; Constructions

Topic check. Correct.

## Method

Resolve each journey into east and north components.

From  $A$  to  $B$ :

$$\text{east} = 8.4 \sin 67^\circ = 7.7322 \dots$$

$$\text{north} = 8.4 \cos 67^\circ = 3.2821 \dots$$

From  $B$  to  $C$ :

$$\text{east} = 9.2 \sin 129^\circ = 7.1497 \dots$$

$$\text{north} = 9.2 \cos 129^\circ = -5.7897 \dots$$

So from  $A$  to  $C$ :

$$\text{east} = 14.8820 \dots$$

$$\text{north} = -2.5076 \dots$$

Therefore from  $C$  to  $A$ :

$$\text{east} = -14.8820 \dots, \quad \text{north} = 2.5076 \dots$$

This direction is west of north.

$$\theta = \tan^{-1} \left( \frac{14.8820 \dots}{2.5076 \dots} \right) = 80.4355 \dots^\circ$$

Bearing of  $A$  from  $C$ :

$$360^\circ - 80.4355 \dots^\circ = 279.5644 \dots^\circ$$

## FINAL ANSWER

280°



# Circle Theorems

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**CLASSIFIED PROBLEM**

Paper: P2HR

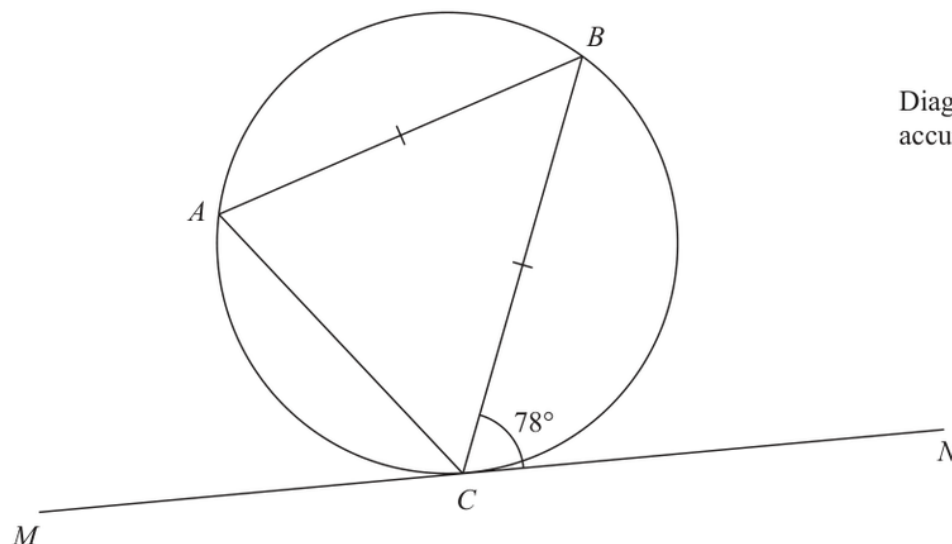
Session: Jun 2023

Q#: 20

Marks: 2

TOPIC: Circle Theorems

Source: Jun 2023 P2HR

**20**  $A, B$  and  $C$  are points on a circle.Diagram **NOT**  
accurately drawn $MN$  is the tangent to the circle at  $C$  $AB = CB$ Angle  $BCN = 78^\circ$ Find the size of angle  $ABC$ 

o

(Total for Question 20 is 2 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2023 P2HR** | **Question 20**  
Circle Theorems

**Topic check.** Correct.

**Method**

The angle between the tangent  $CN$  and the chord  $CB$  is  $78^\circ$ .  
By the alternate segment theorem,

$$\angle BAC = 78^\circ$$

Also,

$$AB = CB$$

so triangle  $ABC$  is isosceles and the base angles are equal:

$$\angle BAC = \angle ACB = 78^\circ$$

Using the angle sum of a triangle:

$$\angle ABC = 180^\circ - 78^\circ - 78^\circ$$

$$\angle ABC = 24^\circ$$

**FINAL ANSWER**

$24^\circ$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jun 2025

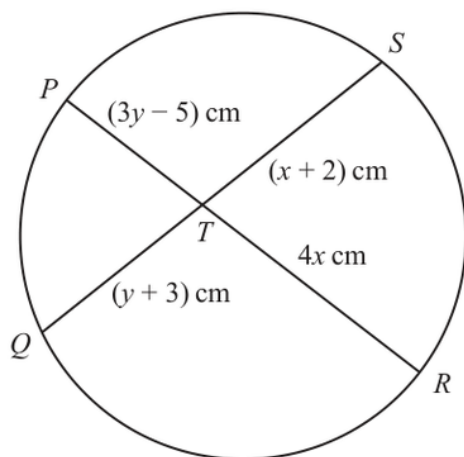
Q#: 23

Marks: 5

TOPIC: Circle Theorems

Source: Jun 2025 P1HR

23

Diagram **NOT**  
accurately drawn $PTR$  and  $QTS$  are chords of a circle.

$$PT = (3y - 5) \text{ cm} \quad QT = (y + 3) \text{ cm} \quad RT = 4x \text{ cm} \quad ST = (x + 2) \text{ cm}$$

Find an expression for  $y$  in terms of  $x$ 

$$y = \dots\dots\dots$$

**Total for Question 23 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

**Jun 2025 P1HR** | **Question 23**  
Circle Theorems

**Topic check.** Correct.

**Method**

For two intersecting chords in a circle,

$$PT \times TR = QT \times TS$$

Substitute the given lengths:

$$(3y - 5)(4x) = (y + 3)(x + 2)$$

Expand:

$$12xy - 20x = xy + 2y + 3x + 6$$

Collect the  $y$  terms on one side:

$$12xy - xy - 2y = 20x + 3x + 6$$

$$11xy - 2y = 23x + 6$$

Factorise:

$$y(11x - 2) = 23x + 6$$

$$y = \frac{23x + 6}{11x - 2}$$

**FINAL ANSWER**

$$y = \frac{23x + 6}{11x - 2}$$

# Volume & Surface Area

---

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: May2025

Q#: 23

Marks: 3

TOPIC: Volume &amp; Surface Area

Source: May 2025 4WM2H

- 23 The diagram shows a cuboid.

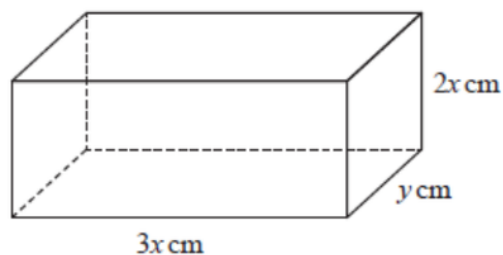


Diagram NOT  
accurately drawn

The cuboid measures  $3x$  cm by  $2x$  cm by  $y$  cm

The volume of the cuboid is  $1014 \text{ cm}^3$

The total surface area of the cuboid is  $A \text{ cm}^2$

Show that  $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 23 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM2H | Question 23

Volume &amp; Surface Area

## Solution

$$3x \times 2x \times y = 1014$$

$$6x^2y = 1014$$

$$y = \frac{169}{x^2}$$

Surface area:

$$A = 2(3x \cdot 2x + 3x \cdot y + 2x \cdot y)$$

$$= 12x^2 + 10xy$$

$$= 12x^2 + 10x \left( \frac{169}{x^2} \right)$$

$$= 12x^2 + \frac{1690}{x}$$

**FINAL ANSWER**

$$A = 12x^2 + \frac{1690}{x}.$$



**CLASSIFIED PROBLEM**

Paper: 4WM2HR

Session: May2025

Q#: 21

Marks: 5

TOPIC: Volume &amp; Surface Area

Source: May 2025 4WM2HR

21 The diagram shows a solid cuboid.

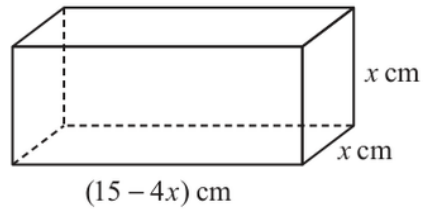


Diagram **NOT**  
accurately drawn

The volume of the cuboid is  $V \text{ cm}^3$

Find the maximum value of  $V$

---

(Total for Question 21 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM2HR | Question 21

Volume &amp; Surface Area

**Required:**

Cuboid has a square cross-section of side  $x$  cm and length  $(15-4x)$  cm. Find the maximum value of the volume  $V$ .

**Step:**

Write  $V$  as a function of  $x$ :

$$V = x \cdot x \cdot (15-4x) = x^2(15-4x) = 15x^2 - 4x^3$$

**Step:**

Differentiate:

$$\frac{dV}{dx} = 30x - 12x^2$$

**Step:**

Set  $\frac{dV}{dx} = 0$ :

$$30x - 12x^2 = 0 \implies 6x(5 - 2x) = 0 \implies x = 0 \text{ or } x = \frac{5}{2} = 2.5$$

( $x = 0$  gives zero volume, so the meaningful solution is  $x = 2.5$ .)

**Step:**

Second derivative test to confirm maximum:

$$\frac{d^2V}{dx^2} = 30 - 24x \Bigg|_{x=2.5} = 30 - 60 = -30 < 0 \quad \checkmark \text{ (maximum)}$$

**Step:**

Calculate  $V_{\max}$ :

$$V_{\max} = (2.5)^2 \times (15-4 \times 2.5) = 6.25 \times 5 = 31.25 \text{ cm}^3$$

**FINAL ANSWER**

$$\text{Maximum volume } V = 31.25 \text{ cm}^3$$

**Note:**

Calculus optimisation: always verify the nature of the stationary point using  $\frac{d^2V}{dx^2}$ . Also check  $x > 0$  and  $15 - 4x > 0$  (i.e.  $x < 3.75$ ) to ensure a valid cuboid.

**CLASSIFIED PROBLEM**

Paper: 4WM2HR

Session: May2025

Q#: 25

Marks: 5

TOPIC: Volume &amp; Surface Area

Source: May 2025 4WM2HR

25 The diagram shows a solid hemisphere,  $H$

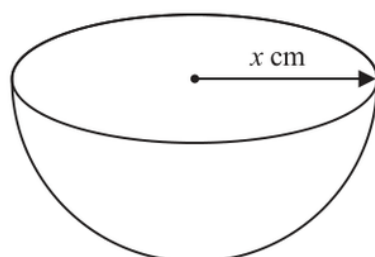


Diagram **NOT**  
accurately drawn

The radius of  $H$  is  $x$  cm

The volume of  $H$  is  $6174\pi \text{ cm}^3$

A bowl is made by removing a solid hemisphere from  $H$  such that the uniform thickness of the bowl is 2 cm

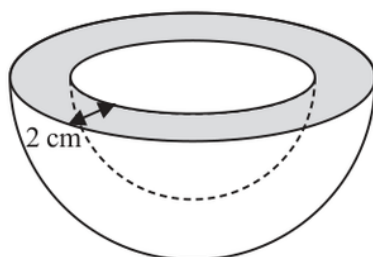


Diagram **NOT**  
accurately drawn

Work out the **total** surface area of the bowl.  
Give your answer in terms of  $\pi$

---



---

.....  $\text{cm}^2$

(Total for Question 25 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM2HR | Question 25

Volume &amp; Surface Area

**Required:**

Hemisphere H has radius  $x$  cm and volume  $6174\pi$  cm<sup>3</sup>. A bowl is made by removing an inner hemisphere, leaving 2 cm thickness. Find total surface area in terms of  $\pi$ .

**Step:**

Find outer radius  $x$ . Volume of hemisphere =  $\frac{2}{3}\pi r^3$ :

$$\frac{2}{3}\pi x^3 = 6174\pi \implies x^3 = \frac{6174 \times 3}{2} = 9261 \implies x = \sqrt[3]{9261} = 21 \text{ cm}$$

**Step:**

Find inner radius:

$$r_{\text{inner}} = 21 - 2 = 19 \text{ cm}$$

**Step:**

Three surfaces to add:

**1. Outer curved surface** (half-sphere, radius 21):

$$A_1 = 2\pi(21)^2 = 2\pi \times 441 = 882\pi$$

**2. Inner curved surface** (half-sphere, radius 19):

$$A_2 = 2\pi(19)^2 = 2\pi \times 361 = 722\pi$$

**3. Flat annular ring** at the top rim:

$$A_3 = \pi(21)^2 - \pi(19)^2 = \pi(441 - 361) = 80\pi$$

**Step:**

Total surface area:

$$A_{\text{total}} = 882\pi + 722\pi + 80\pi = 1684\pi$$

**FINAL ANSWER**

$$\text{Total surface area} = 1684\pi \text{ cm}^2$$

**Note:**

Three separate surfaces! Missing the flat annular ring is the most common error on this type of question.

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: Specimen

Q#: 22

Marks: 5

TOPIC: Volume &amp; Surface Area

Source: Specimen 4WM2H

22 A solid is made from a cone and a hemisphere.

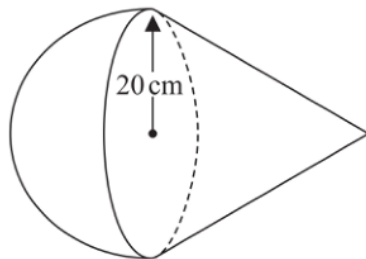


Diagram **NOT**  
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.  
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm

The curved surface area of the cone is  $580\pi \text{ cm}^2$

The volume of the solid is  $k\pi \text{ cm}^3$

Work out the exact value of  $k$

$k = \dots\dots\dots$

(Total for Question 22 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Specimen 4WM2H | Question 22

Volume &amp; Surface Area

**Topic check.** Volume & Surface Area. The tag is correct.**Solution**

The curved surface area of a cone is

$$\pi r l$$

Here  $r = 20$  and  $\pi r l = 580\pi$ , so

$$20l = 580$$

$$l = 29$$

Use Pythagoras to find the height of the cone:

$$h^2 = 29^2 - 20^2$$

$$h^2 = 841 - 400 = 441$$

$$h = 21$$

Volume of the cone:

$$\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(20)^2(21) = 2800\pi$$

Volume of the hemisphere:

$$\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(20)^3 = \frac{16000}{3}\pi$$

Total volume:

$$2800\pi + \frac{16000}{3}\pi = \frac{24400}{3}\pi$$

Since the volume is  $k\pi$ ,

$$k = \frac{24400}{3}$$

**FINAL ANSWER**

$$k = \frac{24400}{3}.$$

# Area & Volume of Similar Shapes

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**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: May2025

Q#: 25

Marks: 4

TOPIC: Area &amp; Volume of Similar Shapes

Source: May 2025 4WM2H

25 **A** and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm<sup>3</sup>

**(Total for Question 25 is 4 marks)**



## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2025 4WM2H | Question 25

Area &amp; Volume of Similar Shapes

**Solution**Linear scale factor from solid  $A$  to solid  $B$ :

$$\frac{18.6}{31} = 0.6$$

So the volume scale factor is

$$0.6^3 = 0.216$$

Let the volume of  $A$  be  $V$ . Then

$$V - 0.216V = 735$$

$$0.784V = 735$$

$$V = 937.5$$

**FINAL ANSWER**937.5 cm<sup>3</sup>.

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: Nov2025

Q#: 23

Marks: 3

TOPIC: Area &amp; Volume of Similar Shapes

Source: Nov 2025 4WM2H

**23** Shape **P** is similar to shape **Q**The table shows some information about shape **P** and shape **Q**

|                | Surface area (cm <sup>2</sup> ) | Volume (cm <sup>3</sup> ) |
|----------------|---------------------------------|---------------------------|
| Shape <b>P</b> | 200                             | 672                       |
| Shape <b>Q</b> | 450                             |                           |

Work out the volume of shape **Q**..... cm<sup>3</sup>

(Total for Question 23 is 3 marks)

**WORKED SOLUTION***Dr. Eslam Ahmed***Nov 2025 4WM2H | Question 23**

Area &amp; Volume of Similar Shapes

**1 Find the Area Scale Factor (ASF).**

$$\text{ASF} = \frac{\text{Area of Q}}{\text{Area of P}} = \frac{450}{200} = \frac{9}{4} = 2.25$$

**2 Find the Length Scale Factor (LSF).**

$$\text{LSF} = \sqrt{\text{ASF}} = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1.5$$

**3 Find the Volume Scale Factor (VSF).**

$$\text{VSF} = (\text{LSF})^3 = \left(\frac{3}{2}\right)^3 = \frac{27}{8} = 3.375$$

**4 Calculate the volume of Shape Q.**

$$\begin{aligned} \text{Volume of Q} &= \text{Volume of P} \times \text{VSF} = 672 \times \frac{27}{8} \\ &= 84 \times 27 = 2268 \end{aligned}$$

**FINAL ANSWER**2268 cm<sup>3</sup>

# Volume & Surface Area

---

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jan 2020

Q#: 26

Marks: 6

TOPIC: Volume &amp; Surface Area

Source: Jan 2020 P2H

26 Here is a sector,  $AOB$ , of a circle with centre  $O$  and angle  $AOB = x^\circ$

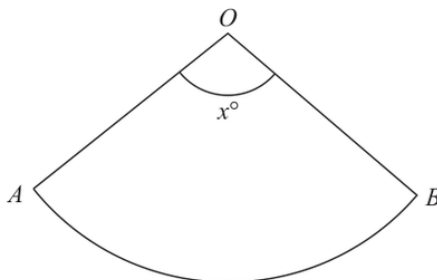


Diagram **NOT**  
accurately drawn

The sector can form the curved surface of a cone by joining  $OA$  to  $OB$ .

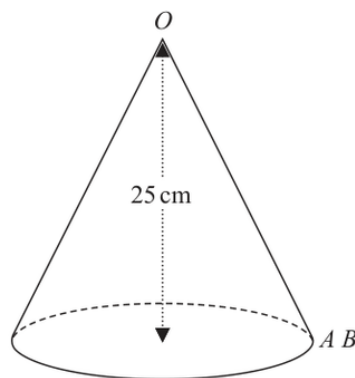


Diagram **NOT**  
accurately drawn

The height of the cone is 25 cm.

The volume of the cone is  $1600 \text{ cm}^3$

Work out the value of  $x$ .

Give your answer correct to the nearest whole number.

---



---

,

$x =$  .....

(Total for Question 26 is 6 marks)

---

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2020 P2H | Question 26

Volume &amp; Surface Area

**Topic check.** Moved to Volume & Surface Area.**Method**Let the radius of the base of the cone be  $r$ , and let the slant height be  $l$ .

The volume of the cone is

$$\frac{1}{3}\pi r^2 h = 1600$$

The height is 25 cm, so

$$\frac{1}{3}\pi r^2 (25) = 1600$$

$$r^2 = \frac{4800}{25\pi}$$

$$r = 7.8176\dots$$

The slant height is found using Pythagoras:

$$l^2 = r^2 + 25^2$$

$$l = 26.1938\dots$$

When the sector is joined to form the cone, the arc length of the sector becomes the circumference of the cone base.

$$\frac{x}{360} \times 2\pi l = 2\pi r$$

Cancel  $2\pi$ :

$$\frac{x}{360} l = r$$

$$x = \frac{360r}{l}$$

$$x = \frac{360(7.8176\dots)}{26.1938\dots} = 107.4433\dots$$

**FINAL ANSWER**

107°

**CLASSIFIED PROBLEM**

Paper: P1H

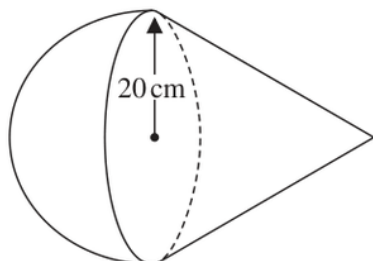
Session: Jun 2022

Q#: 22

Marks: 5

TOPIC: Volume &amp; Surface Area

Source: Jun 2022 P1H

**22** A solid is made from a cone and a hemisphere.Diagram **NOT**  
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.  
 The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is  $580\pi\text{ cm}^2$

The volume of the solid is  $k\pi\text{ cm}^3$

Work out the exact value of  $k$

 $k = \dots\dots\dots$ 

(Total for Question 22 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2022 P1H | Question 22

Volume &amp; Surface Area

**Topic check.** Correct.**Method**

The cone has radius 20 cm.

The curved surface area of the cone is

$$\pi r l$$

So

$$\pi(20)l = 580\pi$$

$$l = 29$$

Use Pythagoras to find the height of the cone:

$$h^2 + 20^2 = 29^2$$

$$h^2 = 841 - 400 = 441$$

$$h = 21$$

Volume of the cone:

$$\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(20)^2(21) = 2800\pi$$

Volume of the hemisphere:

$$\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(20)^3 = \frac{16000}{3}\pi$$

Total volume:

$$2800\pi + \frac{16000}{3}\pi = \frac{24400}{3}\pi$$

The volume is  $k\pi$ , so

$$k = \frac{24400}{3}$$

**FINAL ANSWER**

|                   |
|-------------------|
| $\frac{24400}{3}$ |
|-------------------|



**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2023

Q#: 24

Marks: 5

TOPIC: **Volume & Surface Area**

Source: Jun 2023 P2HR

- 24** The surface area of sphere **A** is nine times the surface area of sphere **B**  
The difference between the volume of sphere **A** and the volume of sphere **B** is  $117\pi \text{ cm}^3$

Find the radius of the smaller sphere.

Show your working clearly.

..... cm

---

**(Total for Question 24 is 5 marks)**

---

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2023 P2HR | Question 24

Volume &amp; Surface Area

**Topic check.** Correct.**Method**Let the radius of the smaller sphere be  $r$ .Let the radius of the larger sphere be  $R$ .

Surface area of a sphere is

$$4\pi r^2$$

The surface area of sphere  $A$  is nine times the surface area of sphere  $B$ , so

$$4\pi R^2 = 9(4\pi r^2)$$

$$R^2 = 9r^2$$

$$R = 3r$$

The difference between the volumes is  $117\pi$ :

$$\frac{4}{3}\pi(3r)^3 - \frac{4}{3}\pi r^3 = 117\pi$$

$$\frac{4}{3}\pi(27r^3 - r^3) = 117\pi$$

$$\frac{104}{3}\pi r^3 = 117\pi$$

Cancel  $\pi$ :

$$\frac{104}{3}r^3 = 117$$

$$r^3 = \frac{117 \times 3}{104}$$

$$r^3 = \frac{27}{8}$$

$$r = \frac{3}{2}$$

**FINAL ANSWER**

1.5 cm

**CLASSIFIED PROBLEM**

Paper: P2H

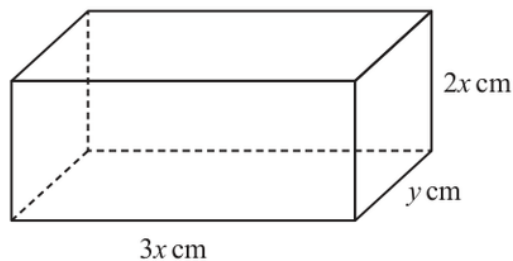
Session: Jun 2025

Q#: 21

Marks: 3

TOPIC: Volume &amp; Surface Area

Source: Jun 2025 P2H

**21** The diagram shows a cuboid.Diagram **NOT**  
accurately drawnThe cuboid measures  $3x$  cm by  $2x$  cm by  $y$  cmThe volume of the cuboid is  $1014 \text{ cm}^3$ The total surface area of the cuboid is  $A \text{ cm}^2$ Show that  $A = 12x^2 + \frac{1690}{x}$ 

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2025 P2H | Question 21

Volume &amp; Surface Area

**Topic check.** Correct.**Method**

The cuboid has dimensions

$$3x, \quad 2x, \quad y$$

Its volume is  $1014 \text{ cm}^3$ , so

$$(3x)(2x)(y) = 1014$$

$$6x^2y = 1014$$

$$y = \frac{1014}{6x^2}$$

$$y = \frac{169}{x^2}$$

The total surface area is

$$A = 2(3x \cdot 2x + 3x \cdot y + 2x \cdot y)$$

$$A = 2(6x^2 + 5xy)$$

$$A = 12x^2 + 10xy$$

Substitute  $y = \frac{169}{x^2}$ :

$$A = 12x^2 + 10x \left( \frac{169}{x^2} \right)$$

$$A = 12x^2 + \frac{1690}{x}$$

**FINAL ANSWER**

$$A = 12x^2 + \frac{1690}{x}$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2025

Q#: 26

Marks: 5

TOPIC: Volume &amp; Surface Area

Source: Jun 2025 P2HR

26 The diagram shows a solid hemisphere,  $H$

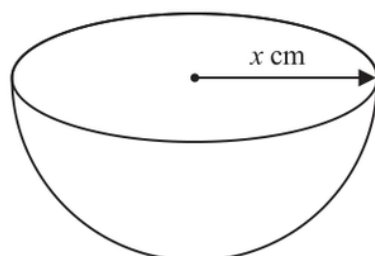


Diagram **NOT**  
accurately drawn

The radius of  $H$  is  $x$  cm

The volume of  $H$  is  $6174\pi \text{ cm}^3$

A bowl is made by removing a solid hemisphere from  $H$  such that the uniform thickness of the bowl is 2 cm

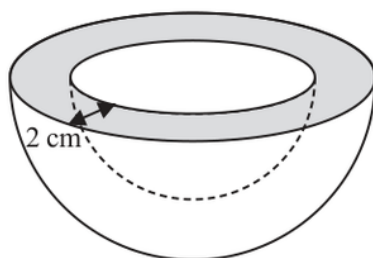


Diagram **NOT**  
accurately drawn

Work out the **total** surface area of the bowl.

Give your answer in terms of  $\pi$

---



---

.....  $\text{cm}^2$

(Total for Question 26 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2025 P2HR | Question 26

Volume &amp; Surface Area

**Topic check.** Correct.**Method**

The volume of a hemisphere is

$$\frac{2}{3}\pi r^3$$

For the original hemisphere  $H$ ,

$$\frac{2}{3}\pi x^3 = 6174\pi$$

Cancel  $\pi$ :

$$\frac{2}{3}x^3 = 6174$$

$$x^3 = 9261$$

$$x = 21$$

The outer radius of the bowl is 21 cm.

The thickness is 2 cm, so the inner radius is

$$21 - 2 = 19$$

Total surface area of the bowl is:

- outer curved hemisphere
- inner curved hemisphere
- flat annular rim

Outer curved surface area:

$$2\pi(21)^2 = 882\pi$$

Inner curved surface area:

$$2\pi(19)^2 = 722\pi$$

Flat rim area:

$$\pi(21^2 - 19^2) = 80\pi$$

Total:

$$882\pi + 722\pi + 80\pi = 1684\pi$$

**FINAL ANSWER**

$$1684\pi \text{ cm}^2$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2024

Q#: 23

Marks: 5

TOPIC: Volume &amp; Surface Area

Source: May 2024 P1H

- 23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

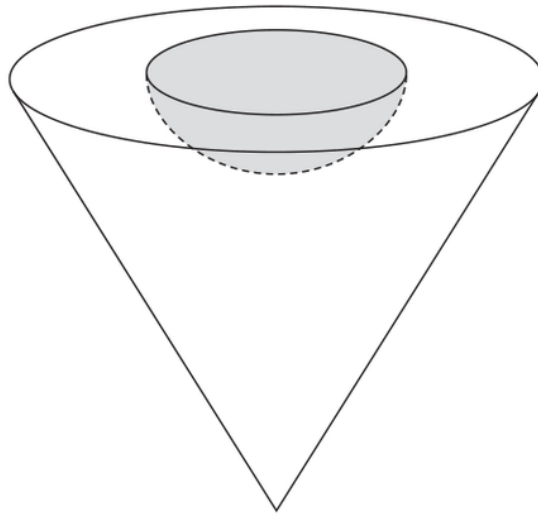


Diagram **NOT**  
accurately drawn

The radius of the hemisphere is  $2x$  cm

The radius of the base of the cone is  $5x$  cm

The vertical height of the cone is  $6x$  cm

The volume of the solid shape is  $6948\pi$  cm<sup>3</sup>

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

---



---

..... cm<sup>2</sup>

(Total for Question 23 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2024 P1H | Question 23

Volume &amp; Surface Area

**Topic check.** Correct.**Method**The cone has radius  $5x$  and height  $6x$ .

Volume of the cone:

$$\begin{aligned}\frac{1}{3}\pi(5x)^2(6x) \\ = 50\pi x^3\end{aligned}$$

The removed hemisphere has radius  $2x$ .

Volume of the hemisphere:

$$\begin{aligned}\frac{2}{3}\pi(2x)^3 \\ = \frac{16}{3}\pi x^3\end{aligned}$$

The remaining solid has volume  $6948\pi$ , so

$$\begin{aligned}50\pi x^3 - \frac{16}{3}\pi x^3 &= 6948\pi \\ \frac{134}{3}\pi x^3 &= 6948\pi\end{aligned}$$

Cancel  $\pi$ :

$$\begin{aligned}\frac{134}{3}x^3 &= 6948 \\ x^3 &= \frac{10422}{67}\end{aligned}$$

$$x = 5.3781\dots$$

The solid hemisphere removed has radius  $2x$ .

Total surface area of a solid hemisphere is

$$3\pi r^2$$

So

$$3\pi(2x)^2 = 12\pi x^2$$

$$12\pi(5.3781\dots)^2 = 1090.3903\dots$$

**FINAL ANSWER**

$$1090 \text{ cm}^2$$



**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2021

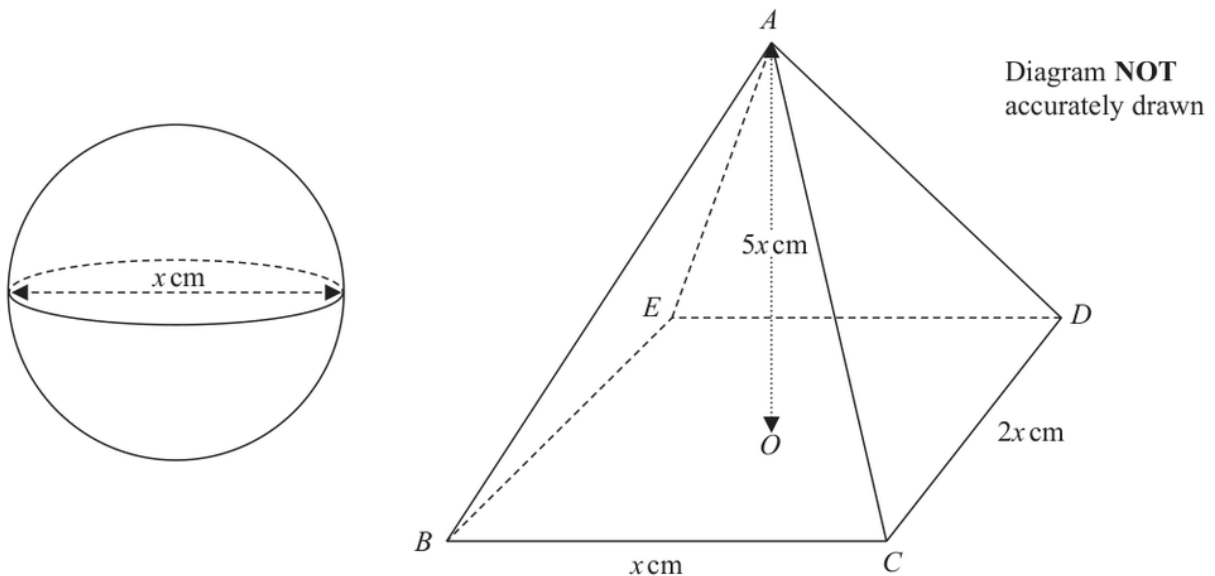
Q#: 22

Marks: 6

TOPIC: Volume &amp; Surface Area

Source: Nov 2021 P2H

- 22 The diagram shows a sphere of diameter  $x$  cm and a pyramid  $ABCDE$  with a horizontal rectangular base  $BCDE$ .



The vertex  $A$  of the pyramid is vertically above the centre  $O$  of the base so that  $AB = AC = AD = AE$ .

$BC = x$  cm,  $CD = 2x$  cm and  $AO = 5x$  cm.

The volume of the sphere is  $288\pi$  cm<sup>3</sup>

Calculate the total surface area of the pyramid.

Give your answer correct to the nearest cm<sup>2</sup>

..... cm<sup>2</sup>

(Total for Question 22 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2021 P2H | Question 22

Volume &amp; Surface Area

**Topic check.** Corrected from Right-Angled Triangles to Volume & Surface Area.**Method**The sphere has diameter  $x$ , so its radius is

$$\frac{x}{2}$$

The volume of the sphere is  $288\pi \text{ cm}^3$ :

$$\frac{4}{3}\pi \left(\frac{x}{2}\right)^3 = 288\pi$$

$$\frac{\pi x^3}{6} = 288\pi$$

$$x^3 = 1728$$

$$x = 12$$

The rectangular base of the pyramid has dimensions  $x$  and  $2x$ , so its area is

$$x(2x) = 2x^2$$

For the two faces with base  $x$ , the slant height is

$$\sqrt{(5x)^2 + x^2} = x\sqrt{26}$$

Their total area is

$$2 \left( \frac{1}{2}x \cdot x\sqrt{26} \right) = x^2\sqrt{26}$$

For the two faces with base  $2x$ , the slant height is

$$\sqrt{(5x)^2 + \left(\frac{x}{2}\right)^2} = \frac{x\sqrt{101}}{2}$$

Their total area is

$$2 \left( \frac{1}{2}(2x) \cdot \frac{x\sqrt{101}}{2} \right) = x^2\sqrt{101}$$

Total surface area:

$$2x^2 + x^2\sqrt{26} + x^2\sqrt{101}$$

Substitute  $x = 12$ :

$$144(2 + \sqrt{26} + \sqrt{101}) = 2469.440 \dots$$

Correct to the nearest  $\text{cm}^2$ :

2469

**FINAL ANSWER**

2469 cm<sup>2</sup>

**CLASSIFIED PROBLEM**

Paper: P1H

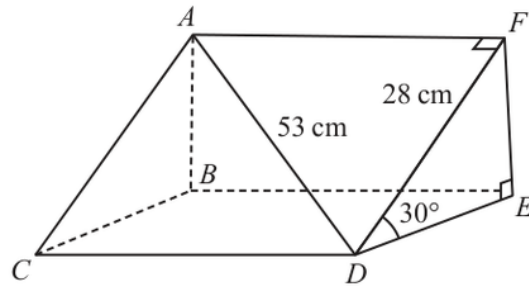
Session: Nov 2023

Q#: 21

Marks: 5

TOPIC: Volume &amp; Surface Area

Source: Nov 2023 P1H

**21** Here is a triangular prism  $ABCDEF$ Diagram **NOT**  
accurately drawn

$$AD = 53 \text{ cm}$$

$$DF = 28 \text{ cm}$$

$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

.....  $\text{cm}^3$ 

(Total for Question 21 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2023 P1H | Question 21

Volume &amp; Surface Area

Topic check. Correct.

**Method**The triangular cross-section is triangle  $DEF$ .It is right-angled at  $E$ , with

$$DF = 28$$

and

$$\angle FDE = 30^\circ$$

So

$$FE = 28 \sin 30^\circ = 14$$

and

$$DE = 28 \cos 30^\circ = 14\sqrt{3}$$

Area of triangle  $DEF$ :

$$\frac{1}{2}(14)(14\sqrt{3}) = 98\sqrt{3}$$

The length of the prism is

$$AD = 53$$

Volume of the prism:

$$98\sqrt{3} \times 53$$

$$= 5194\sqrt{3}$$

$$= 8996.2719 \dots$$

**FINAL ANSWER**

$$8996 \text{ cm}^3$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2024

Q#: 22

Marks: 3

TOPIC: Volume &amp; Surface Area

Source: Nov 2024 P1H

- 22 The diagram shows a bowl in the shape of a hemisphere.  
The bowl is made from metal.

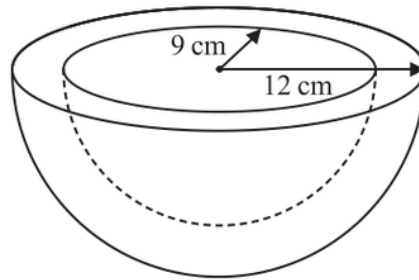


Diagram **NOT**  
accurately drawn

The outer radius of the bowl is 12 cm

The inner radius of the bowl is 9 cm

The thickness of the bowl is uniform.

Work out the volume of the metal.

Give your answer correct to the nearest whole number.

..... cm<sup>3</sup>

(Total for Question 22 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2024 P1H | Question 22

Volume &amp; Surface Area

Topic check. Correct.

**Method**

The bowl is a hemispherical shell.

The volume of the metal is:

outer hemisphere volume – inner hemisphere volume

Volume of a hemisphere:

$$\frac{2}{3}\pi r^3$$

So

$$\text{metal volume} = \frac{2}{3}\pi(12)^3 - \frac{2}{3}\pi(9)^3$$

$$= \frac{2}{3}\pi(12^3 - 9^3)$$

$$= \frac{2}{3}\pi(1728 - 729)$$

$$= 666\pi$$

$$= 2092.3007\dots$$

**FINAL ANSWER**

$$2092 \text{ cm}^3$$

# Area & Volume of Similar Shapes

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**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jan 2020

Q#: 20

Marks: 5

TOPIC: Area &amp; Volume of Similar Shapes

Source: Jan 2020 P1HR

20 The diagram shows a frustum of a cone and a sphere.

The frustum is made by removing a small cone from a large cone.  
The cones are similar.

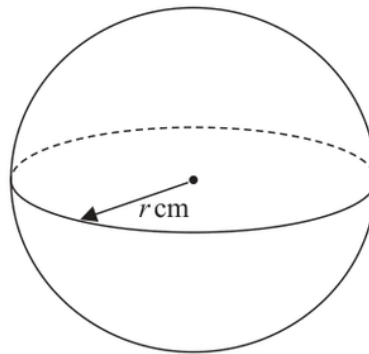
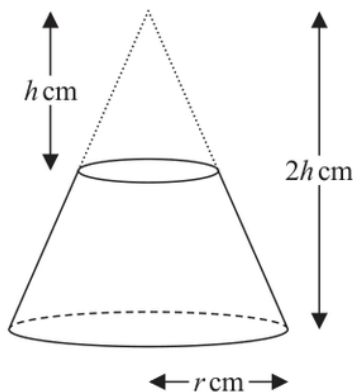


Diagram **NOT**  
accurately drawn

The height of the small cone is  $h$  cm.  
The height of the large cone is  $2h$  cm.  
The radius of the base of the large cone is  $r$  cm.

The radius of the sphere is  $r$  cm.

Given that the volume of the frustum is equal to the volume of the sphere,

find an expression for  $r$  in terms of  $h$ .

Give your expression in its simplest form.

$r = \dots\dots\dots$

(Total for Question 20 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jan 2020 P1HR | Question 20

Area &amp; Volume of Similar Shapes

**Topic check.** Correct.**Method**

The two cones are similar.

The large cone has height  $2h$  and base radius  $r$ .The small cone has height  $h$ , so its linear scale factor compared with the large cone is

$$\frac{h}{2h} = \frac{1}{2}$$

So the small cone has radius

$$\frac{r}{2}$$

Volume of the large cone:

$$\frac{1}{3}\pi r^2(2h) = \frac{2}{3}\pi r^2h$$

Volume of the small cone:

$$\frac{1}{3}\pi \left(\frac{r}{2}\right)^2 h = \frac{1}{12}\pi r^2h$$

So the volume of the frustum is

$$\frac{2}{3}\pi r^2h - \frac{1}{12}\pi r^2h = \frac{7}{12}\pi r^2h$$

The sphere has radius  $r$ , so its volume is

$$\frac{4}{3}\pi r^3$$

The two volumes are equal:

$$\frac{7}{12}\pi r^2h = \frac{4}{3}\pi r^3$$

Divide by  $\pi r^2$ :

$$\frac{7}{12}h = \frac{4}{3}r$$

$$7h = 16r$$

$$r = \frac{7h}{16}$$

**FINAL ANSWER**

$$r = \frac{7h}{16}$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jan 2020

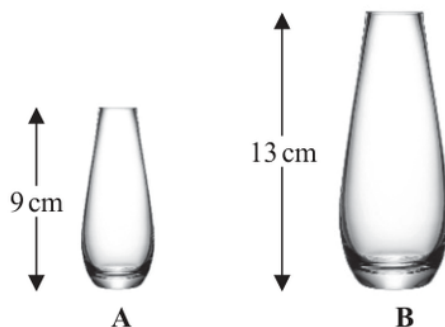
Q#: 20

Marks: 4

TOPIC: Area &amp; Volume of Similar Shapes

Source: Jan 2020 P2HR

20 The diagram shows two similar vases, **A** and **B**.



The height of vase **A** is 9 cm and the height of vase **B** is 13 cm.

Given that

$$\text{surface area of vase A} + \text{surface area of vase B} = 1800 \text{ cm}^2$$

calculate the surface area of vase **A**.

.....  $\text{cm}^2$

(Total for Question 20 is 4 marks)

**WORKED SOLUTION***Dr. Eslam Ahmed***Jan 2020 P2HR | Question 20**

Area &amp; Volume of Similar Shapes

**Topic check.** Correct.**Method**

The two vases are similar.

Their heights are in the ratio

$$9 : 13$$

So their surface areas are in the ratio

$$9^2 : 13^2 = 81 : 169$$

The total surface area is  $1800 \text{ cm}^2$ , so there are

$$81 + 169 = 250$$

parts altogether.

Surface area of vase A:

$$1800 \times \frac{81}{250} = 583.2$$

**FINAL ANSWER**

$$583.2 \text{ cm}^2$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Jan 2021

Q#: 20

Marks: 3

TOPIC: Area &amp; Volume of Similar Shapes

Source: Jan 2021 P1H

20 **A** and **B** are two similar solids.

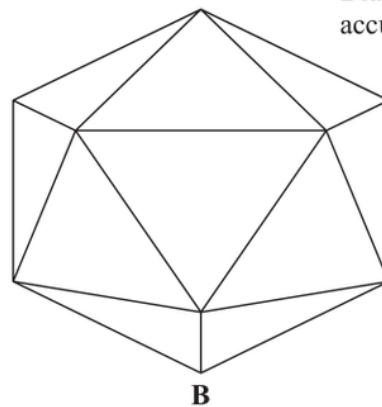
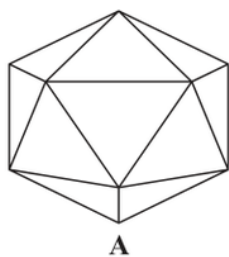


Diagram **NOT**  
accurately drawn

**A** has a volume of  $1836\text{ cm}^3$

**B** has a volume of  $4352\text{ cm}^3$

**B** has a total surface area of  $1120\text{ cm}^2$

Work out the total surface area of **A**.

.....  $\text{cm}^2$

(Total for Question 20 is 3 marks)

**WORKED SOLUTION***Dr. Eslam Ahmed***Jan 2021 P1H | Question 20**

Area &amp; Volume of Similar Shapes

**Topic check.** Correct.**Method**

The volume ratio of solid A to solid B is

$$1836 : 4352$$

Simplify:

$$\frac{1836}{4352} = \frac{27}{64}$$

So the linear ratio is

$$\sqrt[3]{27} : \sqrt[3]{64} = 3 : 4$$

Surface area ratio:

$$3^2 : 4^2 = 9 : 16$$

Solid B has surface area  $1120 \text{ cm}^2$ , so solid A has surface area

$$1120 \times \frac{9}{16} = 630$$

**FINAL ANSWER**

$$630 \text{ cm}^2$$

**CLASSIFIED PROBLEM**

Paper: P1HR

Session: Jun 2022

Q#: 20

Marks: 4

TOPIC: Area &amp; Volume of Similar Shapes

Source: Jun 2022 P1HR

**20** The diagram shows two similar metal statues.

**A****B**

Diagram **NOT**  
accurately drawn

The volume of statue **B** is 20% less than the volume of statue **A**

The surface area of statue **B** is  $k\%$  less than the surface area of statue **A**

Work out the value of  $k$

Give your answer correct to 3 significant figures.

$k = \dots\dots\dots$

(Total for Question 20 is 4 marks)

**WORKED SOLUTION***Dr. Eslam Ahmed***Jun 2022 P1HR | Question 20**

Area &amp; Volume of Similar Shapes

**Topic check.** Corrected from Percentages to Area & Volume of Similar Shapes, because the key idea is the relationship between volume scale factor and surface area scale factor for similar solids.

**Method**

The volume of statue  $B$  is 20% less than statue  $A$ , so the volume scale factor is 0.8.

For similar solids, the linear scale factor is the cube root of the volume scale factor:

linear scale factor =  $\sqrt[3]{0.8}$

Surface area scale factor is the square of the linear scale factor:

surface area scale factor =  $(\sqrt[3]{0.8})^2 = 0.8^{2/3} = 0.861773\dots$

So statue  $B$  has 86.1773...% of the surface area of statue  $A$ .

$k = 100 - 86.1773\dots = 13.8226\dots$

**FINAL ANSWER**

$k = 13.8$  to 3 significant figures.



**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2022

Q#: 24

Marks: 6

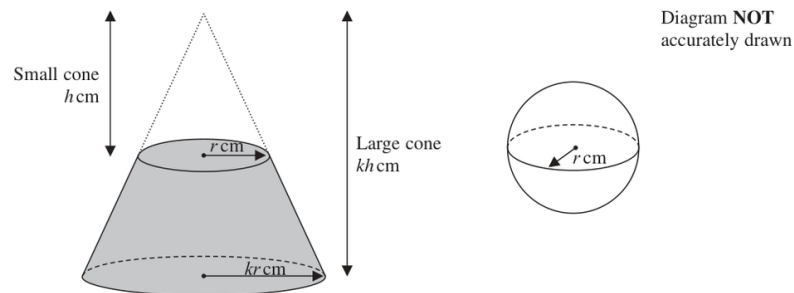
TOPIC: Area & Volume of Similar Shapes

Source: Jun 2022 P2HR

24 The diagram shows a frustum of a cone, and a sphere.

The frustum, shown shaded in the diagram, is made by removing the small cone from the large cone.

The small cone and the large cone are similar.



The height of the small cone is  $h$  cm and the radius of the base of the small cone is  $r$  cm.  
The height of the large cone is  $kh$  cm and the radius of the base of the large cone is  $kr$  cm.  
The radius of the sphere is  $r$  cm.

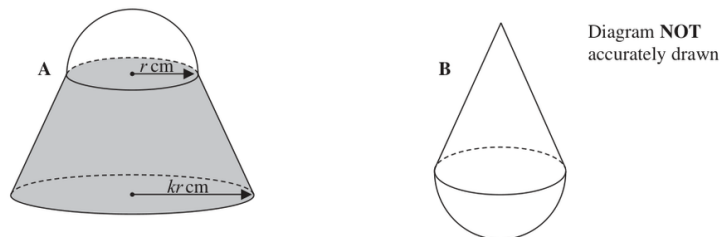
The sphere is divided into two hemispheres, each of radius  $r$  cm.

Solid A is formed by joining one of the hemispheres to the frustum.

The plane face of the hemisphere coincides with the upper plane face of the frustum, as shown in the diagram below.

Solid B is formed by joining the other hemisphere to the small cone that was removed from the large cone.

The plane face of the hemisphere coincides with the plane face of the base of the small cone, as shown in the diagram below.



The volume of solid A is 6 times the volume of solid B.

Given that  $k > \sqrt[3]{7}$

find an expression for  $h$  in terms of  $k$  and  $r$

$h = \dots\dots\dots$

(Total for Question 24 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2022 P2HR | Question 24

Area &amp; Volume of Similar Shapes

**Topic check.** Correct.**Method**

Small cone volume:

$$V_s = \frac{1}{3}\pi r^2 h$$

The larger similar cone has radius  $kr$  and height  $kh$ , so its volume is

$$V_L = \frac{1}{3}\pi(kr)^2(kh) = \frac{1}{3}\pi k^3 r^2 h$$

So the frustum volume is

$$V_L - V_s = \frac{1}{3}\pi r^2 h(k^3 - 1)$$

The volume of one hemisphere of radius  $r$  is

$$\frac{2}{3}\pi r^3$$

Solid A is the frustum plus one hemisphere:

$$V_A = \frac{1}{3}\pi r^2 h(k^3 - 1) + \frac{2}{3}\pi r^3$$

Solid B is the small cone plus one hemisphere:

$$V_B = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3$$

Given  $V_A = 6V_B$ :

$$\frac{1}{3}\pi r^2 h(k^3 - 1) + \frac{2}{3}\pi r^3 = 6\left(\frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3\right)$$

Divide by  $\frac{1}{3}\pi r^2$ :

$$h(k^3 - 1) + 2r = 6h + 12r$$

$$h(k^3 - 7) = 10r$$

$$h = \frac{10r}{k^3 - 7}$$

**FINAL ANSWER**

$$h = \frac{10r}{k^3 - 7}$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2024

Q#: 24

Marks: 5

TOPIC: Area &amp; Volume of Similar Shapes

Source: Jun 2024 P2HR

- 24 The diagram shows a solid, **S**, made from a cone and a hemisphere.

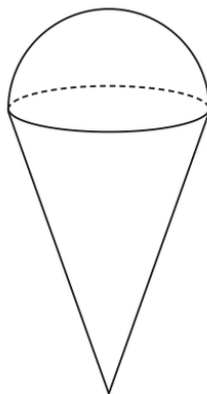


Diagram **NOT**  
accurately drawn

The centre of the circular face of the cone coincides with the centre of the flat surface of the hemisphere.

The radius of the circular face of the cone,  $x$  cm, is equal to the radius of the hemisphere.

The total height of **S** is  $4 \times$  the radius of the hemisphere.

A separate sphere has radius  $kx$  cm.

The volume of this sphere is  $12.5 \times$  the volume of **S**

- (a) Work out the value of  $k$

$$k = \dots\dots\dots (4)$$

A solid, **T**, is similar to solid **S**

The volume of **T** is  $512 \times$  the volume of **S**

The total surface area of **T** is  $d \times$  the total surface area of **S**

- (b) Find the value of  $d$

$$d = \dots\dots\dots (1)$$

(Total for Question 24 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2024 P2HR | Question 24

Area &amp; Volume of Similar Shapes

**Topic check.** Correct.**Method**The hemisphere has radius  $x$ , so its height is  $x$ .The total height of solid S is  $4x$ , so the cone height is

$$4x - x = 3x$$

Volume of the cone:

$$\frac{1}{3}\pi x^2(3x) = \pi x^3$$

Volume of the hemisphere:

$$\frac{2}{3}\pi x^3$$

So the volume of solid S is

$$\pi x^3 + \frac{2}{3}\pi x^3 = \frac{5}{3}\pi x^3$$

The sphere has radius  $kx$ , so its volume is

$$\frac{4}{3}\pi(kx)^3 = \frac{4}{3}\pi k^3 x^3$$

This is 12.5 times the volume of S:

$$\frac{4}{3}\pi k^3 x^3 = 12.5 \left( \frac{5}{3}\pi x^3 \right)$$

Cancel  $\frac{1}{3}\pi x^3$ :

$$4k^3 = 62.5$$

$$k^3 = 15.625 = \frac{125}{8}$$

$$k = 2.5$$

For solid T, the volume scale factor is 512.

$$512 = 8^3$$

So the linear scale factor is 8, and the surface area scale factor is

$$8^2 = 64$$

**FINAL ANSWER**

$$k = 2.5, d = 64$$

**CLASSIFIED PROBLEM**

Paper: P1HR

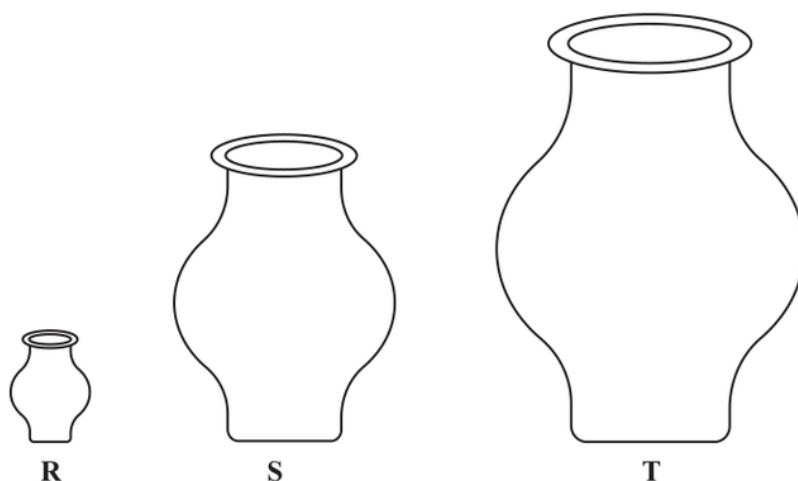
Session: Jun 2025

Q#: 26

Marks: 4

TOPIC: Area &amp; Volume of Similar Shapes

Source: Jun 2025 P1HR

**26** R, S and T are three similar vases.Diagram **NOT**  
accurately drawnThe volume of vase **S** is 72.8% more than the volume of vase **R**The height of vase **R** is  $h$  cmThe height of vase **T** is  $6h$  cmThe surface area of vase **S** is  $A \text{ cm}^2$ The surface area of vase **T** is  $kA \text{ cm}^2$ Work out the value of  $k$  $k = \dots\dots\dots$ **(Total for Question 26 is 4 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2025 P1HR | Question 26

Area &amp; Volume of Similar Shapes

**Topic check.** Corrected from Percentages to Area & Volume of Similar Shapes, because the solution uses volume and surface area scale factors for similar shapes.

**Method**

The volume of vase  $S$  is 72.8% more than vase  $R$ , so

$$\frac{V_S}{V_R} = 1.728$$

For similar solids, volume scale factor is the cube of the linear scale factor:

linear scale factor from  $R$  to  $S = \sqrt[3]{1.728} = 1.2$

The height of vase  $T$  is  $6h$ , so the linear scale factor from  $R$  to  $T$  is 6.

Therefore the linear scale factor from  $S$  to  $T$  is

$$\frac{6}{1.2} = 5$$

Surface area scale factor is the square of the linear scale factor:

$$k = 5^2 = 25$$

**FINAL ANSWER**

$$k = 25$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2025

Q#: 26

Marks: 4

TOPIC: Area &amp; Volume of Similar Shapes

Source: Jun 2025 P2H

**26** **A** and **B** are two similar solids.

The height of solid **A** is 31 cm

The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm<sup>3</sup>

(Total for Question 26 is 4 marks)

**WORKED SOLUTION***Dr. Eslam Ahmed***Jun 2025 P2H | Question 26**

Area &amp; Volume of Similar Shapes

**Topic check.** Correct.**Method**

The heights of solids A and B are 31 cm and 18.6 cm.

$$31 : 18.6 = 5 : 3$$

So the volumes are in the ratio

$$5^3 : 3^3 = 125 : 27$$

The difference in volume is

$$125 - 27 = 98$$

parts.

This difference is  $735 \text{ cm}^3$ , so one part is

$$\frac{735}{98} = 7.5$$

Volume of solid A:

$$125 \times 7.5 = 937.5$$

**FINAL ANSWER**

$$937.5 \text{ cm}^3$$



**CLASSIFIED PROBLEM**

Paper: P2H

Session: May 2021

Q#: 20

Marks: 3

TOPIC: Area &amp; Volume of Similar Shapes

Source: May 2021 P2H

**20** Mathematically similar wooden blocks are made in a workshop.

There are small blocks and there are large blocks.

The volume of each small block is  $300\text{ cm}^3$

Given that

the surface area of each small block : the surface area of each large block =  $25 : 36$

work out the volume of each large block.

.....  $\text{cm}^3$

(Total for Question 20 is 3 marks)

**WORKED SOLUTION***Dr. Eslam Ahmed***May 2021 P2H | Question 20**

Area &amp; Volume of Similar Shapes

**Topic check.** Correct.**Method**

The surface area ratio of the small block to the large block is

$$25 : 36$$

So the linear ratio is

$$\sqrt{25} : \sqrt{36} = 5 : 6$$

Therefore the volume ratio is

$$5^3 : 6^3 = 125 : 216$$

The small block has volume  $300 \text{ cm}^3$ .

Large block volume:

$$300 \times \frac{216}{125} = 518.4$$

**FINAL ANSWER**

$$518.4 \text{ cm}^3$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: May 2023

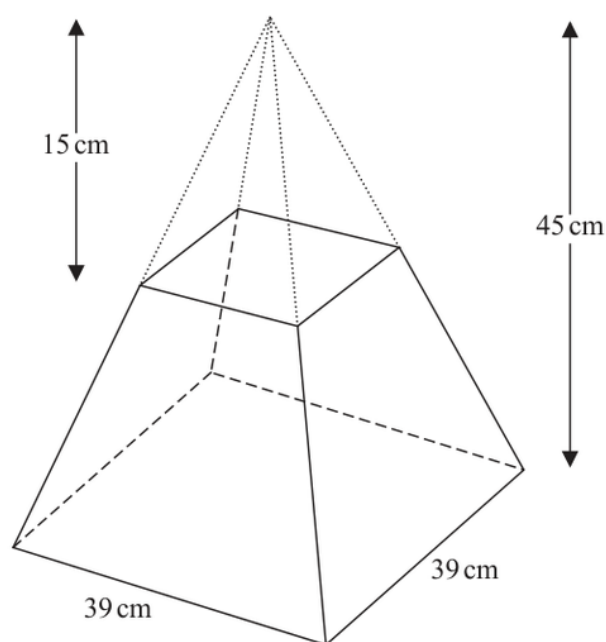
Q#: 23

Marks: 5

TOPIC: Area &amp; Volume of Similar Shapes

Source: May 2023 P1H

- 23 A frustum is made by removing a small square-based pyramid from a similar large squared-based pyramid as shown in the diagram.

Diagram **NOT**  
accurately drawn

The height of the small pyramid is 15 cm.

The height of the large pyramid is 45 cm.

The square base of the large pyramid has side length 39 cm.

Work out the **total** surface area of the frustum.

Give your answer correct to the nearest whole number.

---



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..... cm<sup>2</sup>

(Total for Question 23 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## May 2023 P1H | Question 23

Area &amp; Volume of Similar Shapes

**Topic check.** Correct.**Method**

The small and large square-based pyramids are similar.  
Their heights are

$$15 : 45 = 1 : 3$$

So the side length of the top square of the frustum is

$$\frac{39}{3} = 13 \text{ cm}$$

The vertical height of the frustum is

$$45 - 15 = 30 \text{ cm}$$

For one trapezium face, the difference between the half-side lengths is

$$\frac{39 - 13}{2} = 13 \text{ cm}$$

So the slant height of the trapezium face is

$$\sqrt{30^2 + 13^2} = \sqrt{1069}$$

Area of one trapezium face:

$$\frac{1}{2}(39 + 13)\sqrt{1069} = 26\sqrt{1069}$$

Area of four trapezium faces:

$$4(26\sqrt{1069}) = 104\sqrt{1069}$$

Add the top and bottom square faces:

$$13^2 + 39^2 = 169 + 1521 = 1690$$

Total surface area:

$$1690 + 104\sqrt{1069} = 5090.338 \dots$$

Correct to the nearest whole number:

$$5090$$

**FINAL ANSWER**

$$5090 \text{ cm}^2$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: MayNov 2020

Q#: 20

Marks: 3

TOPIC: **Area & Volume of Similar Shapes**

Source: May-Nov 2020 P1H

**20** **R** and **S** are two similar solid shapes.Shape **R** has surface area  $108\text{ cm}^2$  and volume  $135\text{ cm}^3$ Shape **S** has surface area  $300\text{ cm}^2$ Work out the volume of shape **S**......  $\text{cm}^3$ 

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**(Total for Question 20 is 3 marks)**

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**WORKED SOLUTION***Dr. Eslam Ahmed***May-Nov 2020 P1H | Question 20**

Area &amp; Volume of Similar Shapes

**Topic check.** Correct.**Method**Shape R has surface area  $108 \text{ cm}^2$  and volume  $135 \text{ cm}^3$ .Shape S has surface area  $300 \text{ cm}^2$ .

Surface area ratio S:R:

$$300 : 108 = 25 : 9$$

So the linear ratio S:R is

$$5 : 3$$

Therefore the volume ratio S:R is

$$5^3 : 3^3 = 125 : 27$$

Volume of S:

$$135 \times \frac{125}{27} = 625$$

**FINAL ANSWER**

$$625 \text{ cm}^3$$

**CLASSIFIED PROBLEM**

Paper: P1H

Session: Nov 2023

Q#: 23

Marks: 5

TOPIC: Area &amp; Volume of Similar Shapes

Source: Nov 2023 P1H

**23 Solid A is similar to solid B**

Here is some information about solid A and solid B

|                           | solid A | solid B   |
|---------------------------|---------|-----------|
| Height (cm)               | $3^x$   |           |
| Area (cm <sup>2</sup> )   | 7776    | 486       |
| Volume (cm <sup>3</sup> ) | $8^x$   | $2^{x+4}$ |

Work out the height of solid B  
Give your answer as a decimal.

---



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..... cm

(Total for Question 23 is 5 marks)

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**WORKED SOLUTION***Dr. Eslam Ahmed***Nov 2023 P1H | Question 23**

Area &amp; Volume of Similar Shapes

**Topic check.** Correct.**Method**

The area ratio of solid A to solid B is

$$7776 : 486 = 16 : 1$$

So the linear ratio A:B is

$$\sqrt{16} : 1 = 4 : 1$$

Therefore the volume ratio A:B is

$$4^3 : 1^3 = 64 : 1$$

The volumes are  $8^x$  and  $2^{x+4}$ , so

$$\frac{8^x}{2^{x+4}} = 64$$

Write  $8^x$  as a power of 2:

$$8^x = (2^3)^x = 2^{3x}$$

So

$$\frac{2^{3x}}{2^{x+4}} = 2^6$$

$$2^{3x-(x+4)} = 2^6$$

$$2x - 4 = 6$$

$$x = 5$$

The height of solid A is

$$3^x = 3^5 = 243$$

Since the linear ratio A:B is 4 : 1, the height of solid B is

$$\frac{243}{4} = 60.75$$

**FINAL ANSWER**

60.75 cm



**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2023

Q#: 23

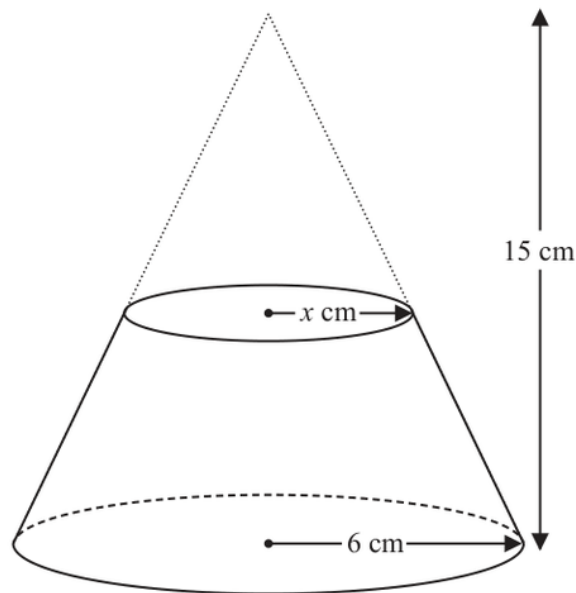
Marks: 5

TOPIC: Area &amp; Volume of Similar Shapes

Source: Nov 2023 P2H

**23** Here is a frustum of a cone.

The frustum is made by removing a small cone from a similar large cone.

Diagram **NOT**  
accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.

The radius of the base of the small cone is  $x$  cm.Given that the volume of the frustum is  $\frac{4212}{25}\pi \text{ cm}^3$ work out the value of  $x$ 

Show clear algebraic working.

 $x = \dots\dots\dots$ **(Total for Question 23 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2023 P2H | Question 23

Area &amp; Volume of Similar Shapes

**Topic check.** Correct.**Method**

The large cone has height 15 cm and radius 6 cm.

The small cone has radius  $x$ . Since the cones are similar,

$$\frac{\text{radius}}{\text{height}} = \frac{6}{15} = \frac{2}{5}$$

So for the small cone,

$$\frac{x}{h} = \frac{2}{5}$$

$$h = \frac{5}{2}x$$

Volume of the large cone:

$$\frac{1}{3}\pi(6)^2(15) = 180\pi$$

Volume of the small cone:

$$\frac{1}{3}\pi x^2 \left(\frac{5}{2}x\right) = \frac{5}{6}\pi x^3$$

The frustum volume is

$$\frac{4212}{25}\pi$$

So

$$180\pi - \frac{5}{6}\pi x^3 = \frac{4212}{25}\pi$$

Cancel  $\pi$ :

$$180 - \frac{5}{6}x^3 = \frac{4212}{25}$$

$$\frac{5}{6}x^3 = 180 - \frac{4212}{25}$$

$$\frac{5}{6}x^3 = \frac{288}{25}$$

$$x^3 = \frac{1728}{125}$$

$$x = \frac{12}{5} = 2.4$$

**FINAL ANSWER**

2.4 cm

**CLASSIFIED PROBLEM****Paper: P2H****Session: Nov 2025****Q#: 23****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

Source: Nov 2025 P2H

**23** Shape **P** is similar to shape **Q**The table shows some information about shape **P** and shape **Q**

|                | Surface area (cm <sup>2</sup> ) | Volume (cm <sup>3</sup> ) |
|----------------|---------------------------------|---------------------------|
| Shape <b>P</b> | 200                             | 672                       |
| Shape <b>Q</b> | 450                             |                           |

Work out the volume of shape **Q**..... cm<sup>3</sup>**(Total for Question 23 is 3 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 P2H | Question 23

Area &amp; Volume of Similar Shapes

Topic check. Correct.

**Method**

Surface area ratio Q:P:

$$450 : 200 = 9 : 4$$

So the linear ratio Q:P is

$$3 : 2$$

Therefore the volume ratio Q:P is

$$3^3 : 2^3 = 27 : 8$$

Shape P has volume  $672 \text{ cm}^3$ .

Volume of Q:

$$672 \times \frac{27}{8} = 2268$$

**FINAL ANSWER**

$$2268 \text{ cm}^3$$

# Vectors

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## CLASSIFIED PROBLEM

Paper: 4WM2H

Session: May2025

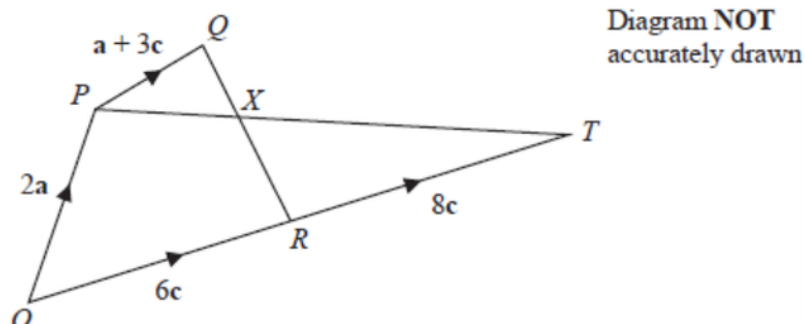
Q#: 26

Marks: 5

TOPIC: **Vectors**

Source: May 2025 4WM2H

26



The diagram shows a quadrilateral  $OPQR$  in which

$$\vec{OP} = 2\mathbf{a} \quad \vec{OR} = 6\mathbf{c} \quad \vec{PQ} = \mathbf{a} + 3\mathbf{c}$$

(a) Find  $\vec{QR}$  in terms of  $\mathbf{a}$  and  $\mathbf{c}$

$$\vec{QR} = \dots\dots\dots (1)$$

The point  $T$  is such that  $\vec{RT} = 8\mathbf{c}$   
 The point  $X$  lies on  $QR$  such that  $PXT$  is a straight line.

(b) Using a vector method, find the ratio  $QX : XR$   
 Give your answer in the form  $m : n$  where  $m$  and  $n$  are integers.  
 Show your working clearly.

P81641A

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$$QX : XR = \dots\dots\dots (4)$$

(Total for Question 26 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM2H | Question 26

Vectors

**Solution**

(a)

$$\overrightarrow{QR} = \overrightarrow{OR} - \overrightarrow{OQ}$$

$$\overrightarrow{OQ} = 2a + (a + 3c) = 3a + 3c$$

$$\overrightarrow{QR} = 6c - (3a + 3c) = 3c - 3a$$

(b) Let  $X$  divide  $QR$  so that

$$\overrightarrow{QX} = \lambda \overrightarrow{QR}$$

Then

$$\overrightarrow{OX} = 3a + 3c + \lambda(3c - 3a)$$

Also  $X$  lies on  $PT$ , where  $\overrightarrow{OT} = 14c$ , so

$$\overrightarrow{OX} = 2a + \mu(14c - 2a)$$

Equating coefficients gives  $\lambda = \frac{5}{9}$ . Therefore

$$QX : XR = \frac{5}{9} : \frac{4}{9} = 5 : 4$$

**FINAL ANSWER**

$$\overrightarrow{QR} = 3c - 3a; QX : XR = 5 : 4.$$



**CLASSIFIED PROBLEM**

Paper: 4WM2H

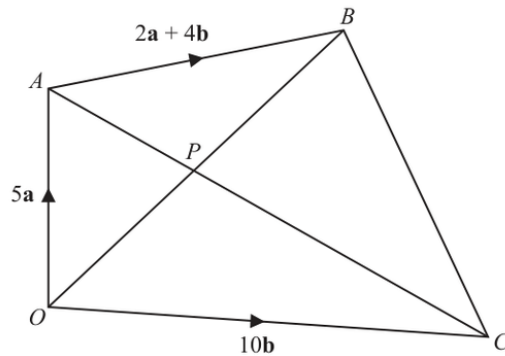
Session: Nov2025

Q#: 24

Marks: 6

TOPIC: **Vectors**

Source: Nov 2025 4WM2H

24 The diagram shows a quadrilateral  $OABC$ Diagram **NOT**  
accurately drawn

$$\vec{OA} = 5\mathbf{a} \quad \vec{OC} = 10\mathbf{b} \quad \vec{AB} = 2\mathbf{a} + 4\mathbf{b}$$

(a) Find  $\vec{AC}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$ 

$$\vec{AC} = \dots\dots\dots (1)$$

(b) Find  $\vec{OB}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$   
Give your answer in its simplest form.

$$\vec{OB} = \dots\dots\dots (1)$$

---



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The point  $P$  is such that  $APC$  and  $OPB$  are straight lines.(c) Using a vector method, find  $\vec{OP}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$   
Give your answer in its simplest form.  
Show your working clearly.

$$\vec{OP} = \dots\dots\dots (4)$$

(Total for Question 24 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM2H | Question 24

Vectors

(a) Find  $\overrightarrow{AC}$ :

$$\overrightarrow{AC} = \overrightarrow{AO} + \overrightarrow{OC} = -\overrightarrow{OA} + \overrightarrow{OC} = -5\mathbf{a} + 10\mathbf{b}$$

**FINAL ANSWER**

$$a - 5\mathbf{a} + 10\mathbf{b}$$

(b) Find  $\overrightarrow{OB}$ :

$$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = 5\mathbf{a} + (2\mathbf{a} + 4\mathbf{b}) = 7\mathbf{a} + 4\mathbf{b}$$

**FINAL ANSWER**

$$b \ 7\mathbf{a} + 4\mathbf{b}$$

**1** Set up vector equations for  $\overrightarrow{OP}$ .

Since  $OPB$  is a straight line,  $\overrightarrow{OP}$  is a scalar multiple of  $\overrightarrow{OB}$ . Let this be  $\lambda$ .

$$\overrightarrow{OP} = \lambda(7\mathbf{a} + 4\mathbf{b}) = 7\lambda\mathbf{a} + 4\lambda\mathbf{b}$$

Since  $P$  lies on the straight line  $AC$ ,  $\overrightarrow{AP} = \mu\overrightarrow{AC}$  for some scalar  $\mu$ .

$$\begin{aligned}\overrightarrow{OP} &= \overrightarrow{OA} + \overrightarrow{AP} = 5\mathbf{a} + \mu(-5\mathbf{a} + 10\mathbf{b}) \\ \overrightarrow{OP} &= (5 - 5\mu)\mathbf{a} + 10\mu\mathbf{b}\end{aligned}$$

**2** Equate the coefficients of

a and b.

$$7\lambda = 5 - 5\mu \quad (1)$$

$$4\lambda = 10\mu \implies \mu = \frac{4}{10}\lambda = 0.4\lambda \quad (2)$$

**3** Solve for  $\lambda$ .

Substitute (2) into (1):

$$7\lambda = 5 - 5(0.4\lambda)$$

$$7\lambda = 5 - 2\lambda$$

$$9\lambda = 5 \implies \lambda = \frac{5}{9}$$

**4** Find  $\overrightarrow{OP}$ .

$$\overrightarrow{OP} = \frac{5}{9}(7\mathbf{a} + 4\mathbf{b}) = \frac{35}{9}\mathbf{a} + \frac{20}{9}\mathbf{b}$$

**FINAL ANSWER**

$$\frac{35}{9}\mathbf{a} + \frac{20}{9}\mathbf{b}$$

**CLASSIFIED PROBLEM**

Paper: 4WM2H

Session: Specimen

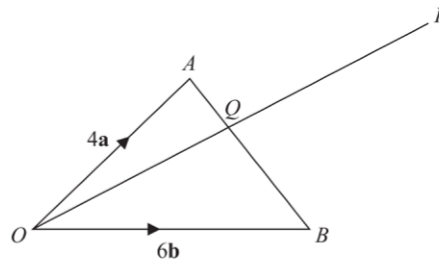
Q#: 23

Marks: 5

TOPIC: **Vectors**

Source: Specimen 4WM2H

23

Diagram **NOT**  
accurately drawn $OAB$  is a triangle. $Q$  is the point on  $AB$  such that  $OQP$  is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio  $AQ : QB$ 

$$AQ : QB = \dots\dots\dots$$

(Total for Question 23 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Specimen 4WM2H | Question 23

Vectors

**Topic check.** Vectors. The tag is correct.**Solution**

$$\overrightarrow{OA} = 4a, \quad \overrightarrow{OB} = 6b, \quad \overrightarrow{AP} = 2a + 8b$$

So

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$$

$$\overrightarrow{OP} = 4a + (2a + 8b) = 6a + 8b$$

Since  $O, Q, P$  are collinear, let

$$\overrightarrow{OQ} = \lambda(6a + 8b)$$

Also  $Q$  lies on  $AB$ . Let

$$\overrightarrow{AQ} = \mu\overrightarrow{AB}$$

Then

$$\overrightarrow{AB} = 6b - 4a$$

$$\overrightarrow{OQ} = 4a + \mu(6b - 4a)$$

$$\overrightarrow{OQ} = (4 - 4\mu)a + 6\mu b$$

Compare coefficients with

$$\lambda(6a + 8b) = 6\lambda a + 8\lambda b$$

So

$$4 - 4\mu = 6\lambda, \quad 6\mu = 8\lambda$$

From  $6\mu = 8\lambda$ ,

$$\lambda = \frac{3}{4}\mu$$

Substitute:

$$4 - 4\mu = 6\left(\frac{3}{4}\mu\right)$$

$$4 - 4\mu = \frac{9}{2}\mu$$

$$8 = 17\mu$$

$$\mu = \frac{8}{17}$$

Therefore

$$AQ : QB = \frac{8}{17} : \frac{9}{17} = 8 : 9$$

**FINAL ANSWER**

$$AQ : QB = 8 : 9.$$

**CLASSIFIED PROBLEM**

Paper: P2H

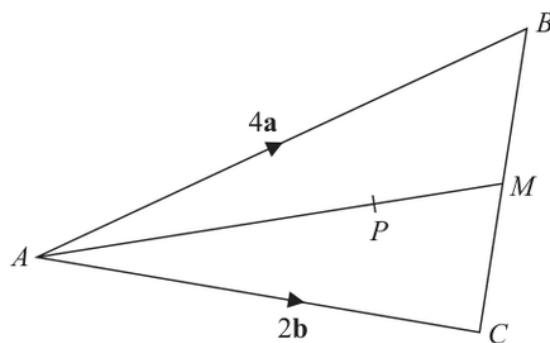
Session: Jan 2020

Q#: 23

Marks: 3

TOPIC: **Vectors**

Source: Jan 2020 P2H

**23**Diagram **NOT**  
accurately drawn

$ABC$  is a triangle.  
The midpoint of  $BC$  is  $M$ .  
 $P$  is a point on  $AM$ .

$$\vec{AB} = 4\mathbf{a}$$

$$\vec{AC} = 2\mathbf{b}$$

$$\vec{AP} = \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$$

Find the ratio  $AP:PM$

(Total for Question 23 is 3 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 23

Vectors

**Topic check.** Corrected from Ratio Toolkit to Vectors.**Method**Since  $M$  is the midpoint of  $BC$ ,

$$\overrightarrow{AM} = \frac{\overrightarrow{AB} + \overrightarrow{AC}}{2}$$

$$\overrightarrow{AM} = \frac{4a + 2b}{2} = 2a + b$$

Given

$$\overrightarrow{AP} = \frac{3}{2}a + \frac{3}{4}b$$

Factorise:

$$\overrightarrow{AP} = \frac{3}{4}(2a + b) = \frac{3}{4}\overrightarrow{AM}$$

So  $AP$  is  $\frac{3}{4}$  of  $AM$ , leaving  $PM$  as  $\frac{1}{4}$  of  $AM$ .**FINAL ANSWER**

$$AP : PM = 3 : 1$$



**CLASSIFIED PROBLEM**

Paper: P2HR

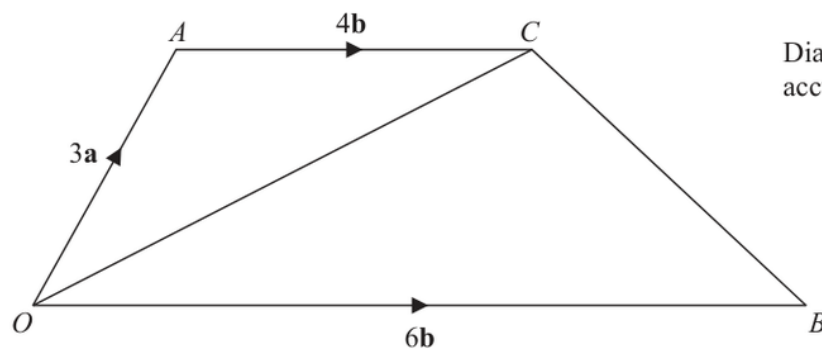
Session: Jan 2020

Q#: 26

Marks: 5

TOPIC: **Vectors**

Source: Jan 2020 P2HR

**26** The diagram shows trapezium  $OACB$ .Diagram **NOT**  
accurately drawn

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AC} = 4\mathbf{b}$$

 $N$  is the point on  $OC$  such that  $ANB$  is a straight line.Find  $\vec{ON}$  as a simplified expression in terms of  $\mathbf{a}$  and  $\mathbf{b}$ .

(Total for Question 26 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 26  
Vectors

**Topic check.** Moved from Linear Graphs to Vectors.

**Method**

$$\vec{OA} = 3\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}, \quad \vec{AC} = 4\mathbf{b}$$

So

$$\vec{OC} = 3\mathbf{a} + 4\mathbf{b}$$

Point  $N$  lies on  $OC$ , so

$$\vec{ON} = t(3\mathbf{a} + 4\mathbf{b})$$

Point  $N$  also lies on  $AB$ , so

$$\vec{ON} = 3\mathbf{a} + s(6\mathbf{b} - 3\mathbf{a})$$

$$\vec{ON} = (3 - 3s)\mathbf{a} + 6s\mathbf{b}$$

Equate coefficients:

$$3t = 3 - 3s$$

$$4t = 6s$$

So

$$t = 1 - s$$

and

$$t = \frac{3}{2}s$$

$$\frac{3}{2}s = 1 - s$$

$$5s = 2$$

$$s = \frac{2}{5}$$

Then

$$t = \frac{3}{5}$$

Therefore

$$\vec{ON} = \frac{3}{5}(3\mathbf{a} + 4\mathbf{b})$$

$$\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

**FINAL ANSWER**

$$\vec{ON} = \frac{9}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

**CLASSIFIED PROBLEM**

Paper: P1H

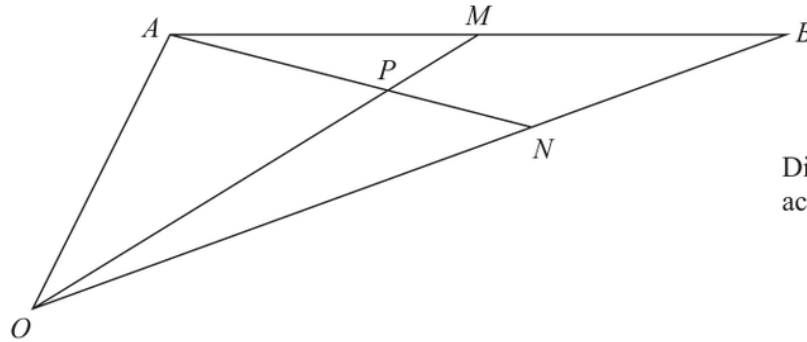
Session: Jan 2021

Q#: 23

Marks: 6

TOPIC: **Vectors**

Source: Jan 2021 P1H

**23**  $OAB$  is a triangle.Diagram **NOT**  
accurately drawn

$$\vec{OA} = 2\mathbf{a} \text{ and } \vec{OB} = 2\mathbf{b}$$

 $M$  is the midpoint of  $AB$ . $N$  is the point on  $OB$  such that  $ON:NB = 2:1$  $P$  is the point on  $AN$  such that  $OPM$  is a straight line.Use a vector method to find  $OP:PM$ 

Show your working clearly.

(Total for Question 23 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1H | Question 23

Vectors

**Topic check.** Moved from Coordinate Geometry to Vectors.**Method**

$$\vec{OA} = 2\mathbf{a}, \quad \vec{OB} = 2\mathbf{b}$$

Since  $M$  is the midpoint of  $AB$ ,

$$\vec{OM} = \frac{2\mathbf{a} + 2\mathbf{b}}{2}$$

$$\vec{OM} = \mathbf{a} + \mathbf{b}$$

Since  $ON : NB = 2 : 1$ ,

$$\vec{ON} = \frac{2}{3}\vec{OB}$$

$$\vec{ON} = \frac{2}{3}(2\mathbf{b}) = \frac{4}{3}\mathbf{b}$$

Point  $P$  lies on  $OM$ , so

$$\vec{OP} = t(\mathbf{a} + \mathbf{b})$$

Point  $P$  also lies on  $AN$ , so

$$\vec{OP} = 2\mathbf{a} + s\left(\frac{4}{3}\mathbf{b} - 2\mathbf{a}\right)$$

$$\vec{OP} = (2 - 2s)\mathbf{a} + \frac{4s}{3}\mathbf{b}$$

Equate coefficients of  $\mathbf{a}$  and  $\mathbf{b}$ :

$$t = 2 - 2s$$

$$t = \frac{4s}{3}$$

So

$$\frac{4s}{3} = 2 - 2s$$

$$4s = 6 - 6s$$

$$10s = 6$$

$$s = \frac{3}{5}$$

Then

$$t = \frac{4}{5}$$

So  $P$  is  $\frac{4}{5}$  of the way from  $O$  to  $M$ .

**FINAL ANSWER**

$$OP : PM = 4 : 1$$

**CLASSIFIED PROBLEM**

Paper: P1H

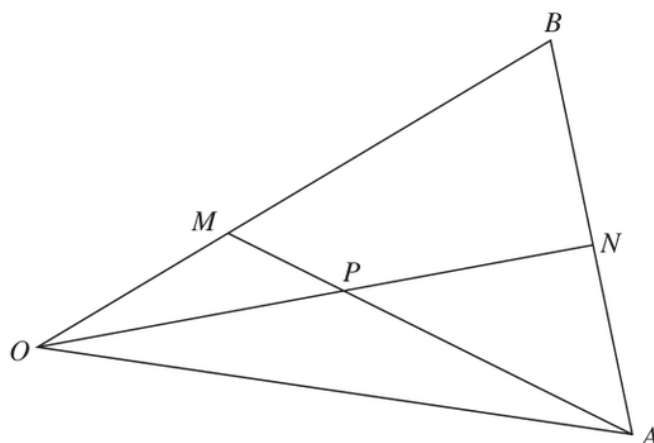
Session: Jan 2022

Q#: 22

Marks: 5

TOPIC: **Vectors**

Source: Jan 2022 P1H

**22** The diagram shows triangle  $OAB$ Diagram **NOT**  
accurately drawn

$$\vec{OA} = 8\mathbf{a} \quad \vec{OB} = 6\mathbf{b}$$

 $M$  is the point on  $OB$  such that  $OM : MB = 1 : 2$  $N$  is the midpoint of  $AB$  $P$  is the point of intersection of  $ON$  and  $AM$ 

Using a vector method, find  $\vec{OP}$  as a simplified expression in terms of  $\mathbf{a}$  and  $\mathbf{b}$   
Show your working clearly.

$$\vec{OP} = \dots\dots\dots$$

(Total for Question 22 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1H | Question 22

Vectors

**Topic check.** Moved from Coordinate Geometry to Vectors.**Method**

$$\vec{OA} = 8\mathbf{a}, \quad \vec{OB} = 6\mathbf{b}$$

Since  $OM : MB = 1 : 2$ ,

$$\vec{OM} = \frac{1}{3}\vec{OB}$$

$$\vec{OM} = 2\mathbf{b}$$

Since  $N$  is the midpoint of  $AB$ ,

$$\vec{ON} = \frac{8\mathbf{a} + 6\mathbf{b}}{2}$$

$$\vec{ON} = 4\mathbf{a} + 3\mathbf{b}$$

Point  $P$  lies on  $ON$ , so

$$\vec{OP} = \lambda(4\mathbf{a} + 3\mathbf{b})$$

Point  $P$  also lies on  $AM$ , so

$$\vec{OP} = 8\mathbf{a} + \mu(2\mathbf{b} - 8\mathbf{a})$$

$$\vec{OP} = (8 - 8\mu)\mathbf{a} + 2\mu\mathbf{b}$$

Equate coefficients:

$$4\lambda = 8 - 8\mu$$

$$3\lambda = 2\mu$$

From  $3\lambda = 2\mu$ ,

$$\mu = \frac{3}{2}\lambda$$

Substitute:

$$4\lambda = 8 - 8\left(\frac{3}{2}\lambda\right)$$

$$4\lambda = 8 - 12\lambda$$

$$16\lambda = 8$$



$$\lambda = \frac{1}{2}$$

Therefore

$$\vec{OP} = \frac{1}{2}(4\mathbf{a} + 3\mathbf{b})$$

$$\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$$

**FINAL ANSWER**

$$\vec{OP} = 2\mathbf{a} + \frac{3}{2}\mathbf{b}$$

**CLASSIFIED PROBLEM**

Paper: P1HR

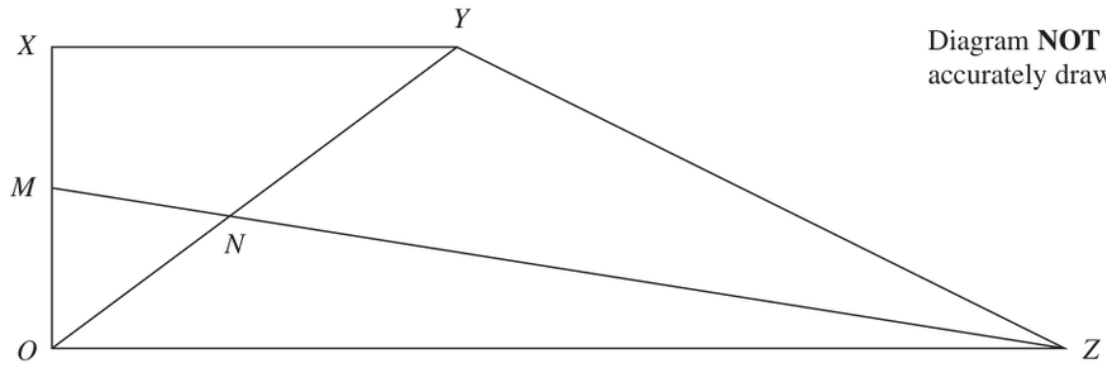
Session: Jan 2022

Q#: 24

Marks: 5

TOPIC: **Vectors**

Source: Jan 2022 P1HR

**24**  $OXYZ$  is a trapezium.

$$\overrightarrow{OX} = \mathbf{a}$$

$$\overrightarrow{XY} = \mathbf{b}$$

$$\overrightarrow{OZ} = 3\mathbf{b}$$

$M$  is the midpoint of  $OX$

$N$  is the point such that  $MNZ$  and  $ONY$  are straight lines.

Given that  $ON : OY = \lambda : 1$

use a vector method to find the value of  $\lambda$

$$\lambda = \dots\dots\dots$$

(Total for Question 24 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1HR | Question 24

Vectors

**Topic check.** Moved from Coordinate Geometry to Vectors.**Method**Take  $O$  as the origin.

$$\vec{OX} = \mathbf{a}, \quad \vec{XY} = \mathbf{b}, \quad \vec{OZ} = 3\mathbf{b}$$

So

$$\vec{OY} = \mathbf{a} + \mathbf{b}$$

Since  $M$  is the midpoint of  $OX$ ,

$$\vec{OM} = \frac{1}{2}\mathbf{a}$$

Given  $ON : OY = \lambda : 1$ ,

$$\vec{ON} = \lambda(\mathbf{a} + \mathbf{b})$$

Point  $N$  also lies on  $MZ$ , so

$$\vec{ON} = \frac{1}{2}\mathbf{a} + t\left(3\mathbf{b} - \frac{1}{2}\mathbf{a}\right)$$

$$\vec{ON} = \left(\frac{1}{2} - \frac{t}{2}\right)\mathbf{a} + 3t\mathbf{b}$$

Equate coefficients with  $\lambda\mathbf{a} + \lambda\mathbf{b}$ :

$$\lambda = \frac{1}{2} - \frac{t}{2}$$

$$\lambda = 3t$$

So

$$t = \frac{\lambda}{3}$$

Substitute:

$$\lambda = \frac{1}{2} - \frac{1}{2}\left(\frac{\lambda}{3}\right)$$

$$\lambda = \frac{1}{2} - \frac{\lambda}{6}$$

$$\frac{7}{6}\lambda = \frac{1}{2}$$

$$\lambda = \frac{3}{7}$$

**FINAL ANSWER**

$$\lambda = \frac{3}{7}$$

**CLASSIFIED PROBLEM**

Paper: P2H

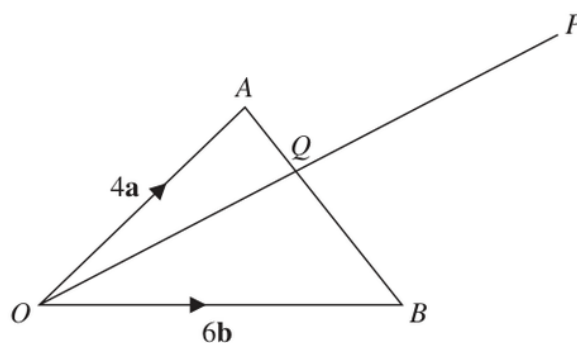
Session: Jun 2022

Q#: 21

Marks: 5

TOPIC: **Vectors**

Source: Jun 2022 P2H

**21**Diagram **NOT**  
accurately drawn $OAB$  is a triangle. $Q$  is the point on  $AB$  such that  $OQP$  is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio  $AQ:QB$ 

$$AQ:QB = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2H | Question 21

Vectors

**Topic check.** Corrected from Ratio Toolkit to Vectors.**Method**

$$\overrightarrow{OA} = 4a, \quad \overrightarrow{OB} = 6b, \quad \overrightarrow{AP} = 2a + 8b$$

So

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 4a + (2a + 8b) = 6a + 8b$$

Since  $Q$  lies on  $OP$ , let

$$\overrightarrow{OQ} = \lambda(6a + 8b)$$

Since  $Q$  lies on  $AB$ , let

$$\overrightarrow{OQ} = \overrightarrow{OA} + \mu\overrightarrow{AB}$$

$$\overrightarrow{AB} = 6b - 4a$$

So

$$\overrightarrow{OQ} = 4a + \mu(6b - 4a) = (4 - 4\mu)a + 6\mu b$$

Equating coefficients:

$$6\lambda = 4 - 4\mu, \quad 8\lambda = 6\mu$$

From  $8\lambda = 6\mu$ ,  $\mu = \frac{4}{3}\lambda$ .

$$6\lambda = 4 - \frac{16}{3}\lambda$$

$$34\lambda = 12, \quad \lambda = \frac{6}{17}$$

$$\mu = \frac{8}{17}$$

So  $AQ : QB = \frac{8}{17} : \frac{9}{17}$ .**FINAL ANSWER**

$$AQ : QB = 8 : 9$$

**CLASSIFIED PROBLEM**

Paper: P2HR

Session: Jun 2022

Q#: 25

Marks: 5

TOPIC: **Vectors**

Source: Jun 2022 P2HR

25  $ABCD$  is a parallelogram and  $ADM$  is a straight line.

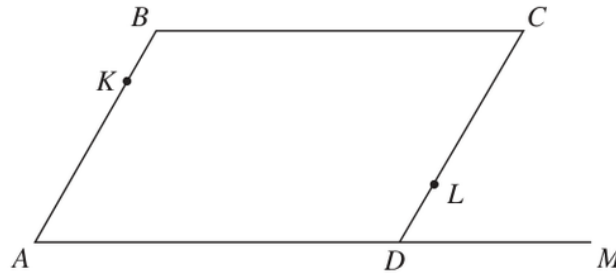


Diagram **NOT**  
accurately drawn

$$\vec{AB} = \mathbf{a} \quad \vec{BC} = \mathbf{b} \quad \vec{DM} = \frac{1}{2} \mathbf{b}$$

$K$  is the point on  $AB$  such that  $AK:AB = \lambda:1$

$L$  is the point on  $CD$  such that  $CL:CD = \mu:1$

$KLM$  is a straight line.

Given that  $\lambda:\mu = 1:2$

use a vector method to find the value of  $\lambda$  and the value of  $\mu$

$$\lambda = \dots\dots\dots$$

$$\mu = \dots\dots\dots$$

(Total for Question 25 is 5 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2HR | Question 25  
Vectors

**Topic check.** Moved from Linear Graphs to Vectors.

**Method**

Let  $A$  be the origin.

$$\vec{AB} = \mathbf{a}, \quad \vec{BC} = \mathbf{b}$$

So

$$B = \mathbf{a}, \quad D = \mathbf{b}, \quad C = \mathbf{a} + \mathbf{b}$$

Also

$$D\vec{M} = \frac{1}{2}\mathbf{b}$$

So

$$M = \frac{3}{2}\mathbf{b}$$

Point  $K$  is on  $AB$ :

$$K = \lambda\mathbf{a}$$

Point  $L$  is on  $CD$ , and  $C\vec{D} = -\mathbf{a}$ :

$$L = C + \mu(D - C)$$

$$L = \mathbf{a} + \mathbf{b} + \mu(-\mathbf{a})$$

$$L = (1 - \mu)\mathbf{a} + \mathbf{b}$$

Given

$$\lambda : \mu = 1 : 2$$

$$\mu = 2\lambda$$

Since  $K, L, M$  are collinear,  $K\vec{M}$  is a multiple of  $K\vec{L}$ .

$$K\vec{M} = -\lambda\mathbf{a} + \frac{3}{2}\mathbf{b}$$

$$K\vec{L} = (1 - \mu - \lambda)\mathbf{a} + \mathbf{b}$$

$$K\vec{L} = (1 - 3\lambda)\mathbf{a} + \mathbf{b}$$

Compare coefficients using



$$K\vec{M} = \frac{3}{2}K\vec{L}$$

$$-\lambda = \frac{3}{2}(1 - 3\lambda)$$

$$-2\lambda = 3 - 9\lambda$$

$$7\lambda = 3$$

$$\lambda = \frac{3}{7}$$

Then

$$\mu = 2\lambda = \frac{6}{7}$$

**FINAL ANSWER**

$$\lambda = \frac{3}{7}, \quad \mu = \frac{6}{7}$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Jun 2024

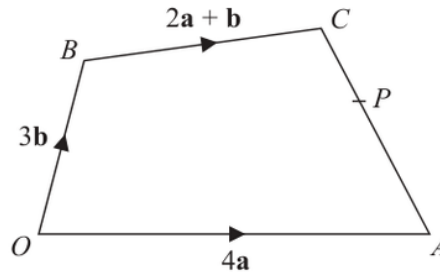
Q#: 24

Marks: 6

TOPIC: **Vectors**

Source: Jun 2024 P2H

24

Diagram **NOT**  
accurately drawnThe diagram shows a quadrilateral  $OACB$  in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

- (a) Find  $\vec{AC}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$   
Give your answer in its simplest form.

$$\vec{AC} = \dots\dots\dots (2)$$

The point  $P$  lies on  $AC$  such that  $AP:PC = 3:2$ The point  $Q$  is such that  $OPQ$  and  $BCQ$  are straight lines.

- (b) Using a vector method, find  $\vec{OQ}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$   
Give your answer in its simplest form.  
Show your working clearly.

$$\vec{OQ} = \dots\dots\dots (4)$$

**(Total for Question 24 is 6 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2024 P2H | Question 24

Vectors

**Topic check.** Moved from Linear Graphs to Vectors.**Method**

$$\vec{OA} = 4\mathbf{a}, \quad \vec{OB} = 3\mathbf{b}, \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

So

$$\vec{OC} = \vec{OB} + \vec{BC}$$

$$\vec{OC} = 2\mathbf{a} + 4\mathbf{b}$$

For part (a),

$$\vec{AC} = \vec{OC} - \vec{OA}$$

$$\vec{AC} = (2\mathbf{a} + 4\mathbf{b}) - 4\mathbf{a}$$

$$\vec{AC} = 4\mathbf{b} - 2\mathbf{a}$$

For part (b),  $AP : PC = 3 : 2$ , so

$$\vec{OP} = \vec{OA} + \frac{3}{5}\vec{AC}$$

$$\vec{OP} = 4\mathbf{a} + \frac{3}{5}(4\mathbf{b} - 2\mathbf{a})$$

$$\vec{OP} = \frac{14}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}$$

Since  $O, P, Q$  are collinear,

$$\vec{OQ} = t\vec{OP}$$

$$\vec{OQ} = t\left(\frac{14}{5}\mathbf{a} + \frac{12}{5}\mathbf{b}\right)$$

Since  $B, C, Q$  are collinear,

$$\vec{OQ} = \vec{OB} + s\vec{BC}$$

$$\vec{OQ} = 3\mathbf{b} + s(2\mathbf{a} + \mathbf{b})$$

$$\vec{OQ} = 2s\mathbf{a} + (3 + s)\mathbf{b}$$

Equate coefficients:

$$\frac{14}{5}t = 2s$$

$$\frac{12}{5}t = 3 + s$$

From the first equation,

$$s = \frac{7}{5}t$$

Substitute into the second:

$$\frac{12}{5}t = 3 + \frac{7}{5}t$$

$$t = 3$$

Therefore

$$\vec{OQ} = 3\vec{OP}$$

$$\vec{OQ} = \frac{42}{5}\mathbf{a} + \frac{36}{5}\mathbf{b}$$

**FINAL ANSWER**

$$\vec{AC} = 4\mathbf{b} - 2\mathbf{a}, \vec{OQ} = \frac{42}{5}\mathbf{a} + \frac{36}{5}\mathbf{b}$$

## CLASSIFIED PROBLEM

Paper: P2HR

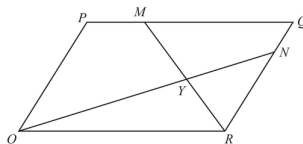
Session: Jun 2024

Q#: 25

Marks: 6

TOPIC: **Vectors**

Source: Jun 2024 P2HR

25  $OPQR$  is a parallelogram.Diagram **NOT**  
accurately drawn

$$\vec{OP} = 2\mathbf{a} \quad \text{and} \quad \vec{OR} = 3\mathbf{b}$$

The point  $M$  lies on  $PQ$  such that  $PM = \frac{1}{4}PQ$

The point  $N$  lies on  $RQ$  such that  $RN = \frac{4}{5}RQ$

(a) Find, in terms of  $\mathbf{a}$  and  $\mathbf{b}$ , giving your answers in simplest form

(i)  $\vec{ON}$

(1)

(ii)  $\vec{MR}$

(1)

$MR$  and  $ON$  intersect at the point  $Y$

Given that

$$OY = k \times ON$$

(b) use a vector method to find the value of  $k$

$$k = \underline{\hspace{2cm}}$$

(4)

(Total for Question 25 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2HR | Question 25

Vectors

Topic check. Correct.

## Method

$$\overrightarrow{OP} = 2a, \quad \overrightarrow{OR} = 3b$$

So

$$\overrightarrow{OQ} = 2a + 3b$$

Since  $PM : PQ = 1 : 4$ ,

$$\overrightarrow{PM} = \frac{1}{4}\overrightarrow{PQ}$$

$$\overrightarrow{PM} = \frac{1}{4}(3b) = \frac{3}{4}b$$

$$\overrightarrow{OM} = 2a + \frac{3}{4}b$$

Since  $RN : RQ = 4 : 5$ ,

$$\overrightarrow{RN} = \frac{4}{5}\overrightarrow{RQ}$$

$$\overrightarrow{RN} = \frac{4}{5}(2a) = \frac{8}{5}a$$

$$\overrightarrow{ON} = 3b + \frac{8}{5}a$$

So

$$\overrightarrow{MR} = \overrightarrow{OR} - \overrightarrow{OM}$$

$$\overrightarrow{MR} = 3b - \left(2a + \frac{3}{4}b\right)$$

$$\overrightarrow{MR} = -2a + \frac{9}{4}b$$

Let  $Y$  be on  $ON$ , so

$$\overrightarrow{OY} = k\overrightarrow{ON} = k\left(\frac{8}{5}a + 3b\right)$$

Also  $Y$  is on  $MR$ , so

$$\overrightarrow{OY} = \overrightarrow{OM} + t\overrightarrow{MR}$$

$$\overrightarrow{OY} = 2a + \frac{3}{4}b + t\left(-2a + \frac{9}{4}b\right)$$

Compare coefficients of  $a$  and  $b$ :

$$\frac{8}{5}k = 2 - 2t$$

$$3k = \frac{3}{4} + \frac{9}{4}t$$

From the first equation,

$$t = 1 - \frac{4}{5}k$$

Substitute into the second equation:

$$3k = \frac{3}{4} + \frac{9}{4} \left( 1 - \frac{4}{5}k \right)$$

$$3k = 3 - \frac{9}{5}k$$

$$\frac{24}{5}k = 3$$

$$k = \frac{5}{8}$$

**FINAL ANSWER**

$$\overrightarrow{ON} = \frac{8}{5}a + 3b, \overrightarrow{MR} = -2a + \frac{9}{4}b, \text{ and } k = \frac{5}{8}$$

**CLASSIFIED PROBLEM**

Paper: P1H

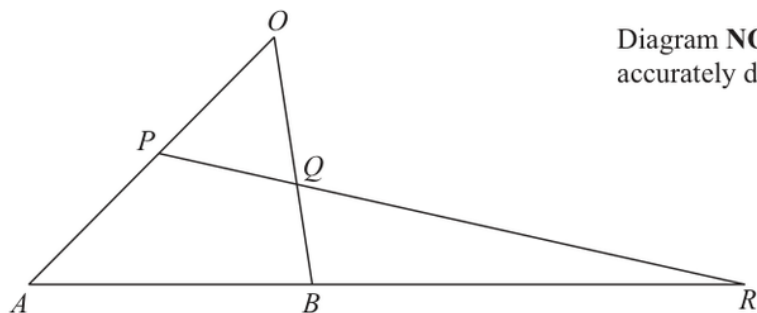
Session: Jun 2025

Q#: 23

Marks: 5

TOPIC: **Vectors**

Source: Jun 2025 P1H

**23**Diagram **NOT**  
accurately drawn $OAB$  is a triangle. $P$  is the midpoint of  $OA$  $Q$  is a point on  $OB$  $ABR$  and  $PQR$  are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

- (a) Express  $\vec{AB}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$

.....  
(1)

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

- (b) Use a vector method to find the value of  $n$

$$n = \text{.....}$$

(4)

**(Total for Question 23 is 5 marks)**



## WORKED SOLUTION

Dr. Eslam Ahmed

## Jun 2025 P1H | Question 23

Vectors

**Topic check.** Moved from Coordinate Geometry to Vectors.**Method**

For part (a),

$$\vec{OA} = 12\mathbf{a}, \quad \vec{OB} = 8\mathbf{b}$$

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$\vec{AB} = 8\mathbf{b} - 12\mathbf{a}$$

For part (b),  $P$  is the midpoint of  $OA$ , so

$$\vec{OP} = 6\mathbf{a}$$

Given

$$AB : BR = 1 : 2$$

$$\vec{BR} = 2\vec{AB}$$

$$\vec{BR} = 16\mathbf{b} - 24\mathbf{a}$$

Therefore

$$\vec{OR} = \vec{OB} + \vec{BR}$$

$$\vec{OR} = 8\mathbf{b} + 16\mathbf{b} - 24\mathbf{a}$$

$$\vec{OR} = -24\mathbf{a} + 24\mathbf{b}$$

Since  $P, Q, R$  are collinear, write  $Q$  on  $PR$ :

$$\vec{OQ} = \vec{OP} + t(\vec{OR} - \vec{OP})$$

$$\vec{OQ} = 6\mathbf{a} + t(-30\mathbf{a} + 24\mathbf{b})$$

$$\vec{OQ} = (6 - 30t)\mathbf{a} + 24t\mathbf{b}$$

But  $Q$  lies on  $OB$ , so there is no  $\mathbf{a}$  component:

$$6 - 30t = 0$$

$$t = \frac{1}{5}$$

Then

$$\vec{OQ} = 24 \left( \frac{1}{5} \right) \mathbf{b}$$

$$\vec{OQ} = \frac{24}{5} \mathbf{b}$$

Since  $\vec{OQ} = n\mathbf{b}$ ,

$$n = \frac{24}{5}$$

**FINAL ANSWER**

$$\vec{AB} = 8\mathbf{b} - 12\mathbf{a}, \quad n = \frac{24}{5}$$

**CLASSIFIED PROBLEM**

Paper: P1H

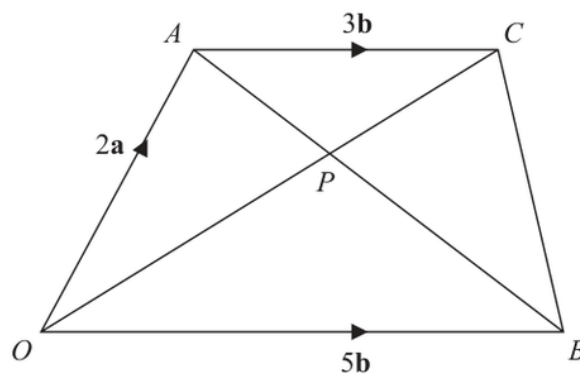
Session: May 2021

Q#: 26

Marks: 5

TOPIC: **Vectors**

Source: May 2021 P1H

**26**  $OACB$  is a trapezium.Diagram **NOT**  
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 5\mathbf{b} \quad \vec{AC} = 3\mathbf{b}$$

The diagonals,  $OC$  and  $AB$ , of the trapezium intersect at the point  $P$ .Find and simplify an expression, in terms of  $\mathbf{a}$  and  $\mathbf{b}$ , for  $\vec{OP}$   
Show your working clearly.

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$$\vec{OP} = \dots\dots\dots$$

**(Total for Question 26 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 26  
Vectors

**Topic check.** Moved to Vectors.

**Method**

$$\overrightarrow{OA} = 2\mathbf{a}, \quad \overrightarrow{OB} = 5\mathbf{b}, \quad \overrightarrow{AC} = 3\mathbf{b}$$

So

$$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = 2\mathbf{a} + 3\mathbf{b}$$

Since  $P$  lies on  $OC$ , let

$$\overrightarrow{OP} = t(2\mathbf{a} + 3\mathbf{b})$$

Since  $P$  also lies on  $AB$ , let

$$\overrightarrow{OP} = \overrightarrow{OA} + s\overrightarrow{AB}$$

Now

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 5\mathbf{b} - 2\mathbf{a}$$

So

$$\overrightarrow{OP} = 2\mathbf{a} + s(5\mathbf{b} - 2\mathbf{a})$$

$$\overrightarrow{OP} = (2 - 2s)\mathbf{a} + 5s\mathbf{b}$$

Compare this with

$$t(2\mathbf{a} + 3\mathbf{b}) = 2t\mathbf{a} + 3t\mathbf{b}$$

Equating coefficients:

$$2t = 2 - 2s$$

$$3t = 5s$$

From  $t = 1 - s$ ,

$$3(1 - s) = 5s$$

$$3 = 8s$$

$$s = \frac{3}{8}$$

So

$$t = 1 - \frac{3}{8} = \frac{5}{8}$$

Therefore

$$\overrightarrow{OP} = \frac{5}{8}(2\mathbf{a} + 3\mathbf{b})$$

$$\overrightarrow{OP} = \frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

**FINAL ANSWER**

$$\frac{5}{4}\mathbf{a} + \frac{15}{8}\mathbf{b}$$

**CLASSIFIED PROBLEM**

Paper: P2H

Session: Nov 2023

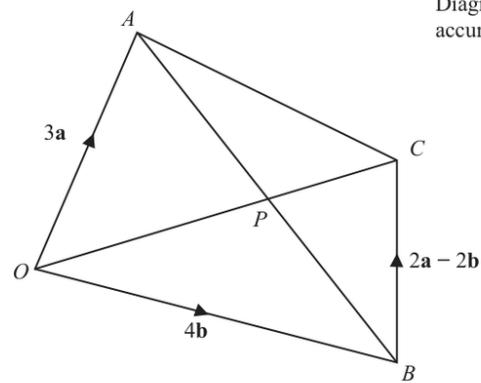
Q#: 22

Marks: 5

TOPIC: **Vectors**

Source: Nov 2023 P2H

22

Diagram **NOT**  
accurately drawn $OACB$  is a quadrilateral.

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 4\mathbf{b} \quad \vec{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector  $\vec{OC}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$   
Simplify your answer.

$$\vec{OC} = \dots\dots\dots (1)$$

- (ii) Find the vector  $\vec{AB}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$

$$\vec{AB} = \dots\dots\dots (1)$$

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The point  $P$  lies on  $AB$  and on  $OC$ 

- (b) Using a vector method, find the ratio  $AP : PB$   
Show your working clearly.

$$\dots\dots\dots (3)$$

**(Total for Question 22 is 5 marks)**

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2023 P2H | Question 22

Vectors

**Topic check.** Corrected from Ratio Toolkit to Vectors.**Method**

$$\overrightarrow{OA} = 3a, \quad \overrightarrow{OB} = 4b, \quad \overrightarrow{BC} = 2a - 2b$$

Part (a):

$$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = 4b + (2a - 2b) = 2a + 2b$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 4b - 3a$$

For part (b), let  $P$  divide  $AB$  so that

$$\overrightarrow{AP} = \mu \overrightarrow{AB}$$

Then

$$\overrightarrow{OP} = 3a + \mu(4b - 3a) = (3 - 3\mu)a + 4\mu b$$

Since  $P$  lies on  $OC$ , also let

$$\overrightarrow{OP} = \lambda(2a + 2b)$$

Equating coefficients:

$$3 - 3\mu = 2\lambda, \quad 4\mu = 2\lambda$$

From  $4\mu = 2\lambda$ ,  $\lambda = 2\mu$ .

$$3 - 3\mu = 4\mu$$

$$\mu = \frac{3}{7}$$

So  $AP : PB = \frac{3}{7} : \frac{4}{7}$ .**FINAL ANSWER**
 $\overrightarrow{OC} = 2a + 2b$ ,  $\overrightarrow{AB} = 4b - 3a$ , and  $AP : PB = 3 : 4$

**CLASSIFIED PROBLEM**

Paper: P1H

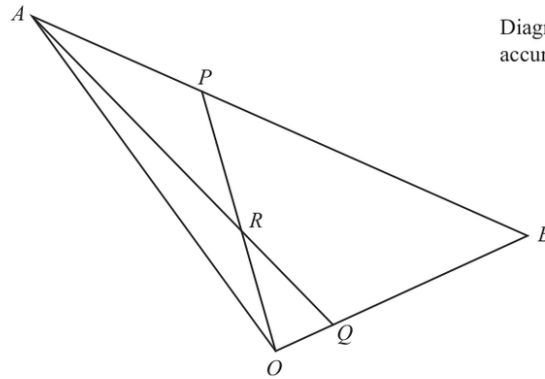
Session: Nov 2024

Q#: 24

Marks: 6

TOPIC: **Vectors**

Source: Nov 2024 P1H

24  $OAB$  is a triangle.Diagram **NOT**  
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

 $ARQ$  and  $ORP$  are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of  $\mathbf{a}$  and  $\mathbf{b}$   
Simplify your answers.

(i)  $\vec{AQ}$

(1)

(ii)  $\vec{OP}$

(1)

(iii)  $\vec{OR}$

(4)

(Total for Question 24 is 6 marks)



## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2024 P1H | Question 24

Vectors

**Topic check.** Moved from Linear Graphs to Vectors.**Method**

$$\vec{OA} = 10\mathbf{a}, \quad \vec{OB} = 10\mathbf{b}$$

Also,

$$\vec{AP} = \frac{1}{4}\vec{AB}$$

and

$$\vec{OQ} = \frac{1}{5}\vec{OB}$$

For part (i),

$$\vec{OQ} = 2\mathbf{b}$$

$$\vec{AQ} = \vec{OQ} - \vec{OA}$$

$$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}$$

For part (ii),

$$\vec{AB} = 10\mathbf{b} - 10\mathbf{a}$$

$$\vec{AP} = \frac{1}{4}(10\mathbf{b} - 10\mathbf{a})$$

$$\vec{AP} = \frac{5}{2}\mathbf{b} - \frac{5}{2}\mathbf{a}$$

$$\vec{OP} = \vec{OA} + \vec{AP}$$

$$\vec{OP} = 10\mathbf{a} - \frac{5}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

$$\vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

For part (iii),  $R$  lies on  $OP$ , so

$$\vec{OR} = t\vec{OP}$$

It also lies on  $AQ$ , so

$$\vec{OR} = \vec{OA} + s\vec{AQ}$$

$$\vec{OR} = 10\mathbf{a} + s(2\mathbf{b} - 10\mathbf{a})$$

$$\vec{OR} = (10 - 10s)\mathbf{a} + 2s\mathbf{b}$$

Equate with

$$t\left(\frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}\right)$$

$$\frac{15}{2}t = 10 - 10s$$

$$\frac{5}{2}t = 2s$$

From the second equation,

$$s = \frac{5}{4}t$$

Substitute:

$$\frac{15}{2}t = 10 - \frac{25}{2}t$$

$$20t = 10$$

$$t = \frac{1}{2}$$

So

$$\vec{OR} = \frac{1}{2}\vec{OP}$$

$$\vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

#### FINAL ANSWER

$$\vec{AQ} = 2\mathbf{b} - 10\mathbf{a}, \vec{OP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}, \vec{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

## CLASSIFIED PROBLEM

Paper: P1H

Session: Nov 2025

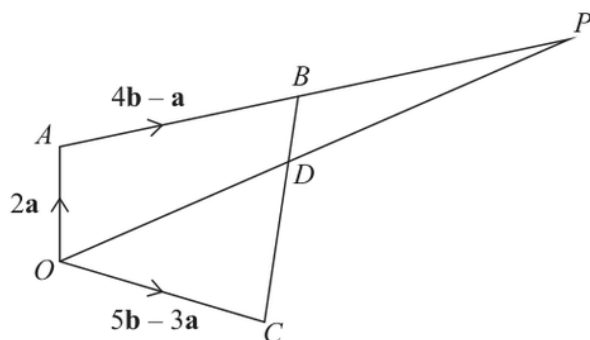
Q#: 27

Marks: 6

TOPIC: **Vectors**

Source: Nov 2025 P1H

27

Diagram NOT  
accurately drawn

$OABC$  is a quadrilateral.  
 $ABP$  and  $ODP$  are straight lines.

$$\vec{OA} = 2\mathbf{a} \quad \vec{AB} = 4\mathbf{b} - \mathbf{a} \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

- (a) Find an expression in terms of  $\mathbf{a}$  and  $\mathbf{b}$  for the vector  $\vec{BC}$   
 Simplify your answer.

(2)

The point  $D$  lies on  $BC$  such that  $BD:DC = 1:3$

Given that  $\vec{OP} = n\vec{OD}$

- (b) use a vector method to find the value of  $n$

$$n = \dots\dots\dots$$

(4)

(Total for Question 27 is 6 marks)

## WORKED SOLUTION

Dr. Eslam Ahmed

## Nov 2025 P1H | Question 27

Vectors

**Topic check.** Moved from Linear Graphs to Vectors.**Method**

$$\vec{OA} = 2\mathbf{a}, \quad \vec{AB} = 4\mathbf{b} - \mathbf{a}, \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

First find  $\vec{OB}$ :

$$\vec{OB} = \vec{OA} + \vec{AB}$$

$$\vec{OB} = 2\mathbf{a} + 4\mathbf{b} - \mathbf{a}$$

$$\vec{OB} = \mathbf{a} + 4\mathbf{b}$$

For part (a),

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$\vec{BC} = (5\mathbf{b} - 3\mathbf{a}) - (\mathbf{a} + 4\mathbf{b})$$

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}$$

For part (b),  $BD : DC = 1 : 3$ , so

$$\vec{OD} = \vec{OB} + \frac{1}{4}\vec{BC}$$

$$\vec{OD} = \mathbf{a} + 4\mathbf{b} + \frac{1}{4}(\mathbf{b} - 4\mathbf{a})$$

$$\vec{OD} = \frac{17}{4}\mathbf{b}$$

Since  $A, B, P$  are collinear, let

$$\vec{OP} = \vec{OA} + s\vec{AB}$$

$$\vec{OP} = 2\mathbf{a} + s(4\mathbf{b} - \mathbf{a})$$

$$\vec{OP} = (2 - s)\mathbf{a} + 4s\mathbf{b}$$

But  $O, D, P$  are collinear, and  $\vec{OD}$  has only a  $\mathbf{b}$  component.  
So

$$2 - s = 0$$

$$s = 2$$

Then

$$\vec{OP} = 8\mathbf{b}$$

Given

$$\vec{OP} = n\vec{OD}$$

$$8\mathbf{b} = n\left(\frac{17}{4}\mathbf{b}\right)$$

$$n = \frac{32}{17}$$

**FINAL ANSWER**

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}, \quad n = \frac{32}{17}$$